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1.1 Scientific Method

Scientists are tasked with discovering and developing the laws that govern nature using what is known as the **scientific method**. This method consists of a series of stages common to any branch of science. These stages are:

1. **Observation**. For example, if you try to dissolve sugar in a glass of water, you might notice that at some point the sugar no longer dissolves, no matter how much more you add.
2. **Problem Statement**. The next step is to ask a series of questions. For instance, in the previous case, you might ask how changing the temperature, the amount of sugar, or the amount of water would affect the outcome.
3. **Hypothesis Formulation**. A hypothesis is a provisional assumption made to explain a specific fact or phenomenon.
4. **Experimentation**. It is necessary to carry out experimental tests to verify the hypothesis.
5. **Analysis of Results**. Once experiments are conducted, you will have a set of data that must be analysed to determine whether the hypothesis is true.
6. **Formulation of Laws, Theories, or Models**. Once the results are analysed, if the hypothesis is correct, it may become a law, theory, or model. If it is not correct, a new hypothesis must be formulated, and the method restarted from that point.

1.2 Graph Representation

The graphical representation of experimental data is a fundamental tool in Physics and Chemistry. Graphs allow us to visualise relationships between variables, identify trends, and analyse physical or chemical behaviour in a clear and precise manner.

Steps for correctly constructing a graph:

1. Identify the variables:

- Independent variable (controlled): placed on the horizontal axis (X).
- Dependent variable (measured): placed on the vertical axis (Y).

2. Choose an appropriate scale:
   - Use proportional scales that cover the full range of the data.
   - Label the axes with regular units and values (e.g., every 2, 5, or 10 units).
3. Label the axes with names and units:
   - Example: Temperature (°C) on the X-axis and Volume (mL) on the Y-axis.
4. Plot the points precisely:
   - Use a ruler or software to plot the experimental points.
   - Do not connect the points with broken lines; instead, draw a trend line or curve if appropriate.
5. Add a title:
   - It should reflect the relationship between the variables. Example: *Relationship between pressure and gas volume*.
   - Another way to name it is by stating the quantity on the Y-axis, followed by the word versus, then the quantity on the X-axis. For example: pressure versus temperature – in this case, pressure is on the Y-axis and temperature on the X-axis.
6. Interpret the graph: you must indicate whether the dependence between the two quantities is directly proportional or inversely proportional (there are more types of dependence, but in this course we will only consider these two).
   - **Directly proportional** These are graphs shaped like straight lines of the form $y = m\,x + n$, where y is the quantity on the Y-axis and x on the X-axis. The letter m represents the gradient (slope) and may be positive or negative. The letter n is the y-intercept, i.e., the point where the line crosses the x-axis.

     R If one quantity increases, the other must also increase.

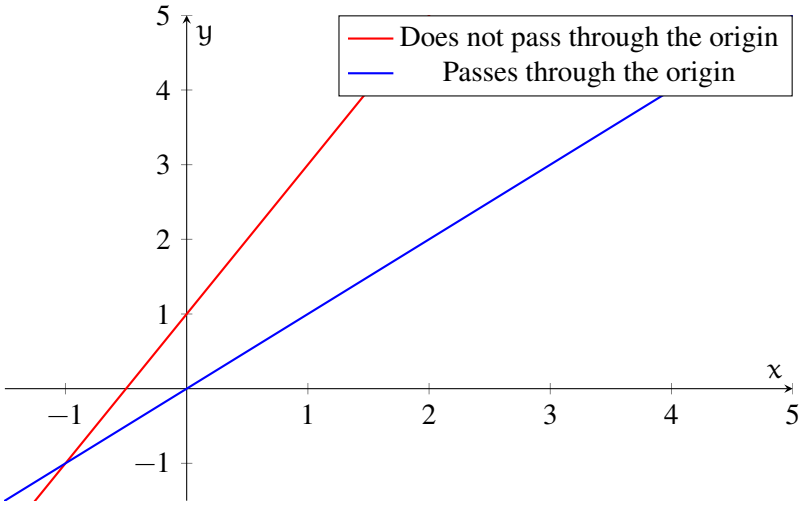


Figure (1.1): Graph of two quantities that are directly proportional

- **Inversely proportional** These are graphs in the form of a curve like $y = \frac{1}{x}$.

  R In this type of graph, if one quantity increases, the other must decrease.

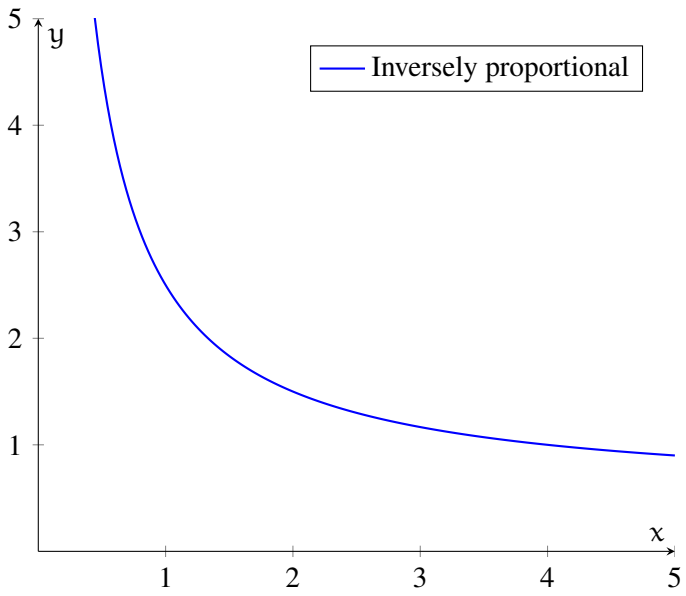


Figure (1.2): Graph of two quantities that are inversely proportional

1.3 Physical Quantities and Their Units

Definition 1.1 — Physical Quantity. Is any property of a body that we can measure. By property, we mean a characteristic of an object.

Every quantity, whether physical or chemical, is assigned a **unit**.

Definition 1.2 — Unit. A unit is a quantity defined by convention to establish a standard.

There are several systems of units, but the most widely used is the International System of Units (SI).

It was created in 1960 by the General Conference on Weights and Measures and consists of 7 base or fundamental units.

Thanks to the SI system, the reliability of object specifications in international trade can be guaranteed.

We classify quantities into two types:

- **Fundamental quantities**. They define these themselves; they are the basic quantities that cannot be broken down into other quantities.
  In the Table **??**, we list the fundamental quantities, their symbols, the SI units, and the unit symbols.

Table (1.1): Fundamental Quantities

Quantity	Quantity Symbol	Unit	Unit Symbol
Length	l	metre	m
Mass	m	kilogram	kg
Temperature	T	kelvin	K
Time	t	second	s
Electric current	I	ampere	A
Amount of substance	n	mole	mol
Luminous intensity	I_v	candela	cd

- **Derived quantities**. These are formed from the fundamental quantities and can be expressed as combinations of them.

  Table **??** lists some derived quantities with their units in the International System.

Table (1.2): Derived Quantities

Quantity	Quantity Symbol	Unit	Unit Symbol
Area	A	square metre	m^2
Volume	V	cubic metre	m^3
Density	ρ	–	$\frac{kg}{m^3}$
Speed	v	–	$\frac{m}{s}$
Force	F	newton	N
Pressure	p	pascal	Pa
Energy	E	joule	J

R If we want to determine whether a quantity is fundamental or derived, the simplest method is to learn the fundamental ones. If it is not fundamental, then it is derived.

Exercise 1.1 In your notebook, indicate which of the following quantities are fundamental and which are derived:

a) Electric current.
b) Pressure.
c) Speed.
d) Mass.
e) Temperature.
f) Density.
g) Energy.
h) Length.
i) Amount of substance

■

1.3.1 Multiples and Submultiples

When defining a system of units, it is also necessary to establish the multiples and submultiples of these units, which allow us to work with more manageable numerical quantities.

The following table shows the relationship between a unit and the various proportionality factors corresponding to its multiples and submultiples.

Table (1.3): Multiples

Order of Magnitude	Name	Symbol
10^{12}	tera-	T
10^{9}	giga-	G
10^{6}	mega-	M
10^{3}	kilo-	k
10^{2}	hecto-	h
10	deca-	da
Base Unit		

Table (1.4): Submultiples

Order of Magnitude	Name	Symbol
Base Unit		
10^{-1}	deci-	d
10^{-2}	centi-	c
10^{-3}	milli-	m
10^{-6}	micro-	μ
10^{-9}	nano-	n
10^{-12}	pico-	p

SI units are not the only ones in use; in fact, almost every quantity has several possible units. The following table lists quantities with various units and their equivalences with other units of the same quantity.

Table (1.5): Quantities with their equivalences

Quantity	Unit	Unit Equivalence
Volume	kL	$1\,\text{kL} = 1\,\text{m}^3$
	L	$1\,\text{L} = 1\,\text{dm}^3$
	mL	$1\,\text{mL} = 1\,\text{cm}^3$
Pressure	Atmosphere (atm)	$1\,\text{atm} = 10^5\,\text{Pa}$
	millimetre of mercury (mmHg)	$1\,\text{atm} = 760\,\text{mmHg}$
Energy	Calorie (cal)	$1\,\text{cal} = 4{,}18\,\text{J}$
Force	Kilopond (kp)	$1\,\text{kp} = 9{,}8\,\text{N}$

1.3.2 Converting Units Using Conversion Factors

We must first keep the following in mind:

a) We can convert between units as long as they belong to the same quantity.

b) We can convert between units with different multiples and submultiples.

c) We can also convert between different units as long as they represent the same quantity, and we know the equivalence between them.

■ **Example 1.1** Let us convert $36\,\frac{\text{km}}{\text{h}}$ to $\frac{\text{m}}{\text{s}}$:

$$36\,\frac{\cancel{\text{km}}}{\cancel{\text{h}}}\cdot\frac{1000\,\text{m}}{1\,\cancel{\text{km}}}\cdot\frac{1\,\cancel{\text{h}}}{3600\,\text{s}} = \frac{36\cdot 1000}{3600} = 10\,\frac{\text{m}}{\text{s}}$$

■

■ **Example 1.2** Convert 13 600 $\frac{\text{kg}}{\text{m}^3}$ to $\frac{\text{g}}{\text{cm}^3}$:

$$13600\ \frac{\cancel{\text{kg}}}{\cancel{\text{m}^3}} \cdot \frac{1000\ \text{g}}{1\ \cancel{\text{kg}}} \cdot \frac{1\ \cancel{\text{m}^3}}{10^6\ \text{cm}^3} = \frac{13600 \cdot 1000}{10^6} = 13{,}6\ \frac{\text{g}}{\text{cm}^3}$$

■

■ **Example 1.3** Convert 12 km to nm:

$$12\ \cancel{\text{km}} \cdot \frac{10^{12}\ \text{nm}}{1\ \cancel{\text{km}}} = 12 \cdot 10^{-12} = 1{,}2 \cdot 10^{-11}\ \text{nm}$$

■

Exercise 1.2 Convert the following speeds into the indicated units using conversion factors.

a) Convert the following speeds into $\frac{\text{m}}{\text{s}}$:

- 90 $\frac{\text{km}}{\text{h}}$
- 100 $\frac{\text{km}}{\text{h}}$
- 120 $\frac{\text{km}}{\text{h}}$
- 50 $\frac{\text{km}}{\text{h}}$

b) Convert the following speeds into $\frac{\text{km}}{\text{h}}$:

- 12 $\frac{\text{m}}{\text{s}}$
- 340 $\frac{\text{m}}{\text{s}}$
- 0,36 $\frac{\text{m}}{\text{s}}$
- $3 \cdot 10^8$ $\frac{\text{m}}{\text{s}}$

■

Exercise 1.3 Convert the following units into the corresponding SI units using conversion factors.

a) 200 cm^2 *b)* 125 cm^3 *c)* 1800 g *d)* 45 min

■

Exercise 1.4 Perform the following unit conversions into SI units using conversion factors

a) 300 mmHg *b)* 25 000 cal *c)* 21 atm *d)* 258 kp

■

Exercise 1.5 If 1 g of pure gold costs 31.32 €, calculate the price of 1 dg and 1 cg of this metal. Resolve the following exercise using factor conversion.

■

Exercise 1.6 If 1 kg of almonds costs 18.40 €, calculate the price of 1 hg and 1 dag of this product. Resolve the following exercise using factor conversion.

■

Exercise 1.7 If 1 L of olive oil costs 9.60 €, calculate the price of 1 dL and 1 cL of this liquid. Resolve the following exercise using factor conversion.

■

1.4 Scientific Notation

Scientific notation allows us to write huge or tiny numbers using powers of 10, with a positive exponent for huge numbers or a negative exponent for tiny ones.

For a number to be correctly expressed in scientific notation, it must have:

- A single non-zero digit to the left of the decimal point.
- A power of 10 raised to a positive or negative exponent.

For example:

- The mass of the Earth is: $6\,000\,000\,000\,000\,000\,000\,000\,000$ kg. As we can see, this number has 24 zeros, so writing it repeatedly is quite tedious. This number in scientific notation would be: $6 \cdot 10^{24}$ kg.
- The mass of the electron is: $0{,}000\,000\,000\,000\,000\,000\,000\,000\,000\,910\,9$ kg. This number in scientific notation is written as $9{,}3109 \cdot 10^{-31}$ kg.

The steps for converting to scientific notation are as follows:

- **Numbers greater than one**. Start at the right of the number. At that point, we place the decimal point and count how many positions we need to move it so that only one non-zero digit remains to its left. The number in scientific notation will be the new number multiplied by a power of 10, raised to the number of positions moved, with a positive exponent. That's why in the earlier case we get $6 \cdot 10^{24}$.
- **Numbers smaller than one**. We start at the decimal point and count how many positions we need to move it so that there is only one non-zero digit to its left. The resulting exponent will be negative.
- **Numbers that are already large or small but already include a power of 10** . We begin at the decimal point. The following cases may occur:
  - If the number is greater than one: We proceed as in the previous case, and the number of positions is *added* to the exponent of the decimal. Examples:
    - $625{,}5 \cdot 10^5$ becomes $6{,}255 \cdot 10^7$.
    - $45\,654{,}21 \cdot 10^{-2}$ becomes $4{,}5654 \cdot 10^{-2}$.
  - If the number is smaller than one: We proceed similarly, but in this case the number of positions is *subtracted* from the exponent. Examples:
    - $0{,}0523 \cdot 10^{-3}$ becomes $5{,}23 \cdot 10^{-5}$.

- $0{,}000\,254 \cdot 10^5$ becomes $2{,}54 \cdot 10^{-1}$.

Exercise 1.8 Write the following quantities in scientific notation:

a) 20 000 000 m
b) 250 000 000 L
c) 7 820 000 000 g
d) 254 830 368 km
e) 0,0001 s
f) 0,000 035 kg
g) 0,000 001 m
h) 0,000 005 001 g
i) 0,000 000 005 g

■

Exercise 1.9 Write the following quantities in scientific notation.

a) 0,000 05 g
b) 0,000 052 g
c) 2 000 000 m
d) 25 000 000 m
e) 301 000 s
f) 0,000 205 A
g) 20 520 000 000 kg
h) 5 000 000 000 kg

■

Exercise 1.10 Express the following measurements in SI units using conversion factors and scientific notation.

a) 0,5 g
b) 0,5 cm^3
c) 200 km^2
d) 2 $\frac{\text{g}}{\text{cm}^3}$
e) 76 hm
f) 3 mg
g) 9 dam
h) 25 ns

■

Exercise 1.11 Express the following volumes in SI units using conversion factors and scientific notation:

a) 0,5 mL
b) 0,001 mL
c) 33,3 kL
d) 10 000 L

■

Exercise 1.12 Make a conceptual map of the unit.

■

1.5 Vocabulary

■ **Vocabulary 1.1 — Scientific Method.** A systematic approach used by scientists to investigate phenomena, acquire new knowledge, or correct and integrate previous knowledge. It involves observation, hypothesis formulation, experimentation, and analysis of results to develop laws, theories, or models.

Translation to Spanish: Método científico

■ **Vocabulary 1.2 — Observation.** The first stage of the scientific method that involves carefully watching and recording phenomena or events in nature using the senses or instruments.
Translation to Spanish: Observación

■ **Vocabulary 1.3 — Hypothesis.** A provisional assumption or proposed explanation made to explain a specific fact or phenomenon that can be tested through experimentation.
Translation to Spanish: Hipótesis

■ **Vocabulary 1.4 — Experimentation.** The process of carrying out controlled experimental tests to verify or refute a hypothesis through systematic observation and measurement.
Translation to Spanish: Experimentación

■ **Vocabulary 1.5 — Law.** A statement based on repeated experimental observations that describes some aspect of nature, always occurring under the same conditions. Laws are often expressed mathematically.
Translation to Spanish: Ley

■ **Vocabulary 1.6 — Theory.** A well-substantiated explanation of some aspect of the natural world that is based on a body of facts that have been repeatedly confirmed through observation and experimentation.
Translation to Spanish: Teoría

■ **Vocabulary 1.7 — Model.** A simplified representation of a system or phenomenon that helps explain and predict its behavior, often using mathematical equations, diagrams, or physical constructs.
Translation to Spanish: Modelo

■ **Vocabulary 1.8 — Variable.** Any factor, trait, or condition that can exist in differing amounts or types in an experiment or observation.
Translation to Spanish: Variable

■ **Vocabulary 1.9 — Independent Variable.** The variable that is deliberately changed or controlled in an experiment to test its effects on the dependent variable. It is plotted on the horizontal axis (X-axis).

Translation to Spanish: Variable independiente

■ **Vocabulary 1.10 — Dependent Variable.** The variable that is measured or observed in an experiment. It depends on the independent variable and is plotted on the vertical axis (Y-axis).
Translation to Spanish: Variable dependiente

■ **Vocabulary 1.11 — Directly Proportional.** A relationship between two quantities where if one quantity increases, the other also increases at a constant rate. Graphically represented by a straight line with equation y = mx + n.
Translation to Spanish: Directamente proporcional

■ **Vocabulary 1.12 — Inversely Proportional.** A relationship between two quantities where if one quantity increases, the other decreases. Graphically represented by a hyperbolic curve of the form y = 1/x.
Translation to Spanish: Inversamente proporcional

■ **Vocabulary 1.13 — Gradient.** The slope of a line in a graph, represented by the letter m in the equation y = mx + n. It indicates the rate of change of the dependent variable with respect to the independent variable.
Translation to Spanish: Pendiente

■ **Vocabulary 1.14 — Physical Quantity.** Any property of a body or system that can be measured and expressed numerically with an appropriate unit.
Translation to Spanish: Magnitud física

■ **Vocabulary 1.15 — Unit.** A quantity defined by convention to establish a standard for measurement. It provides a reference to express the value of physical quantities.
Translation to Spanish: Unidad

■ **Vocabulary 1.16 — International System of Units (SI).** The modern form of the metric system established in 1960 by the General Conference on Weights and Measures, consisting of 7 base units and derived units for measuring physical quantities.
Translation to Spanish: Sistema Internacional de Unidades

■ **Vocabulary 1.17 — Fundamental Quantities.** Basic physical quantities that cannot be broken down into other quantities and are defined independently. Examples include length, mass, time,

temperature, electric current, amount of substance, and luminous intensity.
Translation to Spanish: Magnitudes fundamentales

■ **Vocabulary 1.18 — Derived Quantities.** Physical quantities that are defined in terms of combinations of fundamental quantities. Examples include area, volume, velocity, and density.
Translation to Spanish: Magnitudes derivadas

■ **Vocabulary 1.19 — Length.** A fundamental quantity that measures the distance between two points. The SI unit is the metre (m).
Translation to Spanish: Longitud

■ **Vocabulary 1.20 — Mass.** A fundamental quantity that measures the amount of matter in an object. The SI unit is the kilogram (kg).
Translation to Spanish: Masa

■ **Vocabulary 1.21 — Temperature.** A fundamental quantity that measures the degree of hotness or coldness of a body. The SI unit is the kelvin (K).
Translation to Spanish: Temperatura

■ **Vocabulary 1.22 — Time.** A fundamental quantity that measures the duration of events or the interval between them. The SI unit is the second (s).
Translation to Spanish: Tiempo

■ **Vocabulary 1.23 — Electric Current.** A fundamental quantity that measures the flow of electric charge through a conductor. The SI unit is the ampere (A).
Translation to Spanish: Corriente eléctrica

■ **Vocabulary 1.24 — Amount of Substance.** A fundamental quantity that measures the number of elementary entities (atoms, molecules, ions, etc.) in a sample. The SI unit is the mole (mol).
Translation to Spanish: Cantidad de sustancia

■ **Vocabulary 1.25 — Luminous Intensity.** A fundamental quantity that measures the power emitted by a light source in a particular direction. The SI unit is the candela (cd).
Translation to Spanish: Intensidad luminosa

■ **Vocabulary 1.26 — Scientific Notation.** A method of writing very large or very small numbers

as a product of a number between 1 and 10 and a power of 10. It consists of a single non-zero digit to the left of the decimal point multiplied by 10 raised to an exponent.
Translation to Spanish: Notación científica

■ **Vocabulary 1.27 — Exponent.** The power to which a base number (such as 10 in scientific notation) is raised. It can be positive for large numbers or negative for small numbers.
Translation to Spanish: Exponente

1.6 Supplementary exercises on unit conversions and scientific notation

1.6.1 Scientific Notation

Exercise 1.13

a) 4 569 201
b) 0,0026
c) 235,78
d) 500 000
e) $4055 \cdot 10^{4}$
f) $5687 \cdot 10^{-8}$
g) $253 \cdot 10^{8}$
h) $0{,}0254 \cdot 10^{-2}$

■

1.6.2 Fundamental Units

Exercise 1.14

a) 500 hg a g
b) 60 cm a m
c) 2000 hm a dam
d) 5,7 kg a g
e) 300 dm a hm
f) 563 km a hm
g) 3 dm a dam
h) 7 dm a cm
i) 5 Gm a nm
j) 26 µm a kilom

■

1.6.3 Surface Units

Exercise 1.15

a) $40\,\text{cm}^2$ to dm^2

b) $3\,\text{hm}^2$ to dm^2

c) $50\,\text{cm}^2$ to m^2

d) $0{,}02\,\text{dm}^2$ to dam^2

e) $5\,\text{cm}^2$ to m^2

f) $300\,\text{hm}^2$ to dm^2

g) $653\,\text{cm}^2$ to m^2

h) $3\,\text{dm}^2$ to dam^2

i) $1{,}5\,\text{hm}^2$ to mm^2

j) $5\,\text{Gm}^2$ to nm^2

k) $26\,\mu\text{m}^2$ to km^2

■

1.6.4 Volume Units

Exercise 1.16

a) $40\,\text{cm}^3$ to dm^3

b) $10^3\,\text{km}^3$ to hm^3

c) $3\,\text{dam}^3$ to cm^3

d) $10^2\,\text{cm}^3$ to dm^3

e) $7\,\text{mm}^3$ to dam

f) $2 \cdot 10^{-2}\,\text{m}^3$ to mm^3

g) $5 \cdot 10^2\,\text{km}^3$ to hm^3

h) $4{,}23\,\text{dam}^3$ to cm^3

i) $48\,\text{cm}^3$ to dm^3

j) $0{,}02\,\text{dam}^3$ to mL

k) 100 mL to m^3

l) 300 L to dam^3

■

1.6.5 Time Units

Exercise 1.17 *a)* 340 min to s

b) 6,2 h to min

c) 10 h to s

d) 28 semanas to min

e) 22 semanas to h

■

1.6.6 Derive Units

Exercise 1.18

a) $36\ \frac{\text{m}}{\text{s}}$ a $\frac{\text{km}}{\text{h}}$

b) $100\ \frac{\text{km}}{\text{h}}$ a $\frac{\text{m}}{\text{s}}$

c) $1{,}05\ \frac{\text{kg}}{\text{cm}^3}$ a $\frac{\text{hg}}{\text{mm}^3}$

d) $0{,}25\ \frac{\text{g}}{\text{mL}}$ a $\frac{\text{kg}}{\text{m}^3}$

e) $0{,}05\ \frac{\text{kg}}{\text{cm}^3}$ a $\frac{\text{g}}{\text{mL}}$

■

2.1 Kinematics

Definition 2.1 — Kinematics. Is the branch of Physics that studies motion without taking into account the causes that produce it.

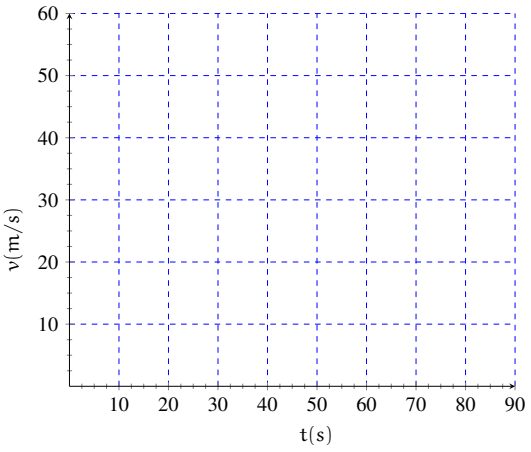


Figure (2.1): Reference system. Velocity vs. time is represented.

We say that an object is in motion when its position changes with respect to a reference system over time.

If its position does not change, we say that the object is at **rest**.

Motion is relative, that is, it depends on the point of reference and therefore, for its study, it is necessary to establish a reference system.

In Physics, the reference system will be a Cartesian coordinate system x , y.

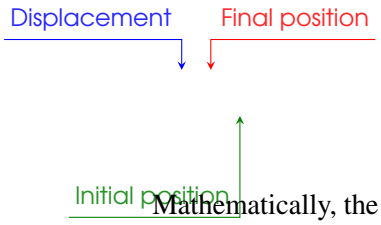
Definition 2.2 — Reference System. Is a system that allows us to fix the position of an object and study its evolution over time.

2.2 Quantities that describe motion

To determine motion, we need to know the following quantities:

1. **Position**. It tells us the exact location of the object. Position is usually represented by the letter x, and its SI unit is the meter (m).
2. **Displacement**. It is the difference between the final position and the initial position of the

motion. Displacement will be represented as Δx:



$$\Delta x = x_f - x_i \tag{2.1}$$

Mathematically, the symbol Δ indicates the difference between the final and the initial values. If this symbol precedes a quantity, it means the final value minus the initial value of that quantity.

For position, we will have $\Delta x = x_f - x_i$, for time $\Delta t = t_f - t_i$, and for velocity $\Delta v = v_f - v_i$.

3. Its unit will also be the meter.

   R Here it is important to note that displacement can be **negative**. This occurs when the moving object goes in the opposite direction.

4. **Trajectory**. It is the path followed during the motion.
5. **Distance travelled**. It is the length of the trajectory. Its unit will also be the meter.

Motion can be classified according to its trajectory. We will therefore have:

a) **Rectilinear**. The trajectory is a straight line. In this type of motion, displacement coincides with the distance traveled.

b) **Circular**. The trajectory is circular.

c) **Curvilinear**. The trajectory describes curves.

Exercise 2.1 In your notebook, identify whether the path (trajectory) of each of the following motions is straight (rectilinear) or curved (circular).

a) a ball dropped from a height

b) a rock attached to a rope and swung in circles

c) the movement of a clock's second hand

d) a sprinter racing 100 m on a straight track

■

Exercise 2.2 Imagine you drive completely around a circular roundabout and arrive back at your starting position. What is your total displacement? ■

Exercise 2.3 If you go all the way around a roundabout and return to the starting point, what's your displacement? ■

2.3 Velocity

Velocity gives us a measure of how fast an object moves. We have two types of velocity:

1. **Instantaneous velocity**. It is the velocity that an object has at a given moment. We can see it in a car's speedometer as the velocity constantly changes.
2. **Average velocity**. This is the measure of velocity that we will use in this course and we will represent it with the letter v.

$$v = \frac{\text{Displacement}}{\text{Elapsed time}} = \frac{\Delta x}{\Delta t} \tag{2.2}$$

3. If we express this in terms of the initial and final position, we have:

$$v = \frac{x_f - x_i}{t_f - t_i} \tag{2.3}$$

- x_i is the initial position.
- t_i is the time measured at the initial position.
- x_f is the final position.
- t_f is the time measured at the final position.

The SI units of velocity are: $\frac{m}{s}$

2.4 Acceleration

Acceleration tells us how velocity changes with time. Just like velocity, we have two types of acceleration:

1. **Instantaneous acceleration**. It is the acceleration at a given instant.
2. **Average acceleration**. It is the acceleration over a period of time. This is the one we will use in this topic. The formula is expressed as:

$$a = \frac{\text{Change in velocity}}{\text{Elapsed time}} = \frac{\Delta v}{\Delta t} \tag{2.4}$$

3. If we express this in terms of the initial and final velocity, we have:

$$a = \frac{v_f - v_i}{t_f - t_i} \tag{2.5}$$

- v_i is the initial velocity.
- t_i is the time measured at the initial velocity.
- v_f is the final velocity.
- t_f is the time measured at the final velocity.

The SI units of acceleration are: $\frac{m}{s^2}$

2.5 Rectilinear Motion

Rectilinear motions are those that have a **straight-line trajectory**. These motions are very interesting because in these cases, **displacement coincides with the distance traveled**.

The previously described equations for velocity and acceleration can be written as:

$$v = \frac{\Delta x}{\Delta t} \longrightarrow v = \frac{x_f - x_i}{t_f - t_i} \tag{2.6}$$

$$a = \frac{\Delta v}{\Delta t} \longrightarrow a = \frac{v_f - v_i}{t_f - t_i} \tag{2.7}$$

Sometimes these expressions can be simplified because:

- Generally, we take $t_i = 0$, so this term can be removed. We can choose the initial time.
- Since there is only one time term left, t_f can be simplified to just t.
- The same applies to x_i, although it is less common. If we set the initial position $x_i = 0$, it can be removed, and the final position x_f can simply be called x.
- The same simplification can also be applied to v_i.

After these simplifications, the equations become:

$$v = \frac{x}{t} \tag{2.8}$$

$$a = \frac{v}{t} \tag{2.9}$$

R Even if we can simplify the expressions, it is important to carefully read the problem to see if simplification is allowed.

2.5.1 Uniform Rectilinear Motion (URM)

This is a specific case of rectilinear motion with the following conditions:

- Its trajectory is a straight line.
- Its velocity does not change during motion. Therefore:
  - $v_f = v_i$
  - $\Delta v = 0$
- Since the velocity does not change, acceleration is zero, i.e., $a = 0\,\frac{m}{s^2}$

Theorem 2.1 The velocity is given by: $v = \frac{x_f - x_i}{t_f - t_i}$. Applying the previous simplifications: $v = \frac{x}{t}$

Exercise 2.4 The Sun is located at an average distance of $1{,}5 \cdot 10^8$ km from Earth, and the speed of light in a vacuum is $3 \cdot 10^8\,\frac{m}{s}$. Assuming light travels at constant velocity (uniform rectilinear motion), calculate how long it takes for light from the Sun to reach Earth.

■

Exercise 2.5 During a Physical Education lesson, you are timing how long it takes students to run 100 m.

a) If one of your classmates completes the 100 m in 15 s, what is her average speed?

b) On average, how many metres does she cover each second?

■

Exercise 2.6 During a family road trip, your car travels along a straight motorway at an average speed of 90 $\frac{\text{km}}{\text{h}}$ for 15 min.

a) Calculate the distance travelled in metres during this time period.

b) Can you be certain that the car's speed was exactly 90 $\frac{\text{km}}{\text{h}}$ at every moment during the journey?

■

Exercise 2.7 A car moving along a straight highway has been travelling for 30 min at an average speed of 100 $\frac{\text{km}}{\text{h}}$. What distance, in metres, has the car covered during this time? ■

Exercise 2.8 A car is moving along a straight road, and its velocity changes as follows: After 1 s, its velocity is 2 $\frac{\text{m}}{\text{s}}$. After 2 s from the start, its velocity is 4 $\frac{\text{m}}{\text{s}}$. After 3 s, its velocity is 6 $\frac{\text{m}}{\text{s}}$.

a) By how much does the car's velocity change during each second?

b) What is the acceleration of this vehicle?

■

Exercise 2.9 A cyclist leaves Almansa on the bike lane at a speed of 8 $\frac{\text{m}}{\text{s}}$. If it takes 100 min to reach his destination:

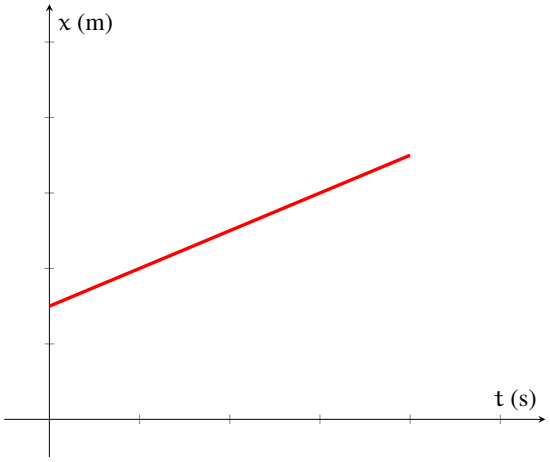
a) What distance has he travelled?

b) Knowing that Bonete is 32 km away, how long would it take him?

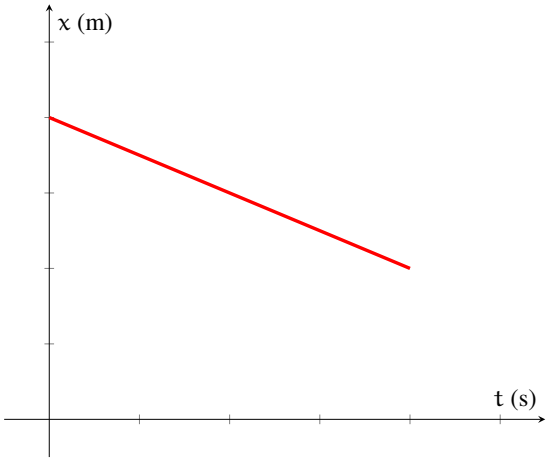
■

2.5.1.1 Graphs of Uniform Rectilinear Motion (URM)

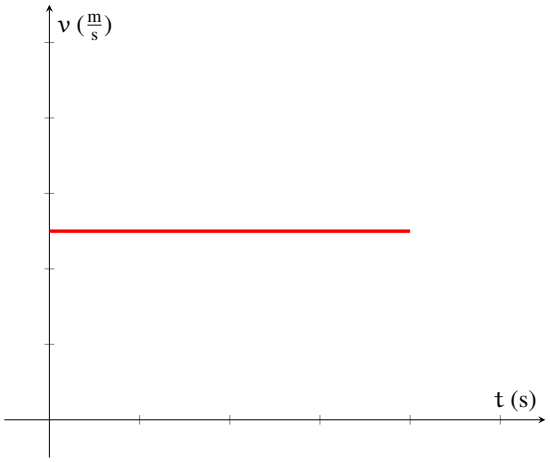
In physics, it is very common to make graphs and from them identify the type of motion. For URM, we will have the following cases.



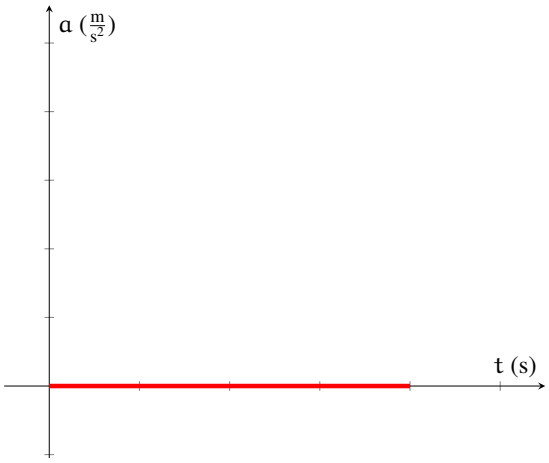
(a) Position vs Time. Moving in the positive direction.



(b) Position vs Time. Moving in the opposite direction.



(a) Velocity vs Time. Moving in the positive direction.



(b) Acceleration vs Time.

2.5.2 Uniformly Accelerated Rectilinear Motion (UARM)

This is another specific case of rectilinear motion that meets the following conditions:

- The trajectory is also in a straight line.
- Now, the velocity changes during the motion.
  - $v_f \neq v_i$
  - $\Delta v \neq 0$
- Since the velocity changes, the acceleration is now different from zero ($a \neq 0$) but constant, meaning the acceleration does not vary.

The equations for this motion are as follows:

- For acceleration we have:

$$a = \frac{v_f - v_i}{t_f - t_i} \tag{2.10}$$

If we apply the simplifications $v_i = 0$ and $t_i = 0$, we get:

$$a = \frac{v}{t} \tag{2.11}$$

- The equation for position is:

$$x_f = x_i + v_i \cdot (t_f - t_i) + \frac{1}{2} \cdot a \cdot (t_f - t_i)^2 \tag{2.12}$$

If we simplify by assuming initial time $t_i = 0$, the equation becomes:

$$x_f = x_i + v_i \cdot t + \frac{1}{2} \cdot a \cdot t^2 \tag{2.13}$$

If we also assume the initial position is zero ($x_i = 0$), this expression simplifies further:

$$x = v_i \cdot t + \frac{1}{2} \cdot a \cdot t^2 \tag{2.14}$$

As we can see, in this equation v_f never appears, which means that the final position does not depend on the final velocity, but it does depend on the initial velocity since it appears in the formula.

- Due to the complexity of solving for time in this type of motion, the expression will not be studied in this course.

Exercise 2.10 You have gone skiing with your classmates and decide to go down a slope. Starting from rest, you accelerate to 10 $\frac{m}{s}$ in 5 s. What was your average acceleration? ■

Exercise 2.11 A vehicle accelerates from rest to 100 $\frac{km}{h}$ in 8 s. Calculate its average acceleration, giving your answer in $\frac{m}{s^2}$. ■

Exercise 2.12 A car travelling at 30 $\frac{m}{s}$ applies the brakes for 5 s, reducing its velocity to 25 $\frac{m}{s}$. Determine its average acceleration during braking. ■

Exercise 2.13 For take-off, an Airbus aircraft needs to reach a speed of 300 $\frac{km}{h}$ in a time of 23,8 s. Calculate:

a) What is its acceleration?

b) What is the minimum length of runway required?

■

Exercise 2.14 A motorcyclist travelling at 20 $\frac{m}{s}$ sees a traffic light turn red ahead. He begins braking and comes to a complete stop in 10 s.

a) Calculate the acceleration during braking.

b) If the traffic light was 90 m away from the motorcyclist when he started braking, does he

manage to stop before reaching it?

■

Exercise 2.15 In an experiment, we measure the position of a cart moving along a straight line at different times. The recorded data shows the times when the cart passes through various positions.

Position (m)	0	3	6	9
Time (s)	0	1	2	3

a) Draw the position-time graph.

b) What position does the cart occupy relative to the origin after 3,5 s?

c) At what instant is the cart 7 m from its initial position?

■

Exercise 2.16 A car at 30 $\frac{\text{m}}{\text{s}}$ brakes for 5 s until it reaches 25 $\frac{\text{m}}{\text{s}}$. Acceleration? ■

Exercise 2.17 An athlete runs along a straight, horizontal track. Her progress is measured and recorded in the table shown below:

Distance (m)	0	30	60	90	120
Time (s)	0	5	10	15	20

a) Draw the position-time graph.

b) Calculate the athlete's average speed. What distance has she covered during the first eight seconds?

c) How long does it take her to travel 66 m?

d) Does the athlete's instantaneous velocity differ from the average speed at any of the measured time points? What type of motion does your answer suggest?

■

Exercise 2.18 In an experiment, we measure different positions of a cart over time and get the results shown in the table below.

Position (m)	0	5	15	25
Time (s)	0	5	10	15

■

Exercise 2.19 A vehicle accelerates from rest to 100 $\frac{\text{km}}{\text{h}}$ in 8 s. Calculate its average acceleration, giving your answer in $\frac{\text{m}}{\text{s}^2}$.

■

Exercise 2.20 From the position-time data table below, determine whether the motion is uniform or accelerated (explain your reasoning) and write the equation of position as a function of time:

Position (cm)	-45	-10	60	77.5
Time (s)	0	2	6	7

■

Exercise 2.21 A car moves at a constant speed of 100 $\frac{\text{km}}{\text{h}}$ along a straight road. If it starts accelerating for 15 seconds, its speed increases to 120 $\frac{\text{km}}{\text{h}}$.

a) What was its acceleration?

b) Calculate the distance travelled.

■

Exercise 2.22 A car is stopped at a traffic light. When the light turns green, it starts with a constant acceleration of 2 $\frac{\text{m}}{\text{s}^2}$. At the moment it starts, it is overtaken by a truck moving at a constant speed of 54 $\frac{\text{km}}{\text{h}}$. Calculate:

a) At what distance from the traffic light will the car catch up with the truck?

b) What speed will the car have at that moment?

■

Exercise 2.23 An airplane begins landing at 240 $\frac{\text{km}}{\text{h}}$. What minimum length (in metres) must the runway have if the braking acceleration is 1,5 $\frac{\text{m}}{\text{s}^2}$? ■

Exercise 2.24 A dog starts running from rest and, following a straight trajectory, reaches a speed of 20 $\frac{\text{m}}{\text{s}}$ in 8 s. From that moment, it maintains this constant speed for 15 s. Calculate:

a) Write the equations of the dog's motion.

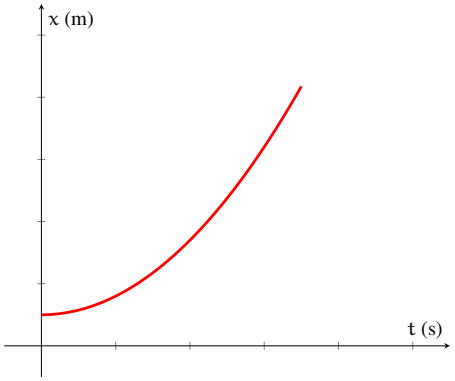
b) The acceleration the dog acquires in those first 8 s.

c) The distance travelled in the 23 s.

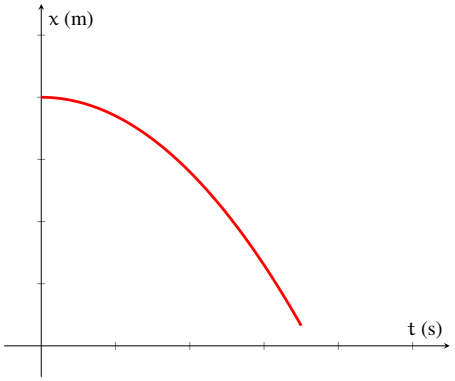
d) The dog's speed in the first four seconds.

■

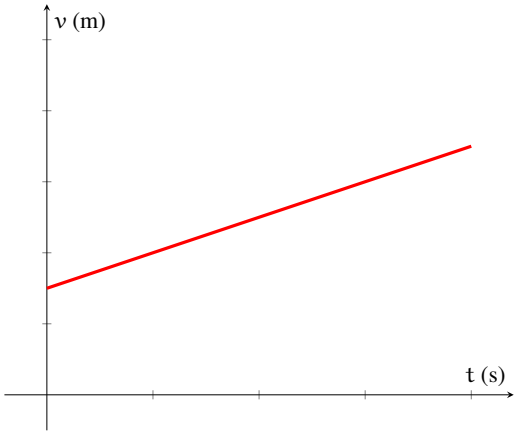
2.5.2.1 Graphs for Uniform Accelerated Rectilinear Motion (UARM)



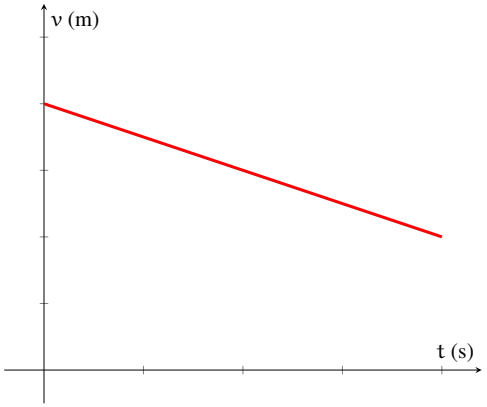
(a) Position vs Time graph with positive acceleration



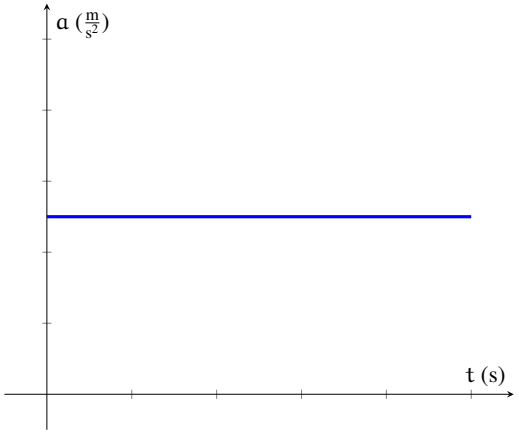
(b) Position vs Time graph with negative acceleration



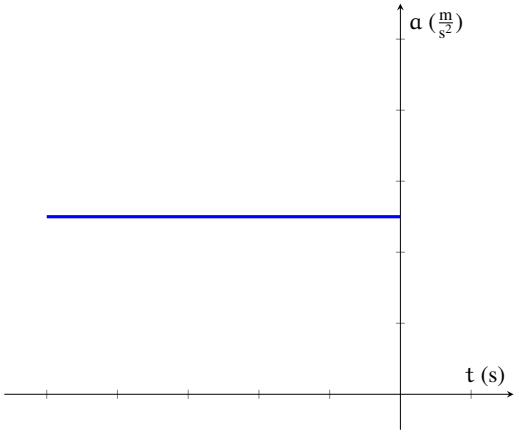
(a) Velocity vs Time graph with positive acceleration



(b) Velocity vs Time graph with negative acceleration



(a) Acceleration vs Time graph. Accelerating ($a > 0$)



(b) Acceleration vs Time graph. Decelerating ($a < 0$)

2.6 Position-Time Graphs

Position-time graphs represent the position of an object over a given time period. An example:

- At initial time $t_0 = 0\,s$, the object is at the origin $x_0 = 0\,m$.
- At time $t_A = 10\,s$, the cyclist is at point A, $x_A = 20\,m$. From $t_0 = 0\,s$ to $t_A = 10\,s$, the displacement is 20 m:

$$\Delta x_{AO} = x_B - x_O = 20 - 0 = 20\,m \quad (2.15)$$

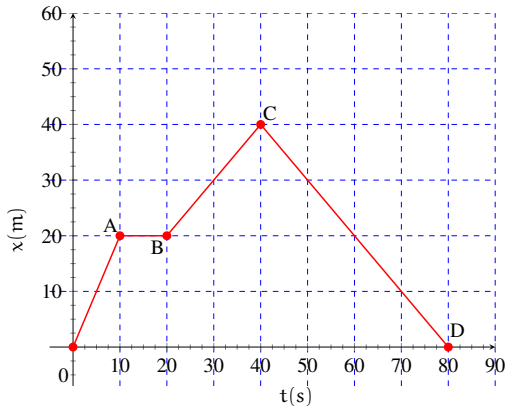


Figure (2.7): Reference system. Position vs time.

- At $t_B = 20\,s$, the cyclist is also at $x_B = 20\,m$. From $t_B = 10$ to $t_C = 20$, the displacement is 0 m, but total displacement from $x_0 = 0$ to $x_B = 20$ is 20 m.
- From $x_C = 40$ to $x_D = 0$, the slope of the line is negative, indicating motion backward. The displacement from $x_C = 40$ to $x_D = 0$ is 40 m backward. From $x_0 = 0$ to $x_D = 80$, the total displacement is 0 m.
- Total distance traveled is the sum of each segment: $20 + 0 + 20 + 40 = 80\,m$.

Motion analysis in each segment:

- Segment 0-A: URM (straight line on position-time graph)
- Segment A-B: No motion (object stationary)
- Segment B-C: URM again
- Segment C-D: URM backward

The graphs show the difference between displacement and distance travelled.

2.7 Velocity-Time Graphs

Velocity-time graphs represent the velocity of an object moving over a given period of time.

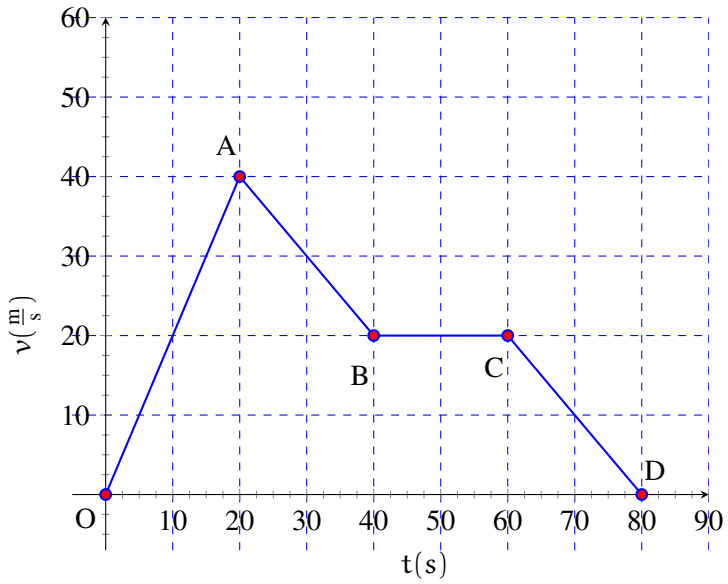


Figure (2.8): Graph of v vs t

Analysis of the type of motion in the different segments:

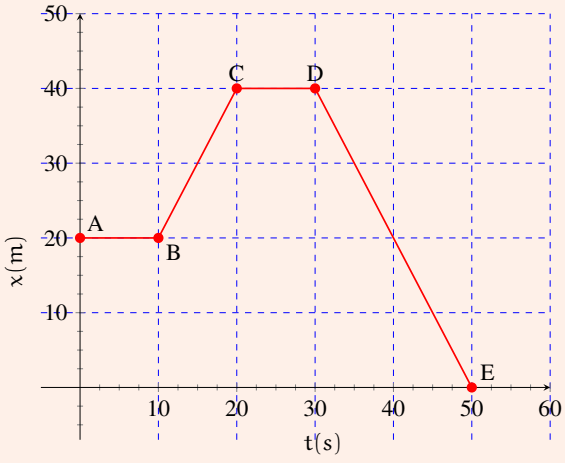
- Segment O-A: We have a uniformly accelerated rectilinear motion (UARM) because the velocity-time graph is a straight line, and it moves forward.
- Segment A-B: We also have UARM, but now moving backward.
- Segment B-C: We have uniform rectilinear motion (URM) again because the velocity does not change, so the graph is horizontal.
- Segment C-D: Another UARM, now moving backward.

Exercise 2.25 The graph below shows the motion of an object. Answer the following questions:

a) What type of motion occurs in each section of the graph?

b) Calculate the acceleration in each section.

c) What is the total distance travelled by the object?



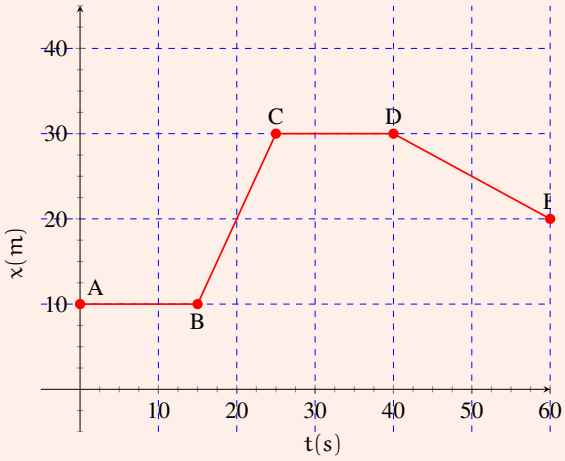
■

Exercise 2.26 The graph below shows the motion of an object. Answer the following questions:

a) What type of motion occurs in each section of the graph?

b) Calculate the acceleration in each section.

c) What is the total distance travelled by the object?

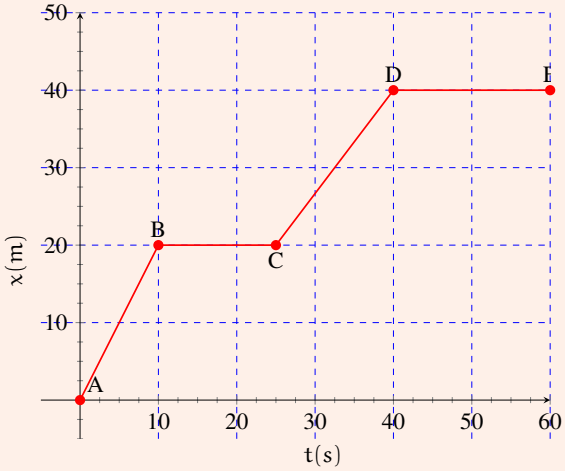


Exercise 2.27 The graph below shows the motion of an object. Answer the following questions:

a) What type of motion occurs in each section of the graph?

b) Calculate the acceleration in each section.

c) What is the total distance travelled by the object?

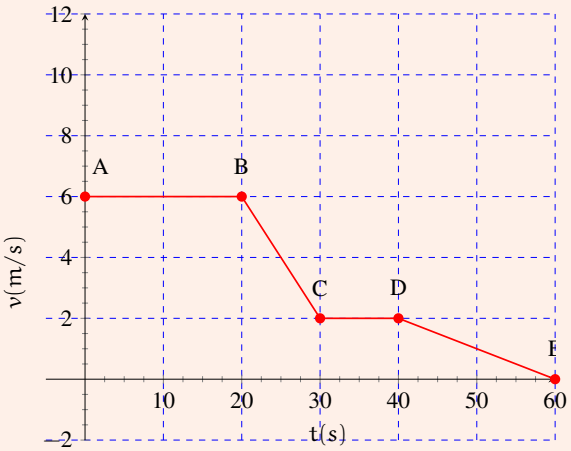


Exercise 2.28 The graph below shows the velocity of an object. Answer the following questions:

a) What type of motion occurs in each section of the graph?

b) Calculate the acceleration in each section.

c) What is the total distance travelled by the object?



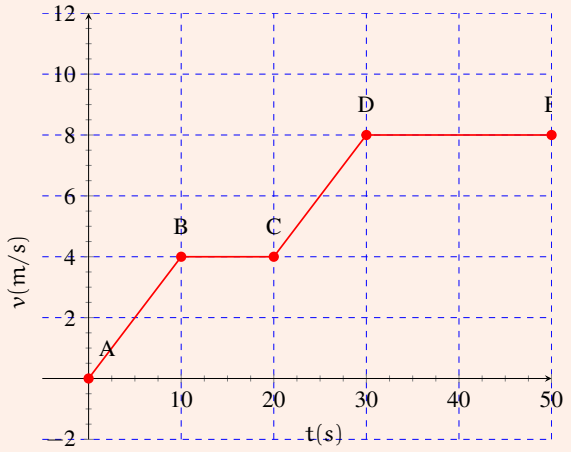
■

Exercise 2.29 The graph below shows the velocity of an object. Answer the following questions:

a) What type of motion occurs in each section of the graph?

b) Calculate the acceleration in each section.

c) What is the total distance travelled by the object?



■

Exercise 2.30 Create a concept map summarising the main topics of this unit. ■

2.8 Vocabulary

■ **Vocabulary 2.1 — Kinematics.** The branch of Physics that studies motion without taking into account the causes that produce it. It focuses on describing how objects move through space and time.

Translation to Spanish: Cinemática

■ **Vocabulary 2.2 — Motion.** The change in position of an object with respect to a reference system over time. An object is in motion when its position changes relative to a chosen reference point.
Translation to Spanish: Movimiento

■ **Vocabulary 2.3 — Rest.** The state of an object when its position does not change with respect to a reference system over time.
Translation to Spanish: Reposo

■ **Vocabulary 2.4 — Reference System.** A system that allows us to fix the position of an object and study its evolution over time. In Physics, it is usually a Cartesian coordinate system (x, y).
Translation to Spanish: Sistema de referencia

■ **Vocabulary 2.5 — Position.** The exact location of an object in space, usually represented by the letter x. The SI unit is the meter (m).
Translation to Spanish: Posición

■ **Vocabulary 2.6 — Displacement.** The difference between the final position and the initial position of the motion, represented as $\Delta x = x_f - x_i$. It is a vector quantity that can be positive or negative. The SI unit is the meter (m).
Translation to Spanish: Desplazamiento

■ **Vocabulary 2.7 — Trajectory.** The path followed by an object during its motion. It can be rectilinear (straight line), circular, or curvilinear (curved).
Translation to Spanish: Trayectoria

■ **Vocabulary 2.8 — Distance Travelled.** The length of the trajectory followed by an object. It is a scalar quantity always positive. The SI unit is the meter (m).
Translation to Spanish: Distancia recorrida

■ **Vocabulary 2.9 — Rectilinear Motion.** Motion in which the trajectory is a straight line. In this type of motion, displacement coincides with the distance traveled.
Translation to Spanish: Movimiento rectilíneo

■ **Vocabulary 2.10 — Circular Motion.** Motion in which the trajectory is circular or describes a

circle.
Translation to Spanish: Movimiento circular

■ **Vocabulary 2.11 — Curvilinear Motion.** Motion in which the trajectory describes curves that are neither straight lines nor perfect circles.
Translation to Spanish: Movimiento curvilíneo

■ **Vocabulary 2.12 — Velocity.** The rate of change of position with respect to time. It is a vector quantity that indicates both speed and direction. The SI unit is meters per second (m/s).
Translation to Spanish: Velocidad

■ **Vocabulary 2.13 — Speed.** The magnitude of velocity, representing how fast an object is moving without regard to direction. It is a scalar quantity. The SI unit is meters per second (m/s).
Translation to Spanish: Rapidez

■ **Vocabulary 2.14 — Average Velocity.** The total displacement divided by the total time taken. It is calculated as $v_{avg} = \Delta x/\Delta t$
Translation to Spanish: Velocidad media

■ **Vocabulary 2.15 — Instantaneous Velocity.** The velocity of an object at a specific instant of time. It is the limit of average velocity as the time interval approaches zero.
Translation to Spanish: Velocidad instantánea

■ **Vocabulary 2.16 — Acceleration.** The rate of change of velocity with respect to time. It is a vector quantity. The SI unit is meters per second squared (m/s²).
Translation to Spanish: Aceleración

■ **Vocabulary 2.17 — Uniform Rectilinear Motion (URM).** Motion in a straight line with constant velocity (zero acceleration). The object covers equal distances in equal time intervals.
Translation to Spanish: Movimiento rectilíneo uniforme (MRU)

■ **Vocabulary 2.18 — Uniformly Accelerated Rectilinear Motion (UARM).** Motion in a straight line with constant acceleration. The velocity changes uniformly over time. Also known as uniformly varied motion.
Translation to Spanish: Movimiento rectilíneo uniformemente acelerado (MRUA)

■ **Vocabulary 2.19 — Coordinate System.** A system used to specify the position of a point in space using numerical coordinates. The most common is the Cartesian coordinate system.
Translation to Spanish: Sistema de coordenadas

■ **Vocabulary 2.20 — Origin.** The reference point in a coordinate system where all coordinates are zero. It serves as the starting point for measuring position.
Translation to Spanish: Origen

■ **Vocabulary 2.21 — Time Interval.** The difference between two moments in time, calculated as $\Delta t = t_f - t_i$. It represents the duration of an event or motion.
Translation to Spanish: Intervalo de tiempo

■ **Vocabulary 2.22 — Position-Time Graph.** A graphical representation showing how an object's position changes over time. The slope of the graph represents velocity.
Translation to Spanish: Gráfica posición-tiempo

■ **Vocabulary 2.23 — Velocity-Time Graph.** A graphical representation showing how an object's velocity changes over time. The slope represents acceleration, and the area under the curve represents displacement.
Translation to Spanish: Gráfica velocidad-tiempo

■ **Vocabulary 2.24 — Slope.** In a graph, the slope represents the rate of change of the vertical quantity with respect to the horizontal quantity. In position-time graphs, slope represents velocity; in velocity-time graphs, it represents acceleration.
Translation to Spanish: Pendiente

■ **Vocabulary 2.25 — Vector Quantity.** A physical quantity that has both magnitude and direction. Examples include displacement, velocity, and acceleration.
Translation to Spanish: Magnitud vectorial

■ **Vocabulary 2.26 — Scalar Quantity.** A physical quantity that has only magnitude but no direction. Examples include distance, speed, and time.
Translation to Spanish: Magnitud escalar

■ **Vocabulary 2.27 — Initial Position.** The position of an object at the beginning of motion or at the start of observation, represented as x_i or v.

Translation to Spanish: Posición inicial

■ **Vocabulary 2.28 — Final Position.** The position of an object at the end of motion or at a specific moment, represented as x_f .
Translation to Spanish: Posición final

■ **Vocabulary 2.29 — Cartesian Coordinate System.** A coordinate system that uses perpendicular axes (typically x and y in 2D, or x, y, and z in 3D) to specify positions in space.
Translation to Spanish: Sistema de coordenadas cartesianas

■ **Vocabulary 2.30 — Frame of Reference.** A perspective from which motion is observed and measured. Motion is relative and depends on the chosen frame of reference.
Translation to Spanish: Sistema de referencia

3.1 Forces

Definition 3.1 — Force. A **force** is a physical quantity that represents any action capable of modifying the shape of a body or changing the state of rest or motion of that body.

The unit of force in the International System is the Newton (N). We can define one Newton as the force that produces on a body of one kilogram an acceleration of one $\frac{\text{m}}{\text{s}^2}$.

In nature there are four fundamental forces:

- Gravitational Force.
- Electromagnetic Force.
- Strong Nuclear Force.
- Weak Nuclear Force.

In this chapter we will study the gravitational force and the electromagnetic force, which we will separate into electric force (within the electromagnetic force).

3.1.1 Parts of a force: representation of a force

To correctly represent a force, it is necessary to know that it has four elements:

- **Point of application**. The location where the force is applied. It is usually indicated with a point, although it is quite common to omit it.
- **Magnitude**. Indicates the amount of force being applied, that is, the numerical value.
- **Direction**. Indicates where the force is applied (horizontal, vertical or inclined) and is represented by a straight line. The greater the magnitude, the longer the line.
- **Sense**. Shows to which side the force points and is represented with an arrowhead. Numerically, the sense is indicated with a positive or negative sign. One direction is always chosen; if the force is positive, it means it is directed towards the chosen direction, while if it is negative, it is directed in the opposite sense.

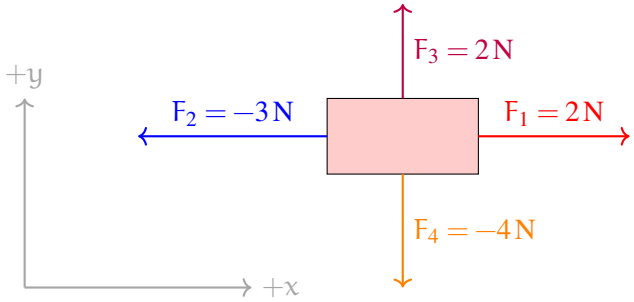


Figure (3.1): Representation of forces

Exercise 3.1 In your notebook, draw the forces below, applied at the origin of the XY coordinates:

a) a force of 5 N, in a horizontal direction to the right.

b) a force equal to the previous one, but in the opposite direction.

c) a vertical force of 10 N, pointing upwards.

■

3.1.2 Operations with forces: sum of forces

In Physics, the symbol Σ indicates that all the present magnitudes must be added. If we have ΣF, it means that we must add all the forces acting on the object.

We can have the following cases:

- **Forces with the same sense**. Both will have a positive sign.

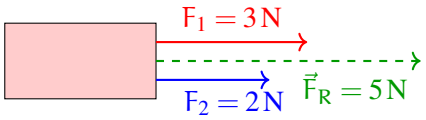


Figure (3.2): Forces with the same sense.

In this case, the resultant force F_R, that is the sum of F_1 and F_2, will be: $\Sigma F_R = F_1 + F_2 = 3\,N + 2\,N = 5\,N$.

- **Forces with opposite sense**. Now one of the forces will have a negative sign; in the following example it will be F_2.

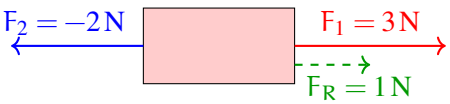


Figure (3.3): Two forces with opposite sense.

In this case, we have: $\Sigma F_R = F_1 + F_2 = 3\,N - 2\,N = 1\,N$. Since F_2 goes in the opposite direction, we assign it the negative sign.

- **Two perpendicular forces**. To obtain the resultant force, we must apply the Pythagorean theorem.

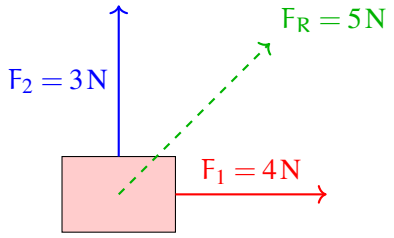


Figure (3.4): Two perpendicular forces.

In this case: $F_R^2 = F_1^2 + F_2^2 \longrightarrow F_R = \sqrt{F_1^2 + F_2^2}$.
$F_R = \sqrt{3^2 + 4^2} = 5\,N$.

Exercise 3.2 Two horses pull a cart in the same direction with a force of 300 N each. Calculate the magnitude and direction of the resultant force.

■

Exercise 3.3 Draw the forces shown below, applied at the origin of the X- and Y-axes:

a) a force of magnitude 6 N, downwards.

b) a force the same as in a., but in the opposite direction.

c) a force of magnitude 7 N, horizontal direction and to the left.

d) a force of magnitude 4 N and whose direction forms an angle of 45° with the positive X-axis.

■

Exercise 3.4 Two colleagues push a table in the same direction with forces of 110 N and 135 N. Find the resultant force with which they push the table. Draw the force diagram.

■

Exercise 3.5 These two dogs pull on each end of a rubber toy. The dog on the right pulls with a force of magnitude 80 N and the one on the left with a force of 95 N.

a) Find the magnitude and direction of the resultant force.

b) Draw a force diagram of the two forces and the resultant force.

■

Exercise 3.6 The result of two perpendicular forces applied to a body is 25 N. If one of the forces has a magnitude of 7 N, find the magnitude of the other force.

■

Exercise 3.7 Two groups of students pull on an object, with the same force of 250 N, using ropes attached to opposite sides of the object.

a) What's the resultant force?

b) If the group on the right now pulls with a force of 325 N, calculate the resultant force and

direction.

■

Exercise 3.8 Two separate forces, each of 100 N, act on the same object at an angle of 90°. Find the magnitude of the resultant force and draw its direction on a force diagram.

■

3.2 Deformation of Bodies

We have previously seen that a force can produce a deformation. We can mainly distinguish two types of deformation:

- **Plastic Deformation**: occurs when the deformed body does not recover its original shape after the force ceases. It is an irreversible and permanent process.
- **Elastic Deformation**: is the deformation that disappears when the external force that produced it stops. In this unit, we will study this type of deformation.

3.2.1 Elastic Force: Hooke's Law

This type of force refers to the forces that arise when we deform an elastic material.

Thus, under the action of a force, a spring may experience elongation or compression until the force disappears.

Hooke's Law: the deformation of an elastic spring is proportional to the deforming force. The elastic force we calculate in a given spring will be equal in magnitude and direction but opposite in sense to the force causing that deformation (later we will see that this corresponds to Newton's 3rd law).

R The elastic force will always have a negative sign because it opposes the direction of the applied force. We will not include the sign here because we are only determining the magnitude, not the direction.

Elastic force — Spring constant specific to each spring

$$F_{elastic} = k \cdot \Delta l \tag{3.1}$$

n stretched or compressed

To calculate Δl, we can use the expression: $\Delta l = l_f - l_i$, where l_f is the final length of the spring and l_i is the initial length of the spring.

R If we are told that a spring has stretched or compressed by a certain distance, this directly gives us the value of Δl.

Each spring has its own spring constant, with units in the International System of $\frac{N}{m}$.

3.2.2 Measurement of Forces

To measure forces, a **dynamometer** is used.
A dynamometer is an instrument with a spring that has a known spring constant. This spring has an attached scale calibrated in newtons, which allows us to measure forces. To calculate these forces, we use Hooke's law studied previously.
Its use is basic: if we want to calculate the force exerted by an object (usually its weight), we hang the object from the hook of the dynamometer.



Figure (3.5): Dynamometer

We said previously that the force exerted by an object is equal to the elastic force, so we can apply:

$$F_{elastic} = F_{object} \tag{3.2}$$

Since we know the spring constant of the dynamometer and measure the elongation of the spring, we can easily calculate the object's force using:

$$F_{object} = k \cdot \Delta l \tag{3.3}$$

Exercise 3.9 When a force of 20 N is applied to a certain spring, it produces an elongation of 10 cm.

a) Calculate the spring constant.
b) Calculate the elongation of the spring when a force of 200 N is applied.
c) Determine the force required to produce an elongation of 5 dm.

■

Exercise 3.10 In an experiment to determine the spring constant, the following results were obtained:

F (N)	0	5	10	15	20
x (cm)	0	2.1	4	6	7.9

a) Plot the data from the table on a graph.
b) What is the value of the spring constant?

■

Exercise 3.11 Find the force required to stretch a spring with a constant of 150 N/m by 3 cm from its initial length.

3.3 Newton's Laws

3.3.1 First Law of Newton: Law of Inertia

Known as the **Law of Inertia**, it states the following:

First Law of Newton: Law of Inertia. Every body remains at rest or in uniform straight-line motion unless acted upon by an external force.

3.3.2 Second Law of Newton: Fundamental Principle of Dynamics

Second Law of Newton: Fundamental Principle of Dynamics. The acceleration of a body is directly proportional to the force applied to that body. Moreover, the change in velocity of the body occurs along the line of action of the applied force.

$$\Sigma F = m\, a \tag{3.4}$$

Exercise 3.12 A force of 5 N is applied to a body of mass 0.2 kg. What acceleration will the body acquire?

Exercise 3.13 A man with a mass of 75 kg is sitting in a car that starts moving with an acceleration of 1.5 m/s². What force does the man experience?

Exercise 3.14 Calculate the horizontal force being applied to a body of mass 25 kg moving at a constant velocity of 25 km/h on a horizontal surface.

Exercise 3.15 Calculate the force that a person must apply to a table of mass 20 kg to give it an acceleration of 1,25 $\frac{m}{s^2}$. If someone is on the opposite side of the table, what force must be applied to prevent it from moving?

Exercise 3.16 Two girls are pushing a 50 kg wooden cabinet across a horizontal floor. Each of them exerts a force of 25 N. Find the cabinet's acceleration. What is its direction relative to the girls?

3.3.3 Third Law of Newton: Action and Reaction Law

Third Law of Newton: Action and Reaction Law: When two bodies interact, they exert forces on each other that are equal in magnitude but opposite in direction.

If we denote the force that object A exerts on object B as (F_{AB}) and the force that object B exerts on object A as (F_{BA}), we have:

$$F_{AB} = -F_{BA} \tag{3.5}$$

3.4 Gravitational Force

It is one of the four fundamental forces and is responsible for the structure of the universe. It is an interaction that occurs between two bodies with mass, is **always attractive**, and has unlimited range.

In the 17th century, Isaac Newton considered that the force the Earth exerts on the Moon is the same as that exerted on any body on the Earth's surface — gravitational attraction. He extended this interaction to all planets, stating that gravity is an attribute of all bodies and is proportional to the amount of matter contained in each.

The intensity of the gravitational force depends on the masses of the bodies and the distance between them. Thus, the greater the mass of a body, the greater the gravitational force it produces. However, the greater the distance between the bodies, the smaller the gravitational force. Gravitational force is usually represented by F_G and is expressed as:

$$F_G = G \cdot \frac{m_1 \cdot m_2}{r^2} \tag{3.6}$$

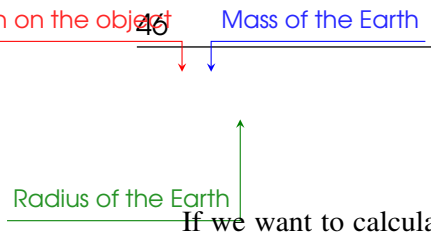
Where G is the universal gravitational constant with a value of $G = 6{,}67 \cdot 10^{-11} \frac{N \cdot m^2}{kg^2}$, m_1 and m_2 are the masses of the two bodies, and r is the distance between them. Since gravitational force is a force, it is measured in Newtons.

R The distance between the two objects is always measured from the centre of each object.

R For planets, measure the distance from the centre of each planet, and because they are large, you must know the radius of each planet.

If we want to calculate the gravitational force that the Earth exerts on an object on its surface with mass m, we apply the following equation:

$$F_G = G \cdot \frac{M_T \cdot m}{R_T^2} \tag{3.7}$$



If we want to calculate the force that the Earth exerts on an object at a height h above the Earth's surface, we must add that height h to the Earth's radius, giving:

$$F_G = G \cdot \frac{M_T \cdot m}{(R_T + h)^2} \tag{3.8}$$

R Remember to convert all quantities to SI units before using them in formulas.

3.4.1 Gravity or Gravitational Field

Definition 3.2 — Gravity or Gravitational Field. Is the attraction generated in space by a mass around it.

For example, the Earth has a gravitational field on its surface of 9,8 $\frac{m}{s^2}$, but this value decreases as we move upward. As we can see, gravity has the same units as acceleration, $\frac{m}{s^2}$ The expression for gravity is:

$$g = G \cdot \frac{m}{r^2} \tag{3.9}$$

Where m is the mass of the body exerting the gravitational force and r is the distance between the centre of that body and the point where we want to calculate the gravity.

The formula to calculate gravity at the Earth's surface is:

$$g_T = G \cdot \frac{M_T}{R_T^2} \tag{3.10}$$

To calculate gravity at a height h above the Earth's surface, the expression is:

$$g_T = G \cdot \frac{M_T}{(R_T + h)^2} \tag{3.11}$$

R Gravity on the surface of planets is different, so to calculate it, we must know the mass and radius of the planet.

$$g_{Planet} = G \cdot \frac{M_{Planet}}{R^2_{Planet}} \tag{3.12}$$

R Remember to convert all quantities to SI units before using them in formulas.

3.4.2 Weight Force (p or w)

In fact, the gravitational force is called **Weight**.

Definition 3.3 — Weight Force. Is the force with which the Earth, or any other planet, attracts the bodies on it. Given the mass of any person or object on the Earth or another planet, the weight of that object or person is given by:

$$w = m \cdot g \quad \text{or} \quad p = m \cdot g \tag{3.13}$$

Where m is the mass of the object and g is the gravity of that planet.

Since weight is a force, its unit is Newton.

As we can see from the weight formula, this force depends on the mass of each object as well as the gravity of the planet we are on.

R Gravity on the Moon is different from that on Earth; therefore, to calculate weight on the Moon, we must first calculate the gravity on the Moon.

Exercise 3.17 European Space Agency Meteosat satellites have a mass of 2000 kg and orbit the Earth in a circular path at a radius of 36000 km. Calculate the force with which the Earth attracts these satellites. What is the effect of this force on the satellites?

Datos: $G = 6{,}67 \cdot 10^{-11}\ \frac{N \cdot m^2}{kg^2}$; $M_T = 5{,}99 \cdot 10^{24}\,kg$; $R_T = 6380\,km$

■

Exercise 3.18 On a certain planet, we measure the weight of a 9 kg mass with a dynamometer and obtain a reading of 108 N. What is the value of g on that planet? ■

Exercise 3.19 From astronomical observations, we know that Jupiter has a radius of $6{,}98 \cdot 10^7$ m and a mass of $1{,}9 \cdot 10^{27}$ kg. Calculate the acceleration due to gravity on this planet.

Datos:: $G = 6{,}67 \cdot 10^{-11}\ \frac{N \cdot m^2}{kg^2}$

■

Exercise 3.20 We hang a 500 g mass on a dynamometer, and its scale reads 4.85 N. What is the value of g? ■

Exercise 3.21 We measure the mass of an object using a laboratory scale and find it is 0.9 kg. Work with a classmate to calculate its weight. ■

Exercise 3.22 A boy weighs 441 N on Earth. In your notebook, answer the following questions:

a) What is the boy's weight on Mars?

b) What is his mass on the Moon?

c) What is his weight on the Moon?

Datos:: $g_{Mars} = 3{,}7\,\frac{m}{s^2}$; $g_{Moon} = 1{,}6\,\frac{m}{s^2}$ ■

Exercise 3.23 Mount Chimborazo is the furthest point from the Earth's centre of gravity, at 6268 m above sea level (which is 6 378 000 metres from the Earth's centre of mass). Find g on Chimborazo's summit and compare it to g at sea level ($9{,}8\,\frac{m}{s^2}$). ■

Exercise 3.24 The Hubble is a telescope that remains in orbit around the Earth at an altitude of 600 km above its surface. Determine the gravitational acceleration the Hubble is subjected to.

Datos:: $G = 6{,}67 \cdot 10^{-11}\,\frac{N \cdot m^2}{kg^2}$; $M_T = 5{,}99 \cdot 10^{24}\,kg$; $R_T = 6380\,km$ ■

Exercise 3.25 Calculate the weight for a person of 80 kg located at the height of Mount Everest (8848 m). And at an altitude of 300 km?

Datos:: $G = 6{,}67 \cdot 10^{-11}\,\frac{N \cdot m^2}{kg^2}$; $M_T = 5{,}99 \cdot 10^{24}\,kg$; $R_T = 6380\,km$ ■

Exercise 3.26 Calculate the gravity on the Moon. What force does the Moon exert on an astronaut with a mass of 95 kg?

Datos:: $G = 6{,}67 \cdot 10^{-11}\,\frac{N \cdot m^2}{kg^2}$; $M_L = 7{,}36 \cdot 10^{22}\,kg$; $R_L = 1740\,km$ ■

3.5 Electric Force

As we studied previously, the atom is composed of two charged particles: electrons, with negative charge, and protons, with positive electric charge. Normally, materials have the same number of electrons as protons and are neutral; however, due to **electrification** processes, there can be a transfer of electrons from one body to another, creating an imbalance between positive and negative charges in a body.

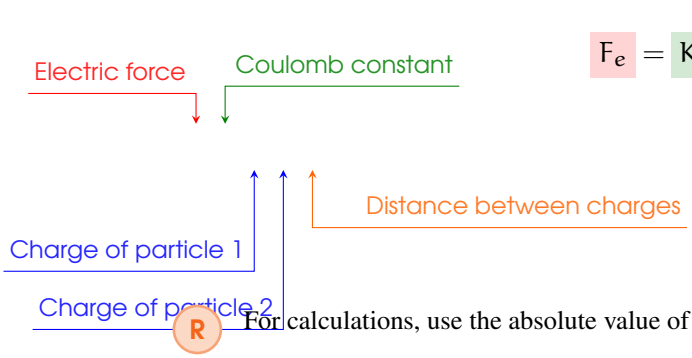
In this type of force, electric charges are involved, and therefore we must define a new quantity that allows us to quantify the value of that charge (equivalent to mass in gravitational force).

Definition 3.4 — Electric Charge. Is a property of matter present in subatomic particles. The unit of electric charge is the Coulomb, represented as C.

Electric charges can be positive or negative.

Electric force is another of the four fundamental forces of nature and becomes clear when we have electric charges. Electric forces can be **attractive** when charges are of the opposite sign, and **repulsive** when charges are of the same sign.

In the 18th century, Charles Coulomb discovered the properties of this force and the relationships between charges and their distances, leading to the expression:



(3.14)

R For calculations, use the absolute value of the charge, i.e. without the sign.

The Coulomb constant (K) has a value of $K = 9 \cdot 10^9 \frac{N \cdot m^2}{C^2}$.

R It is crucial to avoid errors to convert all quantities to SI units before substituting them into the formula.

Exercise 3.27 Calculate the force between two charges:

a) $+5\,\mu C$ and $+3\,\mu C$ located 10 cm apart.

b) $+5\,\mu C$ and $-3\,\mu C$ located 10 cm apart.

Datos:: $K = 9 \cdot 10^9 \frac{N \cdot m^2}{C^2}$

■

Exercise 3.28 What is the magnitude of the electric force exerted on a charge of +7 C by another charge of +3 C located 2 m away? Is the force attractive or repulsive?

Datos:: $K = 9 \cdot 10^9 \frac{N \cdot m^2}{C^2}$

■

Exercise 3.29 The mass of a proton is $1{,}67 \cdot 10^{-27}$ kg and its electric charge is $1{,}6 \cdot 10^{-19}$ C. Compare the electric repulsion force between two protons in a vacuum at a distance of 1 m with the gravitational attraction between them.

Datos:: $K = 9 \cdot 10^9 \frac{N \cdot m^2}{C^2}$; $G = 6{,}67 \cdot 10^{-11} \frac{N \cdot m^2}{kg^2}$

■

Exercise 3.30 Make a conceptual map of the unit. ■

3.6 Vocabulary

■ **Vocabulary 3.1 — Force.** A physical quantity that represents an interaction capable of changing the motion, speed, direction, or shape of an object.
Translation to Spanish: Fuerza

■ **Vocabulary 3.2 — Newton.** The SI unit of force, defined as the force required to accelerate a mass of one kilogram at one meter per second squared.
Translation to Spanish: Newton

■ **Vocabulary 3.3 — Acceleration.** The rate at which the velocity of an object changes with time.
Translation to Spanish: Aceleración

■ **Vocabulary 3.4 — Gravitational Force.** The attractive force that exists between any two masses in the universe.
Translation to Spanish: Fuerza gravitatoria

■ **Vocabulary 3.5 — Electromagnetic Force.** A fundamental force responsible for interactions between electrically charged particles.
Translation to Spanish: Fuerza electromagnética

■ **Vocabulary 3.6 — Strong Nuclear Force.** The fundamental force that holds protons and neutrons together inside the nucleus of an atom.
Translation to Spanish: Fuerza nuclear fuerte

■ **Vocabulary 3.7 — Weak Nuclear Force.** A fundamental force involved in radioactive decay and certain particle interactions.
Translation to Spanish: Fuerza nuclear débil

■ **Vocabulary 3.8 — Vector.** A physical quantity that has both magnitude and direction.
Translation to Spanish: Vector

■ **Vocabulary 3.9 — Equilibrium.** The state in which the net force acting on an object is zero, resulting in no acceleration.
Translation to Spanish: Equilibrio

■ **Vocabulary 3.10 — Weight.** The gravitational force exerted on an object due to the presence of a massive body such as Earth.
Translation to Spanish: Peso

■ **Vocabulary 3.11 — Tension.** A pulling force transmitted through a rope, cable, or string when it is stretched.
Translation to Spanish: Tensión

■ **Vocabulary 3.12 — Inertia.** The tendency of an object to resist changes in its state of motion.
Translation to Spanish: Inercia

■ **Vocabulary 3.13 — Velocity.** A vector quantity that describes the speed of an object together with its direction of motion.
Translation to Spanish: Velocidad

■ **Vocabulary 3.14 — Mass.** A measure of the amount of matter contained in an object.
Translation to Spanish: Masa

■ **Vocabulary 3.15 — Newton's Second Law.** A law stating that the acceleration of an object is proportional to the net force acting on it and inversely proportional to its mass.
Translation to Spanish: Segunda ley de Newton

■ **Vocabulary 3.16 — Net Force.** The overall force acting on an object after all individual forces are combined vectorially.
Translation to Spanish: Fuerza neta

■ **Vocabulary 3.17 — Friction.** A force that opposes the relative motion between two surfaces in contact.
Translation to Spanish: Fricción

■ **Vocabulary 3.18 — Gravity.** The natural phenomenon by which masses attract each other.
Translation to Spanish: Gravedad

■ **Vocabulary 3.19 — Weight Force.** The force exerted on an object due to gravity, usually

directed toward the center of the Earth.

Translation to Spanish: Fuerza peso

■ **Vocabulary 3.20 — Electric Charge.** A physical property of matter that causes it to experience a force in an electromagnetic field.

Translation to Spanish: Carga eléctrica

4.1 Energy

Definition 4.1 — Energy. Is defined as the capacity of a body to produce changes in itself or in its surroundings

The unit of energy in the SI system is the Joule (J). Other forms of energy and their equivalence in joules are:

- **Calorie** (cal). $1\,\text{cal} = 4{,}18\,\text{J}$.
- **Kilowatt-hour** (kWh) $\longrightarrow 1\,\text{kWh} = 3{,}6 \cdot 10^6\,\text{J}$

We have several forms of energy that can be classified mainly into:

a) **Kinetic energy**. It is the energy due to the motion of objects.
b) **Potential energy**. It is the energy due to the change in height of objects.
c) **Mechanical energy**. It is the sum of kinetic energy and potential energy.
d) **Thermal energy**. It is the energy due to heat exchange (later we will see the definition of heat).
e) **Wind energy**. It is the energy due to the wind.
f) **Nuclear energy**. It is the energy due to the nuclear fusion or fission process.
g) **Electrical energy**. It is the energy due to the movement of electrons.

4.2 Work

Definition 4.2 — Work. Is defined as the transformation produced by a force, that is, we talk about work when a force transmits energy

R Energy and work are related.

Mechanical work is represented by W and its expression is:

Work Applied Force

$$W = \mathrm{F} \cdot \mathrm{d} \tag{4.1}$$

Displacement of the object

This distance d can be represented as the final distance minus the initial distance according to the expression: $\mathrm{d} = \mathrm{x_f} - \mathrm{x_i}$

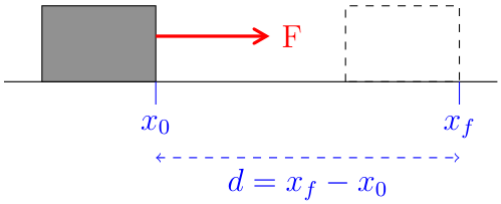


Figure (4.1): Representation of Work

When several forces act on the same body, the work done by those forces is the same as the work done by their resultant (the sum of), that is:

$$\Sigma \mathrm{F} = \mathrm{F_1} + \mathrm{F_2} + \ldots \tag{4.2}$$

Therefore, the previous expression becomes:

$$W = \Sigma \mathrm{F} \cdot \mathrm{d} \tag{4.3}$$

R Work is a form of energy, and therefore its SI unit is the joule.

Exercise 4.1 Calculate the work done by a donkey when it exerts a force of 2000 N on a cart and drags it for 400 meters. ■

Exercise 4.2 A student pushes a box with a force of 80 N and moves it 5 m in the same direction as the force. Calculate the work done. ■

Exercise 4.3 A car is pushed by four people who exert a total force of 600 N. If the car moves 8 meters, what work have they done? ■

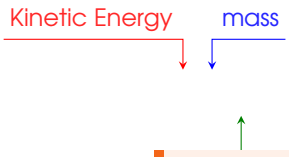
Exercise 4.4 A person drags a suitcase along the floor by applying a force of 45 N over a displacement of 25 m. Calculate the work done. ■

Exercise 4.5 Find the net work required to move a 10 kg block when the body moves 3 meters, with forces $F_1 = 80\,N$ and $F_2 = 15\,N$ acting in the same direction. ■

Exercise 4.6 A builder uses a pulley to lift a 50 kg mass to a height of 10 m. What work has been done? (Data: $g = 9{,}8\,\frac{m}{s^2}$) ■

4.3 Kinetic Energy

Kinetic energy is the energy a body possesses when it is in motion. The mathematical expression for kinetic energy is:



$$E_k = \frac{1}{2} m \cdot v^2 \tag{4.4}$$

Exercise 4.7 A 1000 kg car travels at a speed of 20 m/s. Calculate its kinetic energy. ■

Exercise 4.8 A 0.5 kg ball moves with a speed of 10 m/s. What is its kinetic energy? ■

Exercise 4.9 If a 4 kg object has a kinetic energy of 200 J, at what speed is it moving? ■

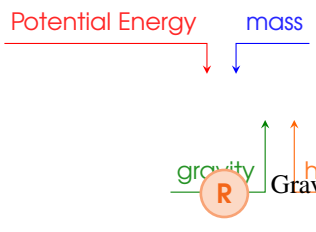
Exercise 4.10 What is the speed of an 80 kg athlete who has a kinetic energy of 8000 J? ■

4.4 Potential Energy

We have two types of potential energy: gravitational potential energy and elastic potential energy.

4.4.1 Gravitational Potential Energy

This energy is a consequence of the position of a body (height). The mathematical expression for potential energy is:



$$E_p = m \cdot g \cdot h \tag{4.5}$$

R Gravitational potential energy is commonly referred to simply as potential energy.

Exercise 4.11 A 2 kg book is on a shelf 1.5 m high. Calculate its gravitational potential energy. (Data: $g = 9.8\ m/s^2$) ■

Exercise 4.12 A 5 kg flowerpot is located on a balcony 12 m from the ground. What is its gravitational potential energy? ■

Exercise 4.13 If an object has a potential energy of 600 J and is located 3 m high, what is its mass? ■

Exercise 4.14 At what height must a 2 kilogram object be raised so that its potential energy is 125 J? ■

4.5 Mechanical Energy

Mechanical energy is the sum of kinetic energy and potential energy.

$$E_m = E_k + E_p \tag{4.6}$$

Exercise 4.15 Calculate the energy of the following objects:

a) A 1200 kg car travelling at a speed of 6 $\frac{m}{s}$

b) A 450 gram picture frame located 1.75 meters above the ground.

c) A 60 gram tennis ball with a speed of 65 $\frac{m}{s}$ and a height of 1.25 meters.

■

Exercise 4.16 Find the mechanical energy of an eagle of 2500 grams that is flying at a speed of 100 $\frac{km}{h}$ and at a height of 250 m. ■

Exercise 4.17 Of the following situations, which will have the greatest mechanical energy?:

a) A 10 kg rock at a height of 5 meters.

b) A 5 kg rock at a height of 10 meters.

c) A 250-gram drone flying close to the ground at a speed of 25 $\frac{m}{s}$

d) A 500-gram stone at a height of 250 meters and a speed of 10 $\frac{m}{s}$

■

Exercise 4.18 A 0.4 kg ball is at a height of 5 m and moves with a speed of 8 m/s. Calculate:

1. Its potential energy
2. Its kinetic energy
3. Its total mechanical energy

■

Exercise 4.19 A 0.2 kg bird flies at a height of 15 m with a speed of 12 m/s. Determine its total mechanical energy. ■

4.6 Principle of Conservation of Mechanical Energy

The principle of conservation of mechanical energy states say: mechanical energy at the beginning (we set an initial point, represented in the formula with i) will be equal to mechanical energy at the end (we set the final point, represented in the formula with f)

Mathematically, this principle can be expressed as:

$$E_{m,i} = E_{m,f} \tag{4.7}$$

This expression can also be written as:

$$E_{p,i} + E_{c,i} = E_{p,f} + E_{c,f} \tag{4.8}$$

If we develop this expression further, we get:

$$m \cdot g \cdot h_i + \frac{1}{2} \cdot m\, v_i^2 = m \cdot g \cdot h_f + \frac{1}{2} \cdot m\, v_f^2 \tag{4.9}$$

Exercise 4.20 A 50 kg metal sphere is dropped from a height of 8 meters to the ground.

a) Calculate the mechanical energy it has at the moment it is dropped.

b) With what speed does it reach the ground?

■

Exercise 4.21 From a height of 15 m, an object of mass 3 kg is thrown vertically downward with an initial speed of 2 $\frac{m}{s}$. If there is no air resistance, find:

a) The kinetic energy at 5 meters above the ground.

b) The speed at that moment and the speed with which it reaches the ground.

■

Exercise 4.22 On a slide 2 meters high, there is a 25 kg child. With what speed would they reach the ground when sliding down? ■

Exercise 4.23 A 0.5 kg ball is dropped from a window 30 m above the street. Calculate:

a) The potential energy relative to the ground at the moment of release.

b) The kinetic energy at the moment it reaches the ground.

c) The speed with which it hits the ground.

■

Exercise 4.24 At the top of a roller coaster, a car and its occupants with a total mass of 1000 kg are at a height of 50 meters above the ground and have a speed of 5 $\frac{m}{s}$.

a) What speed will the car have when it reaches the next hilltop, which is at a height of 25 meters above the ground?

b) If the speed it has at the next hilltop is 20 $\frac{m}{s}$, what is the work done by friction between the wheels and the rails?

■

Exercise 4.25 A 0.3 kg ball is dropped from a height of 10 m. Calculate:

1. The initial mechanical energy
2. The speed with which it reaches the ground
3. The kinetic energy just before touching the ground

■

Exercise 4.26 A pendulum has a mass of 0.5 kg and is released from a height of 2 m. If there is no friction:

1. What is its initial potential energy?
2. What will be its maximum speed at the lowest point?
3. What kinetic energy will it have when it is at a height of 1 m?

■

Exercise 4.27 A 70 kg skier starts from rest from the top of a slope located 20 m high. If we neglect friction, with what speed will they reach the base of the slope? ■

4.7 Power

Another important quantity is power (P). Power can be defined as the energy (E) or work (W) done per unit of time (t), and its mathematical expression is:

$$P = \frac{E}{t} \tag{4.10}$$

or equivalently:

$$P = \frac{W}{t} \tag{4.11}$$

The unit of power in the SI system is the Watt (W). Another widely used unit of power is the horsepower (HP), where $1\ HP = 735\,W$

Exercise 4.28 A crane lifts a load of 500 kg to a height of 20 meters in 10 seconds. Calculate the power developed by the crane. (Use $g = 9.8\,m/s^2$). ■

Exercise 4.29 An electric motor has a power of 1200 W. How long will it take to perform 18,000 J of work? ■

Exercise 4.30 A student with a mass of 60 kg climbs a staircase 5 meters high in 12 seconds. Calculate the power generated. ■

Exercise 4.31 A light bulb has a power of 60 W. If it remains on for 2 hours, how much total energy (in Joules) does it consume? ■

4.8 Domestic Energy Consumption

Figure **??** shows the information usually found in an electricity bill.



CONSUMOS Y FACTURACIÓN

Consumos

Período de facturación 10.01.2023 - 09.02.2023

Potencias contratadas (kW)	P1	P2
	3,45	3,45

* El importe de los cargos viene determinado en el Real Decreto-ley 6/2022.

Concepto		Importe
Potencia		*9,00 €
3,45 kW x 0,084180 €/Kw día x 31 día(s)		
3,45 kW x 0,084180 €/Kw día x 31 día(s)		
* de los cuales corresponden a peajes		6,90 €
* de los cuales corresponden a cargos		0,94 €
Energía		*11,53 €
(10.01.2023 - 31.01.2023) 43 kWh x 0,183002 €/kWh		
(01.02.2023 - 09.02.2023) 20 kWh x 0,183002 €/kWh		
* de los cuales corresponden a peajes		0,44 €
* de los cuales corresponden a cargos		0,52 €
Importe de la energía asociada al mecanismo ibérico regulado por el Real Decreto-ley 10/2022, de 13 de mayo		**0,16 €**
63,00 kWh x 0,002539 €/kWh (*)		
Descuento en energía	**10 %**	1,53 €
Impuesto Eléctrico	0,5 %	0,057 €
Alquiler de contador		**0,82 €**
Base I.V.A.		20,03 €
IVA	5 %	1,00 €
TOTAL		**21,03 €**

Figure (4.2): Information Found on an Electricity Bill

First, we see that we must pay for the contracted power. As we have seen, power is the energy per unit of time supplied, and its SI unit is the watt (W).

In this case, the contracted power is 3,45 kW. F

R Here we see a unit conversion using conversion factors to calculate the price we must pay after 31 days for the contracted power.

■ **Example 4.1** Calculate the price we have to pay for 31 days using conversion factors. Next, it shows the energy consumed as well as the price to pay. In this case, it is $0.084180 \frac{€}{\text{kWh}}$ and we see that 43 kWh was consumed first, followed by 20 kWh. ■

R It is important not to confuse kWh, which is a unit of energy, with W, which is a unit of power.
Calculate the equivalence between kWh and joules.

Exercise 4.32 You should review a home electricity bill and calculate the cost of having three appliances switched on for a certain period of time. ■

4.9 Energy Sources

An **energy source** is a natural resource from which energy can be obtained and transformed into other forms. According to their regeneration capacity, we have:

1. **Renewable**. The regeneration rate equals the consumption rate, and they can be considered inexhaustible.
   - Hydroelectric energy.
   - Solar energy.
   - Geothermal energy.
   - Biomass.
   - Marine energy.
   - Wind energy.
2. **Non-renewable**

4.10 Heat and Temperature

Often, the terms heat and temperature are confused.

Heat is the amount of energy transferred from one body to another due to a temperature difference. For heat to exist, two bodies must be in contact. Since heat is a form of energy, its SI unit is the joule.

Temperature is the measure of the kinetic energy of the molecules of a substance. The SI unit is the Kelvin, although degrees Celsius are widely used.

$$\mathrm{K} = {}^{\circ}\mathrm{C} + 273 \tag{4.12}$$

$${}^{\circ}\mathrm{C} = \mathrm{K} - 273 \tag{4.13}$$

R We can say that the temperature is very high but the heat is low.
Could you explain why?

Exercise 4.33 Explain why the temperature on the surface of the sun is very high, while its heat is not as high in comparison. Find the answer online, including relevant sources. ■

4.11 Vocabulary

■ **Vocabulary 4.1 — Energy.** The capacity to do work or produce change in a physical system.
Translation to Spanish: Energía

■ **Vocabulary 4.2 — Kinetic Energy.** The energy possessed by an object due to its motion.
Translation to Spanish: Energía cinética

■ **Vocabulary 4.3 — Potential Energy.** Stored energy associated with the position or configuration of an object.
Translation to Spanish: Energía potencial

■ **Vocabulary 4.4 — Mechanical Energy.** The sum of kinetic and potential energy in a system.
Translation to Spanish: Energía mecánica

■ **Vocabulary 4.5 — Work.** The transfer of energy that occurs when a force moves an object through a distance.
Translation to Spanish: Trabajo

■ **Vocabulary 4.6 — Power.** The rate at which work is done or energy is transferred.
Translation to Spanish: Potencia

■ **Vocabulary 4.7 — Conservation of Energy.** A principle stating that energy cannot be created or destroyed, only transformed from one form to another.
Translation to Spanish: Conservación de la energía

■ **Vocabulary 4.8 — Thermal Energy.** The internal energy of a system associated with the random motion of its particles.
Translation to Spanish: Energía térmica

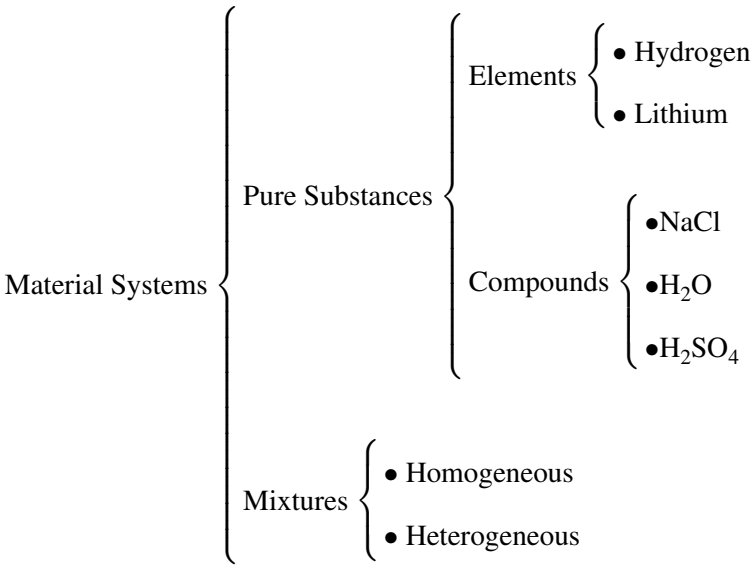
■ **Vocabulary 4.9 — Heat.** The transfer of thermal energy between systems due to a temperature difference.
Translation to Spanish: Calor

■ **Vocabulary 4.10 — Temperature.** A measure of the average kinetic energy of the particles in a substance.

Translation to Spanish: Temperatura

5.1 Material Systems

We can classify material systems into (see diagram):



1. **Pure substances**: a type of matter that has a constant composition and defined properties, and that cannot be separated into other substances by physical methods. Pure substances can be divided into:
   - **Elements**: made of only one type of atom.
   - **Compounds**: made by the union of two or more chemical elements.
2. **Mixtures**: the combination of two or more pure substances that can be separated by physical methods. Mixtures can be divided into:
   - **Homogeneous mixture**: a mixture where the components are evenly distributed and cannot be seen with the naked eye.
   - **Heterogeneous mixture**: a mixture where the components can be distinguished.

5.2 Solutions

Definition 5.1 — Solution. Is a homogeneous system made of two or more pure substances in different proportions.

The components of a solution are:

- **Solvent**. The substance in greater amount.
- **Solute**. The substance in smaller amount. There can be more than one solute.
- **Solution**. The mixture of solute(s) and solvent.

For example, in a solution of water and sugar, the solvent is water and the solute is sugar. In a bottle of mineral water, the label says it contains sodium, calcium, magnesium, etc. In this case, the solvent is water, and the solutes are sodium, calcium, and magnesium.

5.2.1 Concentration of a solution

When we have a solution, it is crucial to know how much solute there is in the solvent.

Definition 5.2 — Concentration. Is the amount of solute dissolved in a certain amount of solvent.

There are different ways to express concentration. The main ones are:

1. **Mass percentage**. It tells us the mass of solute in every 100 g of solution.

$$\text{Mass \% of solute} = \frac{\text{mass of solute}}{\text{mass of solution}} \cdot 100 \tag{5.1}$$

Important:

- If there are several solutes, we calculate the percentage of each one.
- We can also calculate the percentage of solvent.
- All percentages together must add up to 100.

■ **Example 5.1** We have a water solution with sodium and calcium. There are 18 g of sodium, 6 g of calcium, and 100 g of water. Calculate the mass percentage of sodium and calcium.

For both cases we must apply equation (**??**). In both cases we have to:

$$\%\text{ mass of sodium} = \frac{18\,\text{g of sodium}}{18\,\text{g of sodium} + 6\,\text{g of calcium} + 100\,\text{g of water}} \cdot 100 = 14.52\%$$

$$\%\text{ mass of calcium} = \frac{6\,\text{g of calcium}}{18\,\text{g of sodium} + 6\,\text{g of calcium} + 100\,\text{g of water}} \cdot 100 = 4.84\%$$

If we also want to calculate the percentage of solvent, it would be:

$$\%\text{ mass of water} = \frac{100\,\text{g of water}}{18\,\text{g of sodium} + 6\,\text{g of calcium} + 100\,\text{g of water}} \cdot 100 = 80.65\%$$

■

Exercise 5.1 Steel is an alloy composed of iron, carbon and other elements. A 120 g mass of steel contains 1.8% carbon and 1.2% manganese. Find the mass of each element. ■

Exercise 5.2 We dissolve 12 g of sodium chloride and 13 g of potassium chloride in 250 g of distilled water. Find the percentage by mass of each solute in the final solution. ■

Exercise 5.3 We want to prepare 500 g of an aqueous solution containing 0.5% sodium chloride, 0.3% potassium chloride and 0.2% potassium nitrate by mass. How much of each solute is needed? How much water do we need? ■

Exercise 5.4 What does it mean that a solution of iodine in alcohol is 1% by mass? ■

Exercise 5.5 We use a mixture of lead and tin in soldering. If the percentage by mass of lead is 66% and that of tin is 34%, what masses of each metal would you melt to prepare 250 g of this alloy? ■

Exercise 5.6 The percentage by mass of potassium chloride in a certain solution is 10%. How much of this solute is dissolved in 250 g of solution? ■

Exercise 5.7 A bottled water contains 99.8% by mass of water and 0.2% carbon dioxide. Identify the solute and solvent in this solution. What mass of carbon dioxide would 1 kg of water contain? ■

2. **Volume percentage**. It tells us the volume of solute in every 100 L of solution.

$$\text{Volume \% of solute} = \frac{\text{volume of solute}}{\text{volume of solution}} \cdot 100 \tag{5.2}$$

■ **Example 5.2** We prepare a solution with 100 mL of alcohol, 150 mL of acetone, and 250 mL of water. Find the volume percentage of each substance.

$$\%\text{ vol alcohol} = \frac{100\,\text{mL of alcohol}}{100\,\text{mL of alcohol} + 150\,\text{mL of acetone} + 250\,\text{mL of water}} \cdot 100 = 20\%$$

$$\%\text{ vol acetone} = \frac{150\,\text{mL of acetone}}{100\,\text{mL of alcohol} + 150\,\text{mL of acetone} + 250\,\text{mL of water}} \cdot 100 = 30\%$$

$$\%\text{ vol water} = \frac{250\,\text{mL of water}}{100\,\text{mL of alcohol} + 150\,\text{mL of acetone} + 250\,\text{mL of water}} \cdot 100 = 50\%$$

■

Exercise 5.8 A solution of ether and chloroform is prepared by adding 10 mL of ether to 90 mL of chloroform. What is the percentage by volume of each component? ■

Exercise 5.9 The composition by volume of the active ingredients in a 200 mL bottle of body lotion is: 2.14% sweet almond oil; 2.14% corn oil; 4.29% peanut oil and 2.88% hydrogenated copra oil. Calculate the volume of each component. ■

Exercise 5.10 We have dissolved 25 cm^3 of alcohol in distilled water to obtain a total solution volume of 250 cm^3.

a) What is the volume of the solute and the solvent?

b) Calculate the percentage by volume of each component of this solution.

■

3. **Mass concentration**. It tells us the mass of solute per unit of volume of solution. It is usually expressed in $\frac{g}{L}$.

$$\text{Mass concentration} = \frac{m_{\text{solute (g)}}}{V_{\text{solution (L)}}}$$

Do not confuse it with density:

- Density is the mass of a substance per unit of volume.
- Mass concentration is the mass of solute per unit of volume of the **solution**.

■ **Example 5.3** To prepare a potassium nitrate solution with 20 g of nitrate in water to obtain a final solution volume of 200 mL. Calculate:

a) The mass concentration of the solution expressed in $\frac{g}{L}$.

Since the volume of the solution is given in mL, we convert it to L. The volume is then 0,2 L.

If we apply:

$$\text{Mass concentration} = \frac{m_{\text{solute (g)}}}{V_{\text{solution (L)}}} = \frac{20}{0.2} = 100\,\frac{g}{L}$$

b) Calculate the mass of the solution if the density of this solution is $\rho = 1{,}09\,\frac{g}{cm^3}$.

To calculate its mass, we solve the density equation for mass:

$$\rho = \frac{m}{V} \rightarrow m = \rho \cdot V$$

Before substituting into the equation, we must check that both the density and the volume are in the same units, but we know that $1\,\text{mL} = 1\,\text{cm}^3$.

$$m = \rho \cdot V = 1.09 \cdot 200 = 218\,\text{g}$$

■

Exercise 5.11 200 ml of an aqueous solution contains 12 g of sugar. The density of this solution is 1,022 $\frac{\text{g}}{\text{cm}^3}$.

a) Calculate the percentage by mass of the sugar.

b) Find the mass concentration in g/L.

■

Exercise 5.12 An isotonic saline infusion can be prepared by dissolving 0.9 g of sodium chloride in water to give 100 mL of solution. Calculate:

a) the mass concentration in g/L of this solution.

b) the mass of sodium chloride needed to prepare 250 mL of isotonic saline.

■

5.3 Solubility

Definition 5.3 — Solubility. The **solubility** of a solute at a certain temperature and in a certain solvent is the maximum amount of that solute that can dissolve in 100 g of solvent.

From this definition we can see that solubility depends on temperature and on the solvent. This means that solubility changes when temperature changes and when the solvent changes.

For example, if something does not dissolve at room temperature, we can heat it. This shows that when the temperature increases, solubility also increases.

To express solubility correctly, we must give the value of the solubility, say that it is in 100 g of the solvent used, and also give the temperature.

5.3.1 Solubility curves

A **solubility curve** is a graph that shows the solubility data of a pure substance in 100 g of a solvent at different temperatures.

The Figure **??**, shows the solubility curves of different pure substances in water are shown.

Summary

- **Solute:** The substance that dissolves and is usually present in a smaller amount in the mixture.
- **Solvent:** The substance that dissolves the solute. It is usually the main component (for example, water in aqueous solutions).
- **Solution:** The homogeneous mixture formed by the solute and the solvent ($m_{solution} = m_{solute} + m_{solvent}$) or, for volume, $V_{solution} = V_{solute} + V_{solvent}$.
- **Concentration:** It expresses the proportion of solute in a certain amount of solution or solvent.
- **Density** (d)**:** Relationship between the total mass of the solution and the volume it occupies ($d = m/V$). It is used to change volume into mass and mass into volume. It is important not to confuse density with mass/volume concentration.

Exercise 5.13 Use the solubility curves on the last page to answer these questions.

a) What's the solubility of KNO3 at 25 °C and 45 °C?

b) What mass of KNO_3 crystals will form if a saturated solution of 100 g of water is cooled from 45 °C to 25 °C? And from 50 °C to 20 °C?

c) What's the greatest mass of KNO3 that could dissolve in 1 kg of water at 50 °C?

d) How much water is needed to dissolve 100 g of potassium nitrate at 45 °C?

e) At what temperature does potassium nitrate have a solubility of 20 g per 100 g of water?

■

Exercise 5.14 Use the solubility curves of KCl and NaOH on the last page to answer the questions.

a) The solubility of which of these two substances varies less with temperature?

b) What are their solubilities at 10 °C?

c) What's the solubility of KCl at 30 °C? And of NaOH at 80 °C?

d) What mass of KCl will form if a solution saturated in 100 g of water is cooled from 30 °C to 10 °C?

e) What mass of NaOH crystals will form if a solution saturated in 100 g of water is cooled from 80 °C to 10 °C?

f) At what temperature does sodium chlorate have solubility of 90 g in 100 g of water?

■

5.4 Vocabulary

■ **Vocabulary 5.1 — Matter.** Anything that has mass and occupies space.

Translation to Spanish: Materia

■ **Vocabulary 5.2 — Atom.** The smallest unit of an element that retains the chemical properties of that element.
Translation to Spanish: Átomo

■ **Vocabulary 5.3 — Molecule.** A group of atoms chemically bonded together.
Translation to Spanish: Molécula

■ **Vocabulary 5.4 — Element.** A pure substance consisting of only one type of atom.
Translation to Spanish: Elemento

■ **Vocabulary 5.5 — Compound.** A substance formed when two or more chemical elements are chemically combined.
Translation to Spanish: Compuesto

■ **Vocabulary 5.6 — Mixture.** A combination of substances that are not chemically bonded and can be separated physically.
Translation to Spanish: Mezcla

■ **Vocabulary 5.7 — Density.** The mass per unit volume of a substance.
Translation to Spanish: Densidad

■ **Vocabulary 5.8 — Solution.** A homogeneous mixture composed of a solute dissolved in a solvent.
Translation to Spanish: Disolución

■ **Vocabulary 5.9 — Solute.** The substance that is dissolved in a solution.
Translation to Spanish: Soluto

■ **Vocabulary 5.10 — Solvent.** The substance in which the solute dissolves to form a solution.
Translation to Spanish: Disolvente

■ **Vocabulary 5.11 — Chemical Change.** A process in which substances are transformed into new substances with different properties.
Translation to Spanish: Cambio químico

■ **Vocabulary 5.12 — Physical Change.** A change that affects the form or state of matter without altering its chemical composition.
Translation to Spanish: Cambio físico

■ **Vocabulary 5.13 — State of Matter.** One of the distinct forms that matter can exist in, such as solid, liquid, gas, or plasma.
Translation to Spanish: Estado de la materia

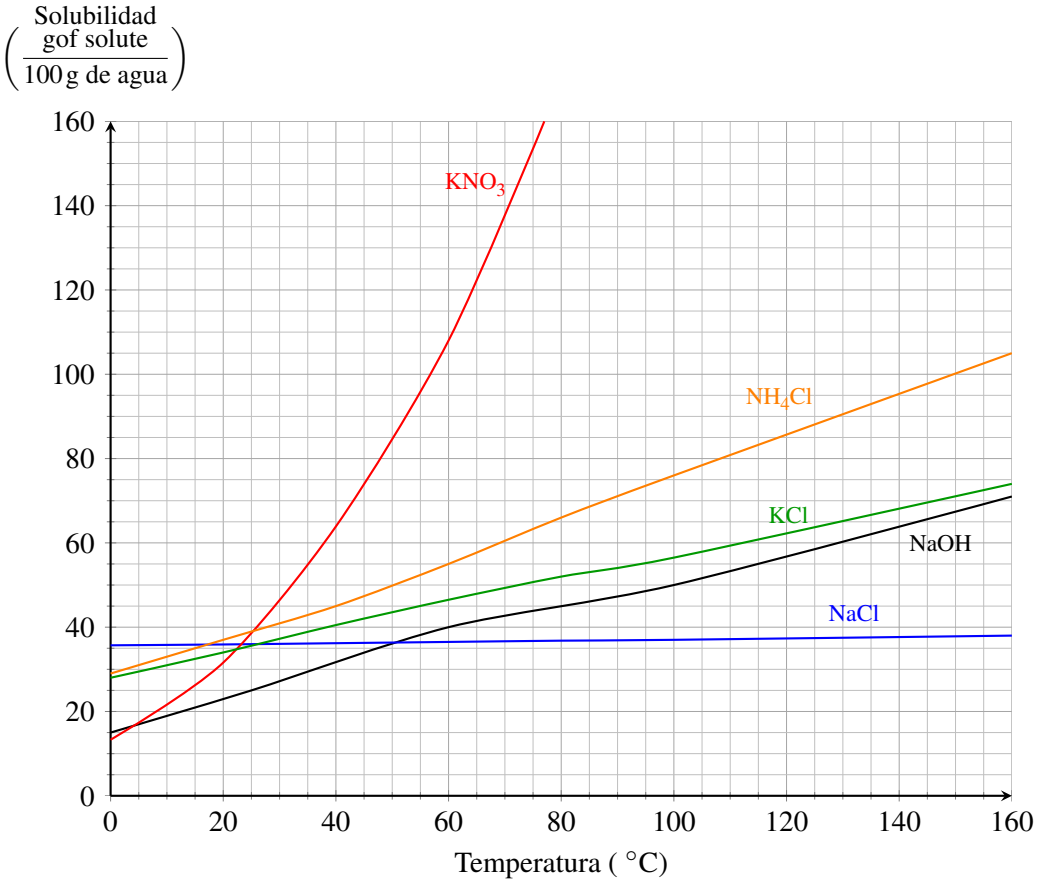


Figure (5.1): Solubility Curve

6.1 Introduction

The word **atom** was first introduced by Greek philosophers over 2000 years ago and means **indivisible**, that is, something that cannot be divided into smaller parts.

The atom is extraordinarily small. To give you an idea of its size: if an atom were the size of a marble, a real marble would be the size of the Earth. The hydrogen atom, which is the smallest of all, has a size on the order of 10^{-10} m (this is one ten-billionth of a meter!).

R **Did you know...?** If we lined up atoms in a row, we would need approximately 10 million atoms to cover the thickness of a sheet of paper.

All the matter around us is made up of atoms: the air we breathe, the water we drink, our own bodies... For this reason, throughout history, different scientists have dedicated their lives to trying to understand what atoms are like and how they are formed.

6.2 Atomic Model

Definition 6.1 — Atomic model. It is the representation of the atom in which its parts and how they are organized are identified. It is like a "map" that helps us understand something that is too small to see directly.

6.2.1 The Particles of the Atom

Although the name "atom" means indivisible, today we know that atoms do have parts. The atom is made up of three types of fundamental particles:

- **Protons (p^+):** are particles with a **positive electric charge** (+). We can imagine them as small balls with a "+" sign.
- **Neutrons (n^0):** are particles **with no electric charge**. They are "neutral," hence their name.
- **Electrons (e^-):** are particles with a **negative electric charge** (-). They are much smaller and

lighter than protons and neutrons.

R p^+ and n^0 have a similar mass (they are like twins in terms of size), while e^- are much lighter. If a proton weighed 1 kg, an electron would weigh barely 0.5 grams. Almost 2000 times less!

6.2.2 Structure of the Atom

The atom is organized into two distinct zones:

- **Nucleus**. It is the center of the atom, like the heart of a city. Here the **protons and neutrons** are found very close together and compact. Although the nucleus is very small compared to the total size of the atom, it contains almost all of its mass.
- **Electron cloud**. It is the zone that surrounds the nucleus, like the outskirts of a city. Here the **electrons** are found moving at high speed around the nucleus. Electrons do not move randomly, but rather occupy specific zones called **orbitals**, which are like "highways" where they can move.

R **An interesting comparison:** If the nucleus of an atom were the size of a tennis ball placed in the center of a soccer stadium, the electrons would be circling around the stands. The atom is mostly empty space!

6.2.3 Atoms Are Neutral

Under normal conditions, atoms are **electrically neutral**, that is, they have no total electric charge. This occurs because they have the same number of positive charges (protons) as negative charges (electrons), and they balance each other out.

Therefore, in a **neutral atom** we will have: Number of protons = Number of electrons

6.3 Atomic Number and Mass Number

To identify and differentiate the different types of atoms, scientists use two very important numbers: the **atomic number** and the **mass number**.

6.3.1 Atomic Number (Z)

The **atomic number**, symbolized by the letter **Z**, indicates the **number of protons** that an atom has in its nucleus.

$$Z = \text{number of protons} \tag{6.1}$$

The atomic number is like the "ID card" of each chemical element. It is what defines which element an atom belongs to:

- All hydrogen atoms have $Z = 1$ (1 proton)
- All carbon atoms have $Z = 6$ (6 protons)
- All oxygen atoms have $Z = 8$ (8 protons)

R As we have already seen, in a neutral atom the number of protons equals the number of electrons. Therefore, the atomic number also tells us how many electrons it has: $e^- = Z$

6.3.2 Mass Number (A)

The **mass number**, symbolized by the letter **A**, indicates the **total number of particles** in the nucleus (protons + neutrons).

$$A = \text{number of protons} + \text{number of neutrons} \tag{6.2}$$

6.3.3 Calculating the Number of Neutrons

If we know the mass number (A) and the atomic number (Z), we can easily calculate how many neutrons an atom has:

$$\text{Number of neutrons} = A - Z \tag{6.3}$$

R **Why?** Because if A is the sum of protons and neutrons, and Z is the number of protons, by subtracting we obtain only the neutrons.

6.3.4 Representation of Atoms

Elements are represented by indicating the element symbol, the atomic number, and the mass number, as shown in the Figure **??**

R If no $+$ or $-$ sign appears in the `upper right` of the element symbol, this indicates that the atom is **neutral** and therefore has the same number of protons as electrons.

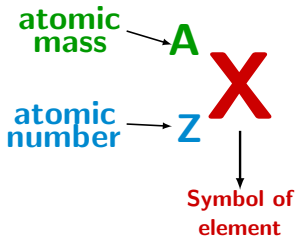


Figure (6.1): Representation of mass number and atomic number

■ **Example 6.1** Let's determine the number of protons, neutrons, and electrons in the following neutral atoms:

- **$^{23}_{11}$Na (Sodium):**
  - Protons: $Z = 11 \rightarrow 11$ protons
  - Electrons: Since it is neutral, 11 electrons
  - Neutrons: $n^0 = A - Z = 23 - 11 = 12$ neutrons
- **$^{210}_{84}$Po (Polonium):**
  - Protons: $Z = 84 \rightarrow 84$ protons
  - Electrons: Since it is neutral, 84 electrons
  - Neutrons: $n^0 = A - Z = 209 - 84 = 125$ neutrons

■

Exercise 6.1 Indicate the number of protons, neutrons, and electrons in a chlorine atom $^{35}_{17}\text{Cl}$. ■

Exercise 6.2 How would you represent an electrically neutral potassium atom if Z = 19 and A = 39? How many protons, neutrons, and electrons does it have? ■

Exercise 6.3 Can we say that Z always indicates the number of electrons? Explain your answer. ■

Exercise 6.4 The number of neutrons in lithium is one more than the number of protons. Knowing that its mass number is 7:

a) What is its atomic number?

b) How many electrons does an electrically neutral lithium atom have? ■

Exercise 6.5 Represent in the notation $^{A}_{Z}\text{X}$ indicating the number of protons, neutrons, and electrons of the following atoms:

a) Boron with 5 protons and 6 neutrons

b) Nitrogen with 7 protons and 7 neutrons

c) Carbon with 6 protons and 6 neutrons

d) Oxygen with 8 protons and 8 neutrons

e) Fluorine with 9 protons and 10 neutrons

f) Helium with 2 protons and 2 neutrons ■

Exercise 6.6 Indicate the number of protons, neutrons, and electrons in the following atoms:

a) $^{4}_{2}\text{He}$ ***b)*** $^{19}_{9}\text{F}$ ***c)*** $^{20}_{10}\text{Ne}$ ***d)*** $^{24}_{12}\text{Mg}$ ***e)*** $^{40}_{18}\text{Ar}$ ■

6.4 Ions

Definition 6.2 — ion. Is an atom (or group of atoms) that has gained or lost electrons and therefore has an electric charge.

When an atom gains or loses electrons, it is no longer neutral and becomes an ion:

- **Cation** (**positive ion**): It is formed when an atom **loses electrons**. By having fewer electrons than protons, it is left with a positive charge. It is represented with the + sign.
  Example: Na^{+} (sodium has lost 1 electron)
- **Anion** (**negative ion**): It is formed when an atom **gains electrons**. By having more electrons

than protons, it is left with a negative charge. It is represented with the − sign.

Example: Cl^- (chlorine has gained 1 electron)

R **Important rule:** The number of protons (Z) **never changes** when an ion is formed, because protons are in the nucleus and cannot be easily gained or lost. Only the number of electrons changes.

■ **Example 6.2** Let's determine the number of protons, neutrons, and electrons in the following ions:

- $^{23}_{11}Na^+$ **(Sodium cation):**
  - Protons: $Z = 11 \rightarrow 11$ protons (does not change)
  - Neutrons: $A - Z = 23 - 11 = 12$ neutrons
  - Electrons: 10 electrons (has lost 1 electron: 11 - 1 = 10)
- $^{35}_{17}Cl^-$ **(Chlorine anion):**
  - Protons: $Z = 17 \rightarrow 17$ protons (does not change)
  - Neutrons: $A - Z = 35 - 17 = 18$ neutrons
  - Electrons: 18 electrons (has gained 1 electron: 17 + 1 = 18)

■

Exercise 6.7 Indicate the number of protons, neutrons, and electrons in the following ions:

a) $^{40}_{20}Ca^{2+}$ *b)* $^{16}_{8}O^{2-}$ *c)* $^{27}_{13}Al^{3+}$ *d)* $^{32}_{16}S^{2-}$

■

Exercise 6.8 How many electrons has an aluminum atom ($Z = 13$) lost or gained to become Al^{3+}? ■

6.5 Isotopes

Definition 6.3 — isotopes. Are atoms of the same chemical element (they have the same atomic number Z) but different numbers of neutrons and therefore different mass numbers (A).

Characteristics of isotopes:

- They have the **same number of protons** (same Z)
- They have **different numbers of neutrons**
- They have **different mass numbers** (A)
- They are of the **same chemical element**
- They have **very similar chemical properties**

■ **Example 6.3** Carbon has several isotopes. The most common are:

- **Carbon-12:** $^{12}_{6}C$
  - 6 protons, 6 neutrons, 6 electrons
  - It is the most abundant isotope (98.9%)

- **Carbon-13:** $^{13}_{6}\mathrm{C}$
  - 6 protons, 7 neutrons, 6 electrons
  - Less abundant (1.1%)
- **Carbon-14:** $^{14}_{6}\mathrm{C}$
  - 6 protons, 8 neutrons, 6 electrons
  - It is radioactive and is used to date fossils

All three are carbon because they have 6 protons, but they differ in the number of neutrons. ■

Exercise 6.9 Which of the following atoms are isotopes of each other?

a) $^{20}_{10}\mathrm{Ne}$ ***b)*** $^{21}_{10}\mathrm{Ne}$ ***c)*** $^{19}_{9}\mathrm{F}$ ***d)*** $^{23}_{11}\mathrm{Na}$

■

Exercise 6.10 Hydrogen has three known isotopes:

- Common hydrogen: 1 proton, 0 neutrons
- Deuterium: 1 proton, 1 neutron
- Tritium: 1 proton, 2 neutrons

Represent them using the notation $^{A}_{Z}\mathrm{X}$. ■

6.6 Electron Configuration

We already know that electrons are distributed around the nucleus at different energy levels, but they don't do so randomly. Electrons are organized in different zones or "floors" around the nucleus.

Definition 6.4 Electron configuration The **electron configuration** is like the "postal address" of electrons. It tells us exactly where each electron is within the atom, using a series of numbers and letters.

6.6.1 Energy levels: the layers of the atom

Imagine the atom as an apartment building where the nucleus is in the center and the electrons live on different floors around it:

- Electrons closest to the nucleus (floor 1) are more strongly attracted to it
- Electrons farther away (upper floors) are less attracted
- Each floor is called an **energy level** and is numbered from 1 to 7

R **Useful analogy:** Energy levels are like the layers of an onion. The first layer (innermost) is level 1, the second layer is level 2, and so on up to a maximum of 7 layers.

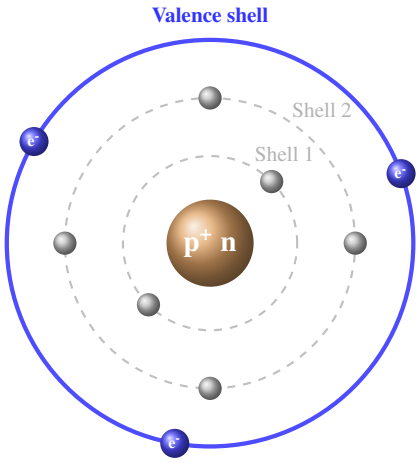


Figure (6.2): Energy levels and valence shell.

6.6.2 Orbitals: rooms within each floor

Within each level (each floor), there are different types of "rooms" where electrons can be. These rooms are called **sublevels** or **orbitals** and are represented by letters: **s, p, d, f**

- **Sublevel s**: The simplest, can hold up to **2 electrons** maximum
- **Sublevel p**: Slightly larger, can hold up to **6 electrons** maximum
- **Sublevel d**: More spacious, can hold up to **10 electrons** maximum
- **Sublevel f**: The largest, can hold up to **14 electrons** maximum

6.6.3 What orbitals are in each level?

Not all levels have all types of orbitals. Here is the complete list:

- **Level 1**: only has orbital **s** $\rightarrow$ we write $1s$
- **Level 2**: has orbitals **s** and **p** $\rightarrow$ we write $2s, 2p$
- **Level 3**: has **s, p** and **d** $\rightarrow$ we write $3s, 3p, 3d$
- **Level 4**: has **s, p, d** and **f** $\rightarrow$ we write $4s, 4p, 4d, 4f$
- **Levels 5, 6 and 7**: same as level 4, they have s, p, d and f

R **Observable pattern:** Each new level adds one more type of orbital. Level 1 only has s, level 2 adds p, level 3 adds d, and level 4 adds f.

6.6.4 How orbitals are written

When we write an orbital, we use:

1. A **number** indicating the level (1, 2, 3, 4...)
2. A **letter** indicating the type of orbital (s, p, d, f)
3. An **exponent** (small number above) indicating how many electrons are in that orbital

Examples:

- $1s^2 \rightarrow$ In level 1, orbital s, there are 2 electrons
- $2p^6 \rightarrow$ In level 2, orbital p, there are 6 electrons
- $3d^{10} \rightarrow$ In level 3, orbital d, there are 10 electrons
- $3s^1 \rightarrow$ In level 3, orbital s, there is 1 electron

6.6.5 The Moeller Diagram: the guide for filling orbitals

Electrons don't fill orbitals in the order we might think (first 1s, then 2s, then 2p...). There is a special order we must follow, and for this we use the **Moeller Diagram**.

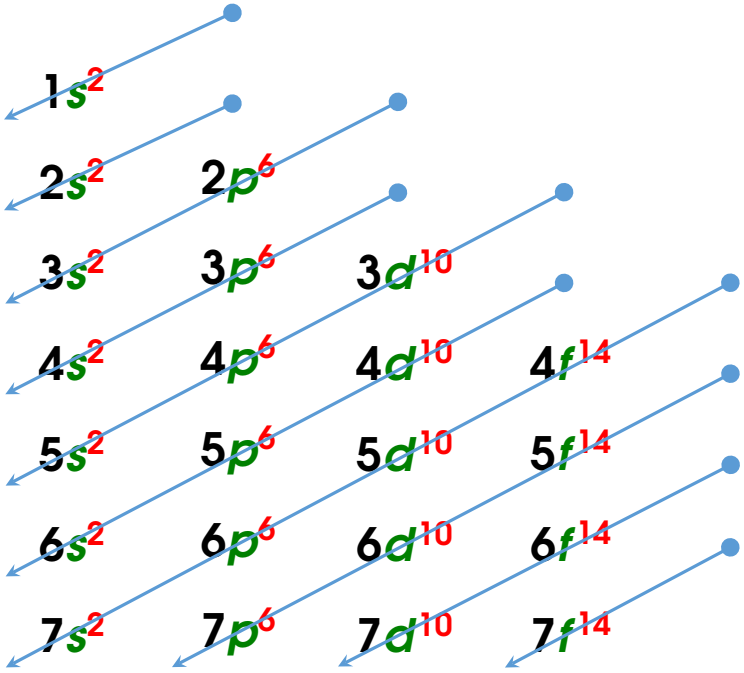


Figure (6.3): Moeller Diagram

R **How to use the Moeller Diagram:** Each arrow tells you which orbitals you should go through. You start with the first arrow (top left) and follow the path of each arrow. The order you get is:

$$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow ...$$

6.6.6 How to write an electron configuration step by step

Let's learn to write the electron configuration of **Sodium (Na)** which has $Z = 11$, that is, 11 electrons.

Step 1: Identify how many electrons the atom has

Sodium has $Z = 11$, therefore it has **11 electrons** that we must place.

Step 2: Follow the Moeller diagram and fill the orbitals in order

- Arrows are always followed from the starting point to the arrowhead, checking the orbitals we pass through until reaching the arrowhead.

- First arrow: In this case it only passes through 1s: This orbital can have a maximum of 2 electrons. We write $1s^2$
  Electrons placed: 2. We have left: 11 - 2 = 9 electrons.
- Second arrow: We have the same situation and it only passes through 2s. This orbital can have a maximum of 2 electrons. We write $2s^2$
  Electrons placed: 2 (from 1s) + 2 (from 2s) = 4. We have left: 11 - 4 = 7 electrons.
- Third arrow: now it passes through 2p and 3s: First we place the electrons in orbital 2p because it is the first to fill and since we can place a maximum of 6 electrons then we write: $2p^6$. Now we have only 1 electron left to place. The next orbital according to the arrow will be 3s, in this case we could put a maximum of 2 electrons but since we only have 1 electron left to place then we must write: $3s^1$
  Electrons placed: 2 (from 1s) + 2 (from 2s) + 6 (from 2p) + 1 (from 3s) = 11. Perfect! We have placed them all
  Electrons placed: 10 + 1 = 11.

Step 3: Write the complete electron configuration

Configuration of Na: $1s^2\ 2s^2\ 2p^6\ 3s^1$

R **Important tip:** Always count the electrons after each step to make sure you don't exceed the total number of electrons in the atom.

■ **Example 6.4 Electron configuration of Rubidium (Rb)** with $Z = 37$: Following the Moeller diagram and counting 37 electrons:

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^1$$

Note that orbital 3d fills **after** 4s, even though 3 is less than 4. This is what the Moeller diagram indicates. **Reordered by levels** (optional, but more organized):

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 5s^1$$

■

6.6.7 Electron configuration of ions

When an atom becomes an ion (gains or loses electrons), its electron configuration changes:

- **For cations** (positive ions, have lost electrons):
  First write the configuration of the neutral atom, then **remove electrons from the end** (from the orbital with the highest number).
  Example: Na^+ (has lost 1 electron)
  - Neutral Na: $1s^2\ 2s^2\ 2p^6\ 3s^1$
  - Na^+: $1s^2\ 2s^2\ 2p^6$ (we remove 1 electron from 3s)

- **For anions** (negative ions, have gained electrons): First write the configuration of the neutral atom, then **add electrons** following the order of the Moeller diagram.

  Example: Cl^- (has gained 1 electron)
  - Neutral Cl ($Z = 17$): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
  - Cl^-: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (we add 1 electron to 3p)

Exercise 6.11 Write the electron configuration of Carbon ($Z = 6$) in its ground state (does not lose or gain e^-). ■

Exercise 6.12 Determine the electron configuration of Phosphorus ($Z = 15$). ■

Exercise 6.13 Indicate the electron configuration of the chloride anion Cl^- ($Z = 17$). ■

Exercise 6.14 Write the electron configuration of the oxide ion O^{2-} ($Z = 8$). ■

Exercise 6.15 Determine the electron configuration of the sulfide anion S^{2-} ($Z = 16$). ■

Exercise 6.16 Write the electron configuration of the sodium cation Na^+ ($Z = 11$). ■

Exercise 6.17 Indicate the electron configuration of the calcium ion Ca^{2+} ($Z = 20$). ■

6.6.8 Valence shell and valence electrons

The outermost shell or level of an atom is very important because it determines how that atom will behave chemically.

Definition 6.5 Valence Shell The **valence shell** is the highest energy level (the largest number) that has electrons. It is the outermost shell of the atom.

Definition 6.6 Valence electrons **Valence electrons** are all the electrons that are in the valence shell. They are the outermost electrons and are the ones that participate in chemical reactions.

R **Why are valence electrons important?** They are like the atom's "hands": with them it can form chemical bonds with other atoms. Inner electrons are too close to the nucleus and do not participate in reactions.

■ **Example 6.5 Sodium (Na):** $1s^2\ 2s^2\ 2p^6\ 3s^1$
- Valence shell: **n = 3** (the highest number that appears)
- Valence electrons: **1 electron** (the one in $3s^1$)

Oxygen (O): $1s^2\ 2s^2\ 2p^4$
- Valence shell: **n = 2**
- Valence electrons: **6 electrons** (2 in 2s + 4 in 2p = 6)

Carbon (C): $1s^2\ 2s^2\ 2p^2$

- Valence shell: **n = 2**
- Valence electrons: **4 electrons** (2 in 2s + 2 in 2p = 4)

■

R **Connection with the periodic table:** Elements in the same group (same column) have the same number of valence electrons. That's why they have similar chemical properties. We'll see this in the next section!

Exercise 6.18 Identify the valence shell and the number of valence electrons of Lithium ($Z = 3$).

■

Exercise 6.19 Determine the valence shell and valence electrons of Nitrogen ($Z = 7$). ■

Exercise 6.20 For the Magnesium atom ($Z = 12$), indicate its configuration and valence electrons. ■

Exercise 6.21 Write the configuration of Aluminum ($Z = 13$) and identify its valence shell. ■

Exercise 6.22 Determine the valence electrons of Sulfur ($Z = 16$). ■

6.7 The Periodic System of Elements

The Periodic System, also called the Periodic Table, is a fundamental tool in chemistry that organizes all known chemical elements.

6.7.1 History and Organization

It was created by the Russian chemist **Dmitri Mendeleev** in 1869. Mendeleev organized the elements in order of increasing atomic mass and observed that chemical properties repeated periodically.

R **Interesting fact:** Mendeleev left gaps in his table for elements that had not yet been discovered. He even predicted their properties! Years later, when those elements were discovered, their properties matched Mendeleev's predictions.

In the current periodic table, elements are arranged in **increasing atomic number** (from lowest to highest Z) and organized into:

- **Horizontal rows** called **periods**. There are 7 periods numbered from 1 to 7.
- **Vertical columns** called **groups** or **families**. There are 18 groups numbered from 1 to 18.

6.7.2 Periods

Periods are the horizontal rows of the periodic table. The period number indicates how many electron shells an atom of that element has.

- Period 1: elements with 1 electron shell (H, He)
- Period 2: elements with 2 electron shells (Li, Be, B, C, N, O, F, Ne)

- Period 3: elements with 3 electron shells (Na, Mg, Al, Si, P, S, Cl, Ar)
- And so on...

6.7.3 Groups or Families

Groups are the vertical columns. Elements in the same group have similar chemical properties because they have the same number of electrons in their outermost shell. Some groups have special names due to their characteristic properties:

- **Group 1 - Alkali metals**: They are very reactive metals (Li, Na, K, Rb, Cs, Fr). They react violently with water.
- **Group 2 - Alkaline earth metals**: They are less reactive metals than alkali metals (Be, Mg, Ca, Sr, Ba, Ra).
- **Groups 3 to 12 - Transition metals**: They include many well-known metals such as iron (Fe), copper (Cu), silver (Ag), gold (Au), etc.
- **Group 13 - Boron group**: Notable members include boron (B) and aluminum (Al).
- **Group 14 - Carbon group**: Includes carbon (C), silicon (Si), etc.
- **Group 15 - Nitrogen group**: Includes nitrogen (N) and phosphorus (P).
- **Group 16 - Chalcogens**: Includes oxygen (O) and sulfur (S).
- **Group 17 - Halogens**: They are very reactive nonmetals (F, Cl, Br, I, At). Chlorine is used to disinfect swimming pool water.
- **Group 18 - Noble gases**: They are gases that hardly react with anything (He, Ne, Ar, Kr, Xe, Rn). Helium is used to inflate balloons.

R **Memory trick:** Elements in the same group (same column) resemble each other, like siblings in a family. That's why they are called "families".

6.7.4 Metals, Nonmetals, and Metalloids

Elements are also classified according to their properties into:

- **Metals**: Shiny, good conductors of heat and electricity, malleable and ductile. Most elements are metals. Examples: Fe, Cu, Au, Ag, Al.
- **Nonmetals**: Not shiny, poor conductors, brittle. They are on the right side of the table. Examples: O, N, C, S, Cl.
- **Metalloids**: They have intermediate properties. They are used in electronics. Examples: Si, Ge, As.

Exercise 6.23 Indicate the group and period in which the following elements are found: nitrogen, aluminum, krypton, silver, rubidium, titanium, cesium, and lead. ■

Exercise 6.24 Write the symbol of the following elements:

a) magnesium	*d)* argon	*g)* copper	*j)* phosphorus
b) carbon	*e)* lithium	*h)* zinc	*k)* palladium
c) aluminum	*f)* silicon	*i)* helium	*l)* rubidium

■

Exercise 6.25 Write the name of the following elements:

a) O	*d)* H	*g)* Ag	*j)* I
b) N	*e)* Ne	*h)* Au	*k)* Cu
c) Ni	*f)* Mn	*i)* Hg	*l)* Pt

■

Exercise 6.26 Write the name and symbol of the following elements given their group and period.

a) Group 1 Period 4	*e)* Group 14 Period 4	*i)* Group 15 Period 5
b) Group 18 Period 2	*f)* Group 12 Period 5	*j)* Group 14 Period 3
c) Group 13 Period 3	*g)* Group 10 Period 4	*k)* Group 18 Period 6
d) Group 2 Period 5	*h)* Group 16 Period 2	*l)* Group 8 Period 4

■

Exercise 6.27 Search the periodic table for three elements that are:

a) Metals	*b)* Nonmetals	*c)* Noble gases	*d)* Halogens

■

6.8 Vocabulary

■ **Vocabulary 6.1 — Atom.** Smallest unit of matter that means "indivisible," referring to something that cannot be divided into smaller parts.
Translation to spanish: Átomo

■ **Vocabulary 6.2 — Atomic model.** Representation of the atom in which its parts and how they are organized are identified.
Translation to spanish: Modelo atómico

■ **Vocabulary 6.3 — Protons.** Fundamental particles of the atom that have a positive electric charge (+).
Translation to spanish: Protones

■ **Vocabulary 6.4 — Neutrons.** Fundamental particles of the atom with no electric charge (neutral).
Translation to spanish: Neutrones

■ **Vocabulary 6.5 — Electrons.** Fundamental particles of the atom with a negative electric charge (-).
Translation to spanish: Electrones

■ **Vocabulary 6.6 — Nucleus.** The small, dense center of the atom where protons and neutrons are found very close together and compact.
Translation to spanish: Núcleo

■ **Vocabulary 6.7 — Electron cloud.** The zone that surrounds the nucleus where the electrons are found moving at high speed.
Translation to spanish: Corteza electrónica

■ **Vocabulary 6.8 — Orbitals.** Specific zones within the electron cloud where electrons move around the nucleus.
Translation to spanish: Orbitales

■ **Vocabulary 6.9 — Neutral atom.** An atom that has no total electric charge because it has the same number of protons and electrons.
Translation to spanish: Átomo neutro

■ **Vocabulary 6.10 — Atomic Number (Z).** Property that indicates the number of protons that an atom has in its nucleus; it identifies each chemical element.
Translation to spanish: Número atómico (Z)

■ **Vocabulary 6.11 — Mass Number (A).** Property that indicates the total number of particles (protons and neutrons) in the nucleus.
Translation to spanish: Número másico (A)

■ **Vocabulary 6.12 — Ion.** An atom or group of atoms that has gained or lost electrons and therefore has an electric charge.
Translation to spanish: Ion

■ **Vocabulary 6.13 — Cation.** A positive ion formed when an atom loses electrons, leaving it with fewer electrons than protons.
Translation to spanish: Catión

■ **Vocabulary 6.14 — Anion.** A negative ion formed when an atom gains electrons, leaving it with more electrons than protons.
Translation to spanish: Anión

■ **Vocabulary 6.15 — Isotopes.** Atoms of the same chemical element (same Z) but different numbers of neutrons and therefore different mass numbers (A).
Translation to spanish: Isótopos

■ **Vocabulary 6.16 — Indivisible.** Term referring to something that cannot be divided into smaller parts, originally used to describe atoms.
Translation to spanish: Indivisible

7.1 Consideraciones previas

7.1.1 Conceptos básicos

- **Fórmula química**: es como la "receta" de un compuesto. Nos dice qué elementos lo forman y en qué cantidad. Usamos el símbolo químico de cada elemento y unos números pequeños (subíndices) que indican cuántos átomos hay de cada uno. Por ejemplo: Fe_2O_3 significa 2 átomos de hierro y 3 átomos de oxígeno.
- **Formulación**: es el proceso de escribir la fórmula química cuando nos dan el nombre del compuesto. Es decir, pasar del nombre a la fórmula.
- **Nomenclatura**: es el proceso inverso. Dada la fórmula química, debemos encontrar el nombre correcto del compuesto.
- **Compuestos binarios**: están formados por dos elementos químicos diferentes (aunque algunos casos especiales como H_2 también se consideran binarios).
- **Compuestos ternarios**: están formados por tres elementos químicos diferentes.
- **Metales**: son los elementos de los grupos del 1 al 12 de la tabla periódica. Ejemplos: sodio (Na), hierro (Fe), cobre (Cu), calcio (Ca).
- **No metales**: son los elementos de los grupos 13 al 18. Ejemplos: oxígeno (O), nitrógeno (N), cloro (Cl), azufre (S).
- **Halógenos**: son los elementos del grupo 17 (flúor, cloro, bromo, yodo).

7.1.2 Estados de oxidación: la clave de la formulación

Cuando dos elementos se unen para formar un compuesto, se produce un intercambio de electrones: uno de los elementos **cede** electrones y el otro los **recibe**.

Definition 7.1 — Estados de oxidación. Los estados de oxidación son como una "puntuación" que indica cuántos electrones cede (+) o recibe (-) un elemento al unirse con otro para formar un compuesto. Un mismo elemento puede tener varios estados de oxidación dependiendo del compuesto que forme.

Regla importante: El elemento con estado de oxidación positivo se escribe a la izquierda, y el que tiene estado de oxidación negativo se escribe a la derecha.

■ **Example 7.1** El cobre (Cu) puede unirse con el oxígeno (O) de dos formas diferentes:

a) **Formando CuO**:
- El Cu cede 2 electrones → estado de oxidación +2
- El O recibe 2 electrones → estado de oxidación -2

b) **Formando Cu_2O**:
- El Cu cede 1 electrón → estado de oxidación +1
- El O recibe 2 electrones → estado de oxidación -2

■

7.1.3 Cómo usar los estados de oxidación para formular

Durante este curso trabajaremos con los estados de oxidación en valor absoluto (sin el signo) para intercambiarlos y crear las fórmulas. El proceso es el siguiente:

■ **Example 7.2** Vamos a formar el compuesto entre el sodio (Na) y el oxígeno (O):

a) Escribimos el Na (estado de oxidación +1) a la izquierda y el O (estado de oxidación -2) a la derecha:
$^{1}\mathrm{Na}\ ^{2}\mathrm{O}$

b) **Intercambiamos los estados de oxidación** (en valor absoluto): el 1 del Na pasa como subíndice al O, y el 2 del O pasa como subíndice al Na:
$\mathrm{Na_2O_1}$

c) Como el subíndice 1 no se escribe, la fórmula final es: $\mathrm{Na_2O}$

d) **Comprobación**: $2\times(+1)+1\times(-2)=+2-2=0$ ✓ (la suma debe dar siempre cero)

■

R ▷ **Regla de oro**: Los estados de oxidación se intercambian en valor absoluto y pasan a ser los subíndices del otro elemento.

Nota importante: Cuando un elemento tiene varios estados de oxidación posibles, el ejercicio nos indicará cuál usar, o lo deduciremos de los subíndices de la fórmula.

7.1.4 Prefijos multiplicadores

Los prefijos son palabras que ponemos delante del nombre del elemento para indicar el número de átomos de ese elemento en la fórmula.

- **Mono**: 1 átomo. *Normalmente se omite*.
- **Di**: 2 átomos
- **Tri**: 3 átomos
- **Tetra**: 4 átomos
- **Penta**: 5 átomos
- **Hexa**: 6 átomos
- **Hepta**: 7 átomos
- **Octa**: 8 átomos

■ **Example 7.3** Aquí tenemos varios ejemplos, el prefijo -mono se suele omitir pero en el caso del CO no se omite, es una de las excepciones.

- CO: monóxido de carbono (1 oxígeno)
- CO_2: dióxido de carbono (2 oxígenos)
- N_2O_3: trióxido de dinitrógeno (3 oxígenos, 2 nitrógenos)

■

Nota: Los nombres de los prefijos son iguales en inglés y en español.

7.1.5 Tipos de nomenclatura

La IUPAC (Unión Internacional de Química Pura y Aplicada) permite nombrar los compuestos de varias formas. En este curso aprenderemos dos nomenclaturas principales:

a) **Nomenclatura de multiplicadores** (o nomenclatura sistemática): usa los prefijos (mono, di, tri...) para indicar el número de átomos.

b) **Nomenclatura con estados de oxidación**: indica el estado de oxidación del elemento mediante números romanos entre paréntesis.

■ **Example 7.4** El compuesto Fe_2O_3 se puede nombrar de dos formas:

- **Nomenclatura de multiplicadores**: trióxido de dihierro
- **Nomenclatura con estados de oxidación**: óxido de hierro (III)

■

7.2 Óxidos

7.2.1 ¿Qué son los óxidos?

Los óxidos son compuestos formados por un elemento (metal o no metal) y oxígeno.

- El oxígeno **siempre** va a la derecha en la fórmula
- El oxígeno tiene estado de oxidación **-2** (en óxidos normales)
- Fórmula general: E_xO_y (donde E es cualquier elemento)

7.2.2 Nomenclatura de multiplicadores

-  : prefijo + óxido + de + prefijo + nombre del elemento

  Ejemplos:

  ❶: Na_2O: óxido de disodio (o monóxido de disodio)

  ❷: SO_3: trióxido de azufre

  ❸: Al_2O_3: trióxido de dialuminio

- : prefix + name of the element + prefix + oxide

  ejemplos:

  ❶: CaO: calcium oxide (or calcium monoxide)

  ❷: CO_2: carbon dioxide

7.2.3 Nomenclatura con estados de oxidación

- : óxido + de + nombre del elemento (estado de oxidación)

  Regla importante: Si el elemento solo tiene un estado de oxidación, **NO** se pone el número romano.

  Ejemplos:

  ❶: MgO: óxido de magnesio (el Mg solo tiene +2, no hace falta indicarlo)

  ❷: FeO: óxido de hierro (II)

  ❸: Fe_2O_3: óxido de hierro (III)
- : name of the element (oxidation state) + oxide

  ejemplos:

  ❶: CuO: copper (II) oxide

  ❷: Cu_2O: copper (I) oxide

7.2.4 Cómo formular óxidos

Pasos para formular:

1. Escribe el símbolo del elemento a la izquierda
2. Escribe el símbolo del oxígeno (O) a la derecha
3. Intercambia los estados de oxidación como subíndices
4. Simplifica los subíndices si es posible (divide por el máximo común divisor)

■ **Example 7.5** **Formular**: Óxido de aluminio

- Al tiene estado +3, O tiene estado -2
- Escribimos: $\overset{3}{\text{Al}}\ \overset{2}{\text{O}}$
- Intercambiamos: Al_2O_3
- No se puede simplificar → **Fórmula final**: Al_2O_3

■

R **¡Atención!** Cuando simplifiquemos, debemos dividir **todos** los subíndices por el mismo número (siempre será 2).

Ejemplo: Ca_2O_2 se simplifica a CaO (dividimos ambos subíndices entre 2)

7.3 Hidruros

7.3.1 ¿Qué son los hidruros?

Los hidruros son compuestos formados por hidrógeno y otro elemento.

- Cuando el hidrógeno se une a un **metal**, tiene estado de oxidación **-1** → Hidruros metálicos
- Cuando el hidrógeno se une a un **no metal**, tiene estado de oxidación **+1** → Hidruros no metálicos

7.3.2 Hidruros metálicos

En los hidruros metálicos:

- El metal va a la **izquierda**
- El hidrógeno va a la **derecha**
- El hidrógeno tiene estado de oxidación **-1**

7.3.2.1 Nomenclatura de multiplicadores

- : prefijo + hidruro + de + prefijo + nombre del metal

  Ejemplos:

  ❶: NaH: hidruro de sodio

  ❷: CaH_2: dihidruro de calcio

  ❸: AlH_3: trihidruro de aluminio
- : prefix + name of the metal + prefix + hydride

  ejemplos:

  ❶: LiH: lithium hydride

  ❷: FeH_3: iron trihydride

7.3.2.2 Nomenclatura con estados de oxidación

- : hidruro + de + nombre del metal $\left(\text{estado de oxidación}\right)$

  Ejemplos:

  ❶: KH: hidruro de potasio (el K solo tiene +1)

  ❷: FeH_2: hidruro de hierro (II)

  ❸: FeH_3: hidruro de hierro (III)
- : name of the metal $\left(\text{oxidation state}\right)$ + hydride

  ejemplos:

  ❶: CuH: copper (I) hydride

  ❷: CuH_2: copper (II) hydride

7.3.2.3 Cómo formular hidruros metálicos

1. Escribe el símbolo del metal a la izquierda
2. Escribe H a la derecha
3. Intercambia los estados de oxidación como subíndices
4. Simplifica si es posible

■ **Example 7.6** **Formular**: Dihidruro de magnesio

- Mg tiene estado +2, H tiene estado -1

- Escribimos: $\overset{2}{Mg}\ \overset{1}{H}$
- Intercambiamos: MgH_2
- No se puede simplificar → **Fórmula final**: MgH_2

■

7.3.3 Hidruros no metálicos (Hidrácidos)

Los hidruros no metálicos son compuestos del hidrógeno con elementos del grupo 16 (S, Se, Te) y del grupo 17 (F, Cl, Br, I).

- El hidrógeno va a la **izquierda**
- El no metal va a la **derecha**
- El hidrógeno tiene estado de oxidación **+1**

Característica especial: Cuando estos compuestos se disuelven en agua, forman **ácidos** y entonces se llaman **hidrácidos**. Para indicar que están en disolución acuosa, añadimos **(ac)** en español o **(aq)** en inglés después de la fórmula.

7.3.3.1 Nomenclatura de multiplicadores

Como compuesto puro (sin disolver en agua):

- : prefijo + nombre del no metal terminado en -uro + de + hidrógeno

  Ejemplos:

  ❶: HF: fluoruro de hidrógeno

  ❷: H_2S: sulfuro de hidrógeno (o sulfuro de dihidrógeno)

  ❸: HCl: cloruro de hidrógeno
- : hydrogen + prefix + name of nonmetal ending in -ide

  ejemplos:

  ❶: HBr: hydrogen bromide

  ❷: H_2Se: hydrogen selenide

Como ácido (disuelto en agua - hidrácido):

- : ácido + raíz del no metal + -hídrico

  Ejemplos:

  ❶: HCl(ac): ácido clorhídrico

  ❷: H_2S(ac): ácido sulfhídrico

  ❸: HF(ac): ácido fluorhídrico
- : hydro + root of nonmetal + -ic + acid

  ejemplos:

  ❶: HCl(aq): hydrochloric acid

  ❷: H_2S(aq): hydrosulfuric acid

R **¡No te confundas!**
- HCl (gas) = cloruro de hidrógeno

- HCl(ac) (en agua) = ácido clorhídrico

¡Es el mismo compuesto, pero el nombre cambia según esté disuelto o no!

7.3.3.2 Nomenclatura con estados de oxidación

Los hidrácidos **NO** se nombran con estados de oxidación. Solo se usa la nomenclatura de multiplicadores.

7.3.3.3 Cómo formular hidruros no metálicos

1. Escribe H a la izquierda
2. Escribe el símbolo del no metal a la derecha
3. Intercambia los estados de oxidación como subíndices
4. Simplifica si es posible
5. Si el compuesto está en disolución acuosa, añade (ac) o (aq)

■ **Example 7.7** **Formular**: Ácido bromhídrico

- Es un hidrácido → H a la izquierda, Br a la derecha
- H tiene estado +1, Br tiene estado -1
- Intercambiamos: $H_1Br_1 \rightarrow HBr$
- Como es un ácido (está en disolución), añadimos (ac)
- **Fórmula final**: HBr(ac)

■

7.4 Sales binarias (combinaciones de metal y no metal)

7.4.1 ¿Qué son las sales binarias?

Las sales binarias son compuestos formados por un metal y un no metal que no sean los casos anteriores (óxidos, hidruros, hidróxidos o peróxidos).

- El metal va a la **izquierda**
- El no metal va a la **derecha**
- El no metal usa su estado de oxidación negativo

7.4.2 Nomenclatura de multiplicadores

-  : prefijo + raíz del no metal + -uro + de + prefijo + nombre del metal

  Ejemplos:

  ❶: NaCl: cloruro de sodio (la sal común)

  ❷: Al_4C_3: tricarburo de tetraaluminio

  ❸: CaF_2: difluoruro de calcio
-  : prefix + name of the metal + prefix + root of nonmetal + -ide

  ejemplos:

  ❶: NaF: sodium fluoride

❷: $CaCl_2$: calcium dichloride

❸: FeS: iron sulfide

7.4.3 Nomenclatura con estados de oxidación

- 🇪🇸: raíz del no metal + -uro + de + nombre del metal $\left(\text{estado de oxidación}\right)$

  Ejemplos:

  ❶: NaCl: cloruro de sodio (el Na solo tiene +1)

  ❷: $FeCl_2$: cloruro de hierro (II)

  ❸: $FeCl_3$: cloruro de hierro (III)
- 🇬🇧: name of the metal $\left(\text{oxidation state}\right)$ + root of nonmetal + -ide

  ejemplos:

  ❶: CuCl: copper (I) chloride

  ❷: $CuCl_2$: copper (II) chloride

7.4.4 Cómo formular sales binarias

1. Escribe el símbolo del metal a la izquierda
2. Escribe el símbolo del no metal a la derecha
3. Intercambia los estados de oxidación como subíndices
4. Simplifica si es posible

■ **Example 7.8** **Formular**: Sulfuro de aluminio

- Al tiene estado +3, S tiene estado -2
- Escribimos: $\overset{3}{\text{Al}}\ \overset{2}{\text{S}}$
- Intercambiamos: Al_2S_3
- No se puede simplificar → **Fórmula final**: Al_2S_3

■

7.5 Hidróxidos

7.5.1 ¿Qué son los hidróxidos?

Los hidróxidos son compuestos ternarios (tres elementos) formados por:

- Un metal (o catión)
- El grupo hidróxido: OH^-

Reglas importantes:

- El grupo OH es **indivisible**, funciona como una unidad
- El grupo OH tiene estado de oxidación **-1**
- Fórmula general: $M(OH)_n$, donde M es el metal y n es el estado de oxidación del metal

7.5.2 Nomenclatura de multiplicadores

- 🇪🇸: prefijo + hidróxido + de + nombre del metal

Ejemplos:

❶: LiOH: hidróxido de litio

❷: $Ca(OH)_2$: dihidróxido de calcio

❸: $Al(OH)_3$: trihidróxido de aluminio

- : name of the metal + prefix + hydroxide

ejemplos:

❶: NaOH: sodium hydroxide

❷: $Ca(OH)_2$: calcium dihydroxide

7.5.3 Nomenclatura con estados de oxidación

- : hidróxido + de + nombre del metal (estado de oxidación)

Ejemplos:

❶: NaOH: hidróxido de sodio (el Na solo tiene +1, se omite)

❷: $Fe(OH)_2$: hidróxido de hierro (II)

❸: $Fe(OH)_3$: hidróxido de hierro (III)

- : name of the metal (oxidation state) + hydroxide

ejemplos:

❶: KOH: potassium hydroxide (potassium (I) hydroxide)

❷: $Fe(OH)_2$: iron (II) hydroxide

7.5.4 Cómo formular hidróxidos

1. Escribe el símbolo del metal a la izquierda
2. Escribe OH a la derecha
3. El estado de oxidación del metal será el subíndice del grupo OH
4. Si el subíndice es mayor que 1, el OH debe ir entre paréntesis: $(OH)_n$
5. Si el subíndice es 1, no hace falta paréntesis

■ **Example 7.9** **Formular**: Trihidróxido de hierro

- "Tri" = 3 grupos OH
- Escribimos: $Fe(OH)_3$
- **Fórmula final**: $Fe(OH)_3$

Formular: Hidróxido de cesio

- No hay prefijo = 1 grupo OH
- Escribimos: CsOH (sin paréntesis)
- **Fórmula final**: CsOH

■

R **¡Atención con los paréntesis!**
- 1 grupo OH → NaOH (sin paréntesis)

- 2 o más grupos OH → $Ca(OH)_2$, $Al(OH)_3$ (con paréntesis)

7.6 Peróxidos

7.6.1 ¿Qué son los peróxidos?

Los peróxidos son compuestos muy parecidos a los óxidos, pero con una característica especial:

- Están formados por un metal y oxígeno
- El oxígeno **siempre** lleva el subíndice 2: O_2
- Este subíndice **NUNCA** se puede simplificar
- Se forman principalmente con metales alcalinos (grupo 1) y alcalinotérreos (grupo 2)

Fórmulas generales:

- **Grupo 1 (Alcalinos)**: M_2O_2
- **Grupo 2 (Alcalinotérreos)**: MO_2

7.6.2 Nomenclatura de peróxidos

Aunque se pueden nombrar con la nomenclatura de multiplicadores, lo más común y recomendado es usar la **nomenclatura específica de peróxidos**:

-  : peróxido + de + nombre del metal

  Ejemplos:

  ❶: Na_2O_2: peróxido de sodio

  ❷: CaO_2: peróxido de calcio

  ❸: K_2O_2: peróxido de potasio
-  : name of the metal + peroxide

  Examples:

  ❶: Li_2O_2: lithium peroxide

  ❷: MgO_2: magnesium peroxide

  ❸: BaO_2: barium peroxide

7.6.3 Cómo formular peróxidos

1. Identifica el grupo del metal (grupo 1 o grupo 2)
2. Escribe O_2 a la derecha (siempre)
3. Escribe el metal a la izquierda:
   - Si es del **grupo 1**: M_2O_2
   - Si es del **grupo 2**: MO_2

■ **Example 7.10** **Formular**: Peróxido de litio

- El litio (Li) está en el grupo 1
- Escribimos: Li_2O_2
- **¡No simplificamos!** → **Fórmula final**: Li_2O_2

Formular: Peróxido de berilio

- El berilio (Be) está en el grupo 2
- Escribimos: BeO_2
- **Fórmula final**: BeO_2

■

R **¡Cuidado!** No confundas:
- Na_2O = óxido de sodio (se puede simplificar si fuera necesario)
- Na_2O_2 = peróxido de sodio (¡NUNCA se simplifica!)

7.7 Trucos y consejos para no equivocarte

7.7.1 Errores más comunes

1. **Confundir óxido con peróxido**:
   - Si tiene O_2 que no se puede simplificar → peróxido
   - Si el oxígeno tiene cualquier otro subíndice → óxido
2. **Olvidar los paréntesis en hidróxidos**:
   - $Ca(OH)_2$ ✓ correcto
   - $CaOH_2$ × incorrecto
3. **No simplificar cuando se puede**:
   - Ca_2O_2 → simplificar a CaO ✓
   - Na_2O_2 → NO simplificar (es peróxido) ✓
4. **Poner (ac) en compuestos que no son ácidos**:
   - HCl(ac) ✓ (es un hidrácido)
   - NaCl(ac) × (no es un ácido)
5. **Confundir hidruro con hidróxido**:
   - Hidruro: solo metal + H → NaH
   - Hidróxido: metal + OH → NaOH

7.7.2 Reglas nemotécnicas

- **"URO" para no metales**: cuando combinamos metal + no metal → termina en -uro (cloruro, sulfuro, bromuro...)
- **"IDO" con oxígeno**: cuando hay oxígeno en el compuesto → termina en -ido (óxido, peróxido, hidróxido)
- **El oxígeno es mandón**: siempre va a la derecha (excepto con el flúor, pero eso no lo veremos este año)
- **Los peróxidos son gemelos**: el O_2 en peróxidos es como gemelos unidos, no se pueden separar (no se simplifica)
- **El grupo OH es un equipo**: siempre van juntos, por eso los ponemos entre paréntesis cuando

hay más de uno

7.8 Tablas de interés

7.8.1 Tabla de estados de oxidación más comunes

Estados de oxidación más comunes		
Elemento	**Símbolo**	**Estados de oxidación**
METALES		
Sodio	Na	+1
Potasio	K	+1
Litio	Li	+1
Plata	Ag	+1
Calcio	Ca	+2
Magnesio	Mg	+2
Bario	Ba	+2
Zinc	Zn	+2
Aluminio	Al	+3
Hierro	Fe	+2, +3
Cobre	Cu	+1, +2
Plomo	Pb	+2, +4
Estaño	Sn	+2, +4
NO METALES		
Oxígeno	O	-2 (casi siempre)
Hidrógeno	H	+1 (con no metales), -1 (con metales)
Flúor	F	-1
Cloro	Cl	-1, +1, +3, +5, +7
Bromo	Br	-1, +1, +3, +5
Yodo	I	-1, +1, +3, +5, +7
Azufre	S	-2, +2, +4, +6
Nitrógeno	N	-3, +1, +2, +3, +4, +5

7.8.2 Tabla resumen de todos los compuestos

Tabla resumen			
Tipo	**Fórmula general**	**Ejemplo**	**Nombre**
Óxido	E_xO_y	Fe_2O_3	Trióxido de dihierro Óxido de hierro (III)
Hidruro metálico	MH_n	CaH_2	Dihidruro de calcio Hidruro de calcio
Hidruro no metálico (compuesto puro)	H_nX	HCl	Cloruro de hidrógeno
Hidrácido (en disolución)	$H_nX(ac)$	HCl(ac)	Ácido clorhídrico
Sal binaria	M_xX_y	NaCl $FeCl_3$	Cloruro de sodio Tricloruro de hierro Cloruro de hierro (III)
Hidróxido	$M(OH)_n$	$Ca(OH)_2$	Dihidróxido de calcio Hidróxido de calcio
Peróxido	M_2O_2 o MO_2	Na_2O_2 CaO_2	Peróxido de sodio Peróxido de calcio

Leyenda:

- E = cualquier elemento
- M = metal
- X = no metal
- n = número que depende del estado de oxidación

Table 7.1: Compuestos binarios

nº	Fórmula	Multiplicadores	Estado de Oxidación
Óxidos			
1	Na_2O		
2	Co_2O_3		
3	CuO		
4	Ag_2O		
5	CdO		
6	NO		
7	CO_2		
8	As_2O_5		
9	SO_3		
10	O_5I_2		
11	O_7Br_2		
12		Óxido de dilitio	
13			Óxido de magnesio
14			Óxido de potasio
15			Óxido de estroncio
16			óxido de hierro (II)
17		Dióxido de plomo	

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
18			óxido de oro (III)
19		trióxido de dialuminio	
20			óxido de estaño (II)
21			óxido de cinc
22			óxido de mercurio (II)
23		tetraóxido de dinitrógeno	
24			óxido de cobalto (III)
25		Pentóxido de dinitrógeno	
26			óxido de bario
27		trióxido de azufre	
28		trióxido de dicromo	
29			óxido de manganeso (IV)
30		monóxido de dinitrógeno	
31			óxido de platino (IV)
32		dibromuro de trioxígeno	
Hidruros			
33	NaH		
34	CaH_2		

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
35	CoH_2		
36	CuH_2		
37	AgH		
38	AlH_3		
39	NH_3		
40	CdH_2		
41	AuH_3		
42	MnH_3		
43	PtH_2		
44		dihidruro de níquel	
45			hidruro de níquel (III)
46			hidruro de sodio
47			hidruro de calcio
48		dihidruro de magnesio	
49			hidruro de mercurio (II)
50		dihidruro de hierro	
51			hidruro de cinc
52		dihidruro de cobre	

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
53			hidruro de plomo (IV)
54			hidruro de estaño (II)
55			hidruro de níquel (III)
56		trihidruro de cromo	
57			hidruro de cromo (II)
58	PH_3		
59	AsH_3		
60	SbH_3		
61	BH_3		
62	SiH_4		
63	CH_4		
64	H_2O		
65	H_2S		
66	H_2Se		
67	H_2Te		
68	HF		
69	HCl		
70	HBr		

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
71	HI		
Sales binarias			
72	NaCl		
73	MgI_2		
74	$BeBr_2$		
75	K_2S		
76	FeF_2		
77	$PbCl_4$		
78	Au_2S_3		
79	SnS		
80	$ZnBr_2$		
81	HgTe		
82	CoI_2		
83	$CaCl_2$		
84	Al_2S_3		
85	Cr_2S_3		
86	$MnBr_2$		
87	Na_2Te		

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
88	$ZnCl_2$		
89	Ag_2S		
90	CI_4		
91	NF_3		
92	IBr		
93	$SeBr_4$		
94	$SeCl_6$		
95	AsF_5		
96	CCl_4		
97	SiF_4		
98			cloruro de litio
99		diyoduro de magnesio	
100			bromuro de berilio
101		sulfuro de dipotasio	
102			fluroro de hierro (II)
103		tetracloruro de plomo	
104			sulfuro de oro (III)
105			sulfuro de estaño (II)

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
106		dibromuro de cinc	
107			teluro de mercurio (II)
108			yoduro de cobalto (II)
109			cloruro de calcio
110		trisulfuro de dialuminio	
111		trisulfuro de dicromo	
112			bromuro de manganeso (II)
113		teluro de disodio	
114			cloruro de cinc
115		sulfuro de diplata	
116		trifluoruro de bromo	
117			cloruro de bromo (I)
118		hexafluoruro de azufre	
119			sulfuro de carbono (IV)
Hidróxidos			
120	$CsOH$		
121	KOH		
122	$Be(OH)_2$		

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
123	$Fe(OH)_3$		
124	AgOH		
125	$Al(OH)_3$		
126	NH_4OH		
127	$Cd(OH)_2$		
128	$Au(OH)_3$		
129	$Mn(OH)_3$		
130	$Pt(OH)_2$		
131		dihidróxido de níquel	
132			hidróxido de níquel (III)
133			hidróxido de sodio
134			hidróxido de calcio
135		dihidróxido de magnesio	
136			hidróxido de mercurio (II)
137		dihidróxido de hierro	
138			hidróxido de cinc
139		dihidróxido de cobre	
140			hidróxido de plomo (IV)

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
141			hidróxido de estaño (II)
142			hidróxido de níquel (III)
143		trihidróxido de cromo	
144			hidróxido de cromo (II)
Peróxidos. En este caso solo hemos dado la nomenclatura de los peróxidos. Podéis poner su nombre en la casilla de multiplicadores dejando el resto en blanco			
145	Na_2O_2		–
146	K_2O_2		–
147	SrO_2		–
148	CaO_2		–
149	Rb_2O_2		–
150	CuO_2		–
151	MgO_2		–
152	BaO_2		–
Hidruros grupos 16 y 17 en disolución acuosa. Solo hemos dado una nomenclatura. Podéis poner su nombre en la casilla de multiplicadores dejando el resto en blanco			
153	$HCl_{(aq)}$		
154	$HBr_{(aq)}$		

Continúa en la siguiente página

Table 7.1: Compuestos binarios (Continuación)

nº	Fórmula	Multiplicadores	Estado de Oxidación
155	$HI_{(aq)}$		
156	$HF_{(aq)}$		
157	$H_2S_{(aq)}$		
158	$H_2Te_{(aq)}$		
159	$H_2Se_{(aq)}$		

Table 7.2: Binary compounds

No.	Formula	Multipliers	Oxidation State
Oxides			
1	Na_2O		
2	Co_2O_3		
3	CuO		
4	Ag_2O		
5	CdO		
6	NO		
7	CO_2		
8	As_2O_5		
9	SO_3		
10	O_5I_2		
11	O_7Br_2		
12		Dilithium oxide	
13			Magnesium oxide
14			Potassium oxide
15			Strontium oxide
16			Iron(II) oxide
17		Lead dioxide	

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
18			Gold(III) oxide
19		Dialuminum trioxide	
20			Tin(II) oxide
21			Zinc oxide
22			Mercury(II) oxide
23		Dinitrogen tetraoxide	
24			Cobalt(III) oxide
25		Dinitrogen pentoxide	
26			Barium oxide
27		Sulfur trioxide	
28		Dichromium trioxide	
29			Manganese(IV) oxide
30		Dinitrogen monoxide	
31			Platinum(IV) oxide
32		Trioxygen dibromide	
Hydrides			
33	NaH		
34	CaH_2		

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
35	CoH_2		
36	CuH_2		
37	SnH_4		
38	LiH		
39	HgH_2		
40	NiH_3		
41	AlH_3		
42	H_4Pb		
43	H_3Mn		
44			Magnesium hydride
45			Sodium hydride
46			Strontium hydride
47			Chromium(II) hydride
48		Iron trihydride	
49			Lead(IV) hydride
50		Zinc dihydride	
51			Cadmium hydride
52			Copper(I) hydride

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
53			Gold(III) hydride
54			Tin(II) hydride
55		Potassium hydride	
56			Cobalt(II) hydride
57			Platinum(IV) hydride
58		Barium dihydride	
59		Disilver hydride	
60			Gold(I) hydride
Acids			
61	HCl		
62	HBr		
63	HI		
64	HF		
65	H_2S		
66	H_2Te		
67	H_2Se		
68		Hydrogen chloride	
69			Hydrogen bromide

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
70			Hydrogen iodide
71			Hydrogen fluoride
72		Dihydrogen sulfide	
73			Hydrogen telluride
74			Hydrogen selenide
		Binary salts	
75	$MgCl_2$		
76	$BaBr_2$		
77	Na_3N		
78	K_2S		
79	CaI_2		
80	$FeBr_3$		
81	PbS_2		
82	Au_2S_3		
83	HgF_2		
84	$AlCl_3$		
85	$SnBr_2$		
86	CuI		

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
87	CI_4		
88	NF_3		
89	IBr		
90	$SeBr_4$		
91	$SeCl_6$		
92	AsF_5		
93	CCl_4		
94	SiF_4		
95			Lithium chloride
96		Magnesium diiodide	
97			Beryllium bromide
98		Dipotassium sulfide	
99			Iron (II) fluoride
100		Lead tetrachloride	
101			Gold (III) sulfide
102			Tin (II) sulfide
103		Zinc dibromide	
104			Mercury (II) telluride

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
105			Cobalt (II) iodide
106			Calcium chloride
107		Dialuminum trisulfide	
108		Dichromium trisulfide	
109			Manganese (II) bromide
110		Disodium telluride	
111			Zinc chloride
112		Disilver sulfide	
113		Bromine trifluoride	
114			Bromine (I) chloride
115		Sulfur hexafluoride	
116			Carbon (IV) sulfide
Hydroxides			
117	CsOH		
118	KOH		
119	$Be(OH)_2$		
120	$Fe(OH)_3$		
121	AgOH		

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
122	$Al(OH)_3$		
123	NH_4OH		
124	$Cd(OH)_2$		
125	$Au(OH)_3$		
126	$Mn(OH)_3$		
127	$Pt(OH)_2$		
128		Nickel dihydroxide	
129			Nickel (III) hydroxide
130			Sodium hydroxide
131			Calcium hydroxide
132		Magnesium dihydroxide	
133			Mercury (II) hydroxide
134		Iron dihydroxide	
135			Zinc hydroxide
136		Copper dihydroxide	
137			Lead (IV) hydroxide
138			Tin (II) hydroxide
139			Nickel (III) hydroxide

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
140		Chromium trihydroxide	
141			Chromium (II) hydroxide
Peroxides. In this case we have only given the nomenclature for peroxides. You can write their name in the multipliers column leaving the rest blank			
142	Na_2O_2		–
143	K_2O_2		–
144	SrO_2		–
145	CaO_2		–
146	Rb_2O_2		–
147	CuO_2		–
148	MgO_2		–
149	BaO_2	–	
Hydrides from groups 16 and 17 in aqueous solution. Only one nomenclature has been given. You can write their name in the multipliers column leaving the rest blank			
150	$HCl_{(aq)}$		
151	$HBr_{(aq)}$		
152	$HI_{(aq)}$		
153	$HF_{(aq)}$		

Continued on next page

Table 7.2: Binary compounds (Continued)

No.	Formula	Multipliers	Oxidation State
154	$H_2S_{(aq)}$		
155	$H_2Te_{(aq)}$		
156	$H_2Se_{(aq)}$		

8.1 Chemical compounds

The electrons in the outermost shell (valence electrons) are responsible for the chemical properties of an element, as shown in Figure (??). Elements that have the same electron configuration in their valence shell belong to the same group in the periodic table, which gives them similar chemical properties.
Elements in the periodic table combine with other elements to achieve greater stability, forming chemical compounds.

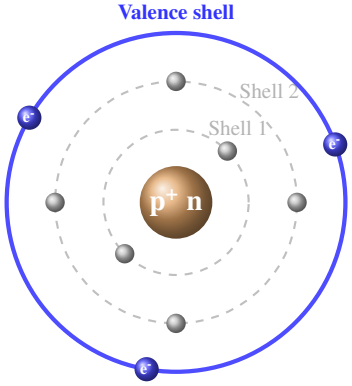


Figure (8.1): Simplified representation of an atom.

Definition 8.1 — Chemical compound. Is a substance formed by two or more different elements that are chemically bonded in fixed proportions.

It is also important to know the definition of molecule.

Definition 8.2 — Molecule. is the smallest unit of a chemical substance that retains its properties.

The different types of structures we can find are:

1. **Isolated atoms**.
   Isolated atoms are found in **noble gases** due to their high stability. Noble gases do not tend to combine with other elements because their valence shell is already complete.
   Isolated atoms have the following **physical and chemical properties**:
   - They are gaseous substances at room temperature.
   - They do not conduct electricity.

- They do not react with other substances.

2. **Homonuclear diatomic molecules**.
   These are molecules formed by two atoms of the same element. The homonuclear diatomic molecules found in nature are: H_2, O_2, N_2, F_2, Cl_2, Br_2, and I_2.
3. **Heteronuclear molecules**.
   These are molecules that contain two or more different elements.
   Examples: HCl, H_2S, H_2O.
4. **Crystals**.
   These are solids whose particles are arranged following a pattern that repeats in all three dimensions of space.
   - **Non-metallic crystals**: such as graphite and diamond, both formed by carbon but with different structures.
   - **Metallic crystals**: such as iron, copper, etc.

   Figure (**??**) shows how carbon can form two very different crystalline structures.

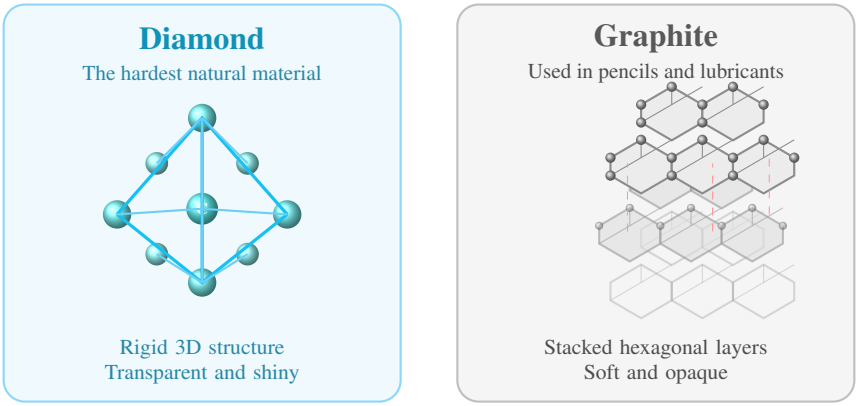


Figure (8.2): Diamond and graphite are both made of carbon, but their different structures give them very different properties.

8.2 Chemical bonding

Atoms of different elements join together to form molecules. This union is called a `chemical bond`. Chemical bonding occurs because the molecule formed is more stable than the isolated atoms.

In nature we can mainly find 3 types of bonds:

1. **Ionic bond**: occurs by electron transfer. One element loses electrons and another gains them, forming a cation (positive ion) and an anion (negative ion). Both ions are held together by the electric force of attraction.
   For example, NaCl forms because Na loses one e^- and becomes Na^+, while Cl gains that e^- and becomes Cl^-, resulting in Na^+Cl^- (Figure (**??**)).
2. **Covalent bond**: this is the bond that forms when two atoms share electrons. The bond formed is represented with a line (—), although it is sometimes omitted for simplicity.

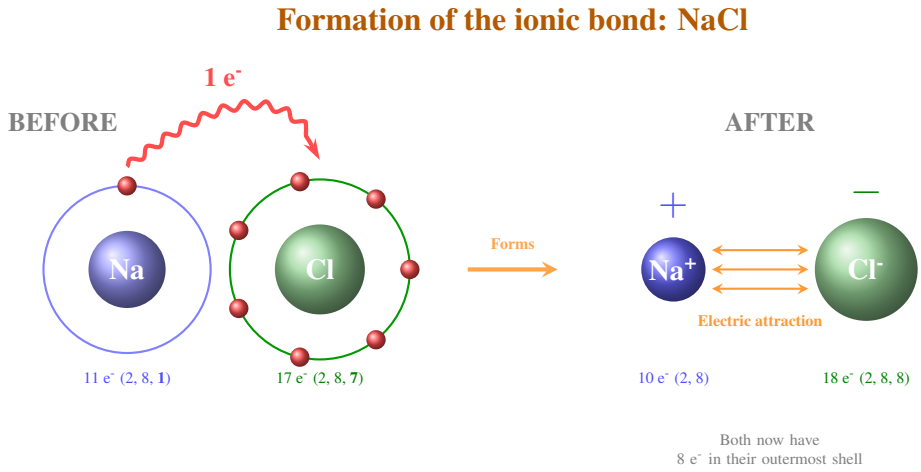


Figure (8.3): Formation of the ionic bond in NaCl. Sodium transfers one electron to chlorine, forming ions that attract each other.

For example, H and Cl share electrons to form a covalent bond: H−Cl (Figure (**??**)).

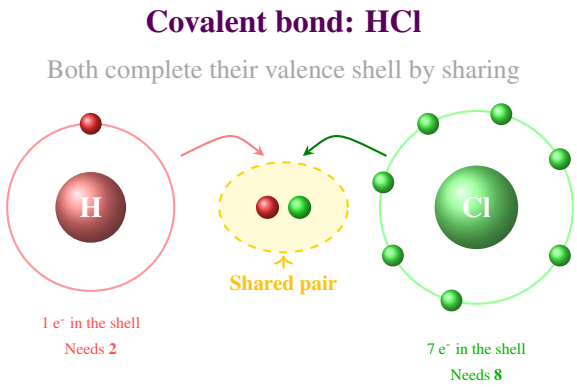


Figure (8.4): Covalent bond in HCl. Both atoms share an electron pair to complete their valence shell.

3. **Metallic bond**: this is the bond that forms between atoms of metallic elements. The atoms of a metal share their valence electrons, which move freely among all the atoms forming an "electron cloud". This explains why metals conduct electricity.
   Figure (**??**) illustrates this type of bond.

8.3 Atomic mass and molecular mass

8.3.1 Atomic mass

Atoms are very small particles, so if we measured their mass in kilograms, we would get extremely small values. That is why a special unit was created that allows us to work with these masses more easily.

This unit is called `atomic mass unit (amu)` and is represented by u.

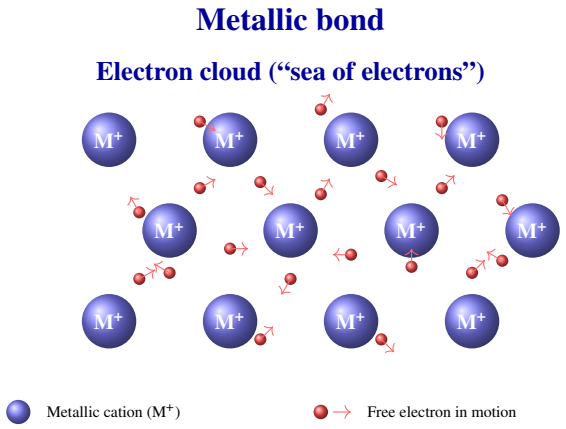


Figure (8.5): In the metallic bond, metal cations are surrounded by a cloud of free-moving electrons, which explains electrical conductivity.

Definition 8.3 — Atomic mass unit. It is one-twelfth ($\frac{1}{12}$) of the mass of a ${}^{12}_{6}C$ atom. We can say that $1\,u \approx 1{,}67 \cdot 10^{-27}\,kg$.

In many cases, the atomic mass unit is omitted and only the number is given.

In the periodic table we can find the atomic masses of each element in amu.

If we wanted to know the mass of an atom in kilograms, we would have to convert amu to kilograms using the equivalence given above.

8.3.2 Molecular mass

If what we have is a molecule, we must calculate its molecular mass, that is, the sum of the atomic masses of all the atoms that make up that molecule. Molecular mass is also expressed in amu.

■ **Example 8.1** Let's calculate the molecular mass of H_2O knowing that the atomic masses of the elements are $H = 1$ and $O = 16$.

$$H_2O \begin{cases} H = 2 \times 1\,u = 2\,u \\ O = 1 \times 16\,u = 16\,u \\ \hline \textbf{Total} = 18\,u \end{cases}$$

■

■ **Example 8.2** Let's calculate the molecular mass of H_2SO_4 knowing that the atomic masses of the elements are $H = 1$, $O = 16$, and $S = 32$.

$$H_2SO_4 \begin{cases} H = 2 \times 1\,u = 2\,u \\ S = 1 \times 32\,u = 32\,u \\ O = 4 \times 16\,u = 64\,u \\ \hline \textbf{Total} = 98\,u \end{cases}$$

■

Exercise 8.1 Indicate which is the correct way to calculate the molecular mass of calcium sulfate, $CaSO_4$:

a) Atomic mass of calcium + atomic mass of sulfur + atomic mass of oxygen.

b) Atomic mass of calcium + atomic mass of sulfur + four times the atomic mass of oxygen.

■

Exercise 8.2 Air, mainly composed of nitrogen (N_2) and oxygen (O_2), contains variable amounts of carbon dioxide (CO_2). Look up the atomic masses in the periodic table and calculate the molecular mass of these three gases. ■

Exercise 8.3 Calculate the molecular mass of zinc oxide (ZnO), potassium nitrate (KNO_3), calcium sulfate ($CaSO_4$), butane (C_4H_{10}), and calcium carbonate ($CaCO_3$). ■

Exercise 8.4 Look up the atomic masses in the periodic table of this unit and arrange the following compounds from highest to lowest oxygen percentage: BaO, K_2O, and CO_2. ■

Exercise 8.5 Indicate which of these two substances has a higher percentage of potassium: potassium chloride (KCl) or potassium nitrate (KNO_3). ■

8.4 Amount of substance: the mole

Since atoms are very small and matter is made up of enormous quantities of them, a unit was created to group many atoms together so we can work with amounts that are easier to handle.

For this purpose, the unit of substance called `mole` was created and is represented by $\mathbf{n}$.

Definition 8.4 — Mole. It is the amount of substance that contains $6{,}022 \cdot 10^{23}$ elementary entities, which can be atoms, molecules, ions, electrons, or other particles.

We can understand the mole as a "box" that always holds $6{,}022 \cdot 10^{23}$ "things".

This very large number is called `Avogadro's constant` and is represented by $\mathbf{N_A}$.

$$N_A = 6{,}022 \cdot 10^{23} \ \frac{\text{particles}}{\text{mol}}$$

Figure (**??**) illustrates this concept.

■ **Example 8.3** Calculate how many moles are in $6{,}022 \cdot 10^{21}$ atoms of Ag. We will do the calculation using conversion factors.

$$6{,}022 \cdot 10^{21} \ \cancel{\text{atoms of Ag}} \cdot \frac{1 \text{ mol of Ag}}{6{,}022 \cdot 10^{23} \ \cancel{\text{atoms of Ag}}} = 0{,}01 \text{ mol of Ag}$$

■

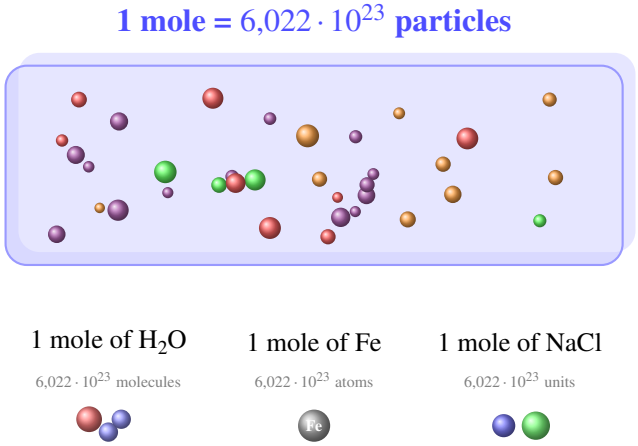


Figure (8.6): One mole always contains $6{,}022 \cdot 10^{23}$ particles, regardless of the substance.

■ **Example 8.4** How many molecules are in 0,5 mol of H_2O molecules?

Now we have a compound (formed by H and O) and not an element, so we will talk about molecules and not atoms.

$$0.5 \text{ mol of } H_2O \cdot \frac{6{,}022 \cdot 10^{23} \text{ molecules of } H_2O}{1 \text{ mol of } H_2O} = 3{,}011 \cdot 10^{23} \text{ molecules of } H_2O$$

■

Exercise 8.6 How many molecules are in 0,5 mol of oxygen molecules (O_2)? What amount of substance, in moles, is in $6{,}022 \cdot 10^{23}$ oxygen atoms? ■

Exercise 8.7 Where are there more molecules, in 1 mole of hydrogen (H_2) or in 1 mole of water (H_2O)? ■

Exercise 8.8 Indicate whether the following statement is true or false and justify your answer: "In 1 mole of helium atoms there is the same number of helium atoms as there are molecules in 1 mole of butane, C_4H_{10}." ■

Exercise 8.9 How many hydrogen atoms are in 2 mol of water (H_2O)? ■

8.5 Molar mass

As its name indicates, molar mass is the mass of 1 mole of a substance (atoms, molecules, ions...) and is expressed in $\frac{g}{mol}$. Molar mass is usually represented by M_w.

In practical terms, we can say that 1 u = 1 $\frac{g}{mol}$. Therefore, if we know that Na has an atomic mass of 23 u, we can say that 1 mole of sodium has a mass of 23 g, that is, the molar mass of sodium is 23 $\frac{g}{mol}$.

This equivalence is very useful if we want to calculate the moles of a substance from its mass,

using conversion factors or the formula:

$$n\ (\text{number of moles}) = \frac{\text{mass in grams}}{\text{molar mass}}$$

If we use abbreviations:

$$n = \frac{m\ (g)}{M_w}$$

Figure (**??**) summarizes the conversions between mass, moles, and particles.

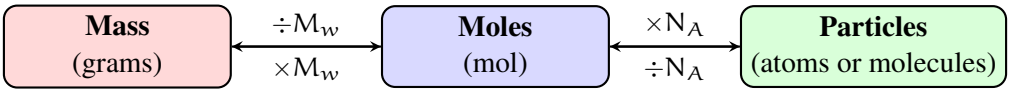


Figure (8.7): Relationship between mass, moles, and number of particles.

■ **Example 8.5** Calculate the moles of CH_4 contained in 48 g.

Datos:: The atomic masses are: H = 1; C = 12.

First we must calculate the molar mass. As we have said, the molar mass has the same numerical value as the molecular mass, but in $\frac{g}{mol}$ instead of amu.

$$CH_4 \begin{cases} H = 4 \times 1\,u = 4\,u \\ C = 1 \times 12\,u = 12\,u \\ \hline \textbf{Total} = 16\,u \end{cases}$$

Therefore, the molar mass of CH_4 is 16 $\frac{g}{mol}$.

Applying the formula:

$$n = \frac{m}{M_w} = \frac{48}{16} = 3\,\text{mol of } CH_4$$

■

Exercise 8.10 A flask contains 100 g of calcium carbonate, $CaCO_3$. Calculate the amount of this substance, in moles, in that container. ■

Exercise 8.11 What amount in moles of gaseous nitrogen, N_2, is in 56 g of this substance? How many molecules are there? And how many atoms? ■

Exercise 8.12 Calculate the molar masses of the following substances: bromine (Br_2), potassium iodide (KI), calcium carbonate ($CaCO_3$), and sodium carbonate (Na_2CO_3). ■

Exercise 8.13 What amount of substance, in moles, is in 66 g of carbon dioxide, CO_2? ■

Exercise 8.14 Calculate the mass of:

a) 2 mol of hydrogen chloride, HCl.

b) 2 mol of water, H_2O.

c) 1,5 mol of sulfuric acid, H_2SO_4.

d) 0,5 mol of chlorine, Cl_2. ■

Exercise 8.15 What amount in moles of substance is in 36,5 g of HCl, in 73 g of HCl, and in 146 g of HCl? ■

8.6 Ideal Gas Law

The three gas laws (Gay-Lussac's Law, Charles's Law, and Boyle-Mariotte's Law) can be combined into a single law called the **Ideal Gas Law**. This law relates pressure, volume, and temperature with the amount of substance in moles.

The mathematical expression of this law is:

Pressure in atm

number of moles

$$p \cdot V = n \cdot R \cdot T \tag{8.1}$$

Volume in L

Temperature in K

gas constant; R = 0

Exercise 8.16 We have 3 mol of an ideal gas at a temperature of 300 K and a pressure of 2 atm. Calculate the volume occupied by the gas. ■

Exercise 8.17 An ideal gas occupies a volume of 10 L at a pressure of 1,5 atm and a temperature of 350 K. How many moles of gas are there? ■

Exercise 8.18 What is the volume in mL occupied by an ideal gas if 0,553 mol are at a temperature of 1226,49 K and a pressure of 4935 mmHg? ■

Exercise 8.19 What is the temperature in K of an ideal gas, if 0,405 mol occupy a volume of 7720 mL at a pressure of 790 mmHg? ■

Exercise 8.20 What is the temperature in °C of an ideal gas, if 0,746 mol occupy a volume of 9890 mL at a pressure of 1,41 atm? ■

Exercise 8.21 What is the pressure in atm of an ideal gas if 0,532 mol occupy a volume of 4390 mL at a temperature of 183,93 K? ■

Exercise 8.22 How many moles of an ideal gas are in a volume of 4220 mL if the temperature is 279,35 K and the pressure is 1,75 atm? ■

8.6.1 Standard conditions (STP)

In chemistry, it is very common to talk about standard conditions. When we talk about standard conditions, we refer to pressure of 1 atm and a temperature of 0 °C (273 K).

It is important to know that 1 mol of **any gas** under standard conditions always occupies a volume of 22,4 L. This volume is called the **molar volume**.

To verify this value, we just need to apply the ideal gas equation with $n = 1$, $p = 1\,\text{atm}$, and $T = 273\,\text{K}$:

$$V = \frac{nRT}{p} = \frac{1 \times 0{,}082 \times 273}{1} = 22{,}4\,\text{L} \tag{8.2}$$

Exercise 8.23 What volume do 2 mol of oxygen (O_2) occupy under standard conditions? ■

Exercise 8.24 How many moles of an ideal gas are in 11,2 L under standard conditions? ■

Exercise 8.25 What is the volume in mL occupied by 0,553 mol of an ideal gas at STP? ■

Exercise 8.26 How many moles of an ideal gas are in a volume of 7720 mL at STP? ■

Exercise 8.27 What is the volume in L occupied by 0,746 mol of an ideal gas at STP? ■

Exercise 8.28 How many moles of an ideal gas are in a volume of 4390 mL at STP? ■

Exercise 8.29 What is the volume in mL occupied by 1,75 mol of an ideal gas at STP? ■

8.7 Vocabulary

■ **Vocabulary 8.1 — Valence electrons.** Electrons in the outermost shell of an atom which are responsible for the chemical properties of an element.
Translation to Spanish: Electrones de valencia

■ **Vocabulary 8.2 — Chemical compound.** Substance formed by two or more different elements that are chemically bonded together in fixed and specific proportions.
Translation to Spanish: Compuesto químico

■ **Vocabulary 8.3 — Molecule.** The smallest possible unit of a chemical substance that still retains all of its unique and characteristic properties.
Translation to Spanish: Molécula

■ **Vocabulary 8.4 — Isolated atoms.** Atoms found in noble gases that do not tend to combine with others because their valence shell is complete.
Translation to Spanish: Átomos aislados

■ **Vocabulary 8.5 — Noble gases.** Highly stable elements that do not conduct electricity or react with other substances due to their complete shells.
Translation to Spanish: Gases nobles

■ **Vocabulary 8.6 — Homonuclear diatomic molecules.** Molecules formed by the union of two atoms of the same chemical element, such as oxygen or hydrogen.
Translation to Spanish: Moléculas diatómicas homonucleares

■ **Vocabulary 8.7 — Heteronuclear molecules.** Specific types of molecules that contain two or more different elements chemically bonded together in a single unit.
Translation to Spanish: Moléculas heteronucleares

■ **Vocabulary 8.8 — Crystals.** Solids whose internal particles are arranged following a pattern that repeats in all three dimensions of physical space.
Translation to Spanish: Cristales

■ **Vocabulary 8.9 — Chemical bond.** The union between atoms of different elements that occurs because the molecule formed is more stable than others.
Translation to Spanish: Enlace químico

■ **Vocabulary 8.10 — Ionic bond.** Bond that occurs by electron transfer where one element loses electrons and another gains them, forming stable ions.
Translation to Spanish: Enlace iónico

■ **Vocabulary 8.11 — Cation.** A positive ion formed when a neutral atom loses one or more electrons during a chemical reaction process.
Translation to Spanish: Catión

■ **Vocabulary 8.12 — Anion.** A negative ion formed when a neutral atom gains one or more electrons during a chemical reaction process.
Translation to Spanish: Anión

■ **Vocabulary 8.13 — Covalent bond.** The specific type of chemical bond that forms when two atoms share electrons to complete their outermost shell.
Translation to Spanish: Enlace covalente

■ **Vocabulary 8.14 — Metallic bond.** Bond between atoms of metallic elements where valence electrons move freely, forming a large and common electron cloud.
Translation to Spanish: Enlace metálico

■ **Vocabulary 8.15 — Electron cloud.** The zone in metallic bonds where valence electrons move freely among all the atoms, allowing electrical conduction.
Translation to Spanish: Nube electrónica

■ **Vocabulary 8.16 — Graphite.** A non-metallic crystal formed by carbon atoms arranged in stacked hexagonal layers that are soft and opaque.
Translation to Spanish: Grafito

9.1 Chemical Reactions

Definition 9.1 — Chemical Reaction. A chemical reaction is a process in which initial substances or compounds are transformed into other final substances or compounds with a different chemical composition from the initial one.

9.1.1 Parts to a chemical reaction

Writing a chemical reaction is similar to writing a mathematical equation, but in this case the symbol $=$ is replaced by a $\longrightarrow$. The parts to a chemical reaction are as follows.

1. We write the compounds or substances using their corresponding chemical formula.
2. An $\longrightarrow$. This separates the initial substances from the final substances.
3. **Reactants**. These are the initial substances written to the left to the arrow.
4. **Products**. These are the final substances written to the right to the arrow.
5. If we have more than one reactant or more than one product, they are separated from each other using $+$.

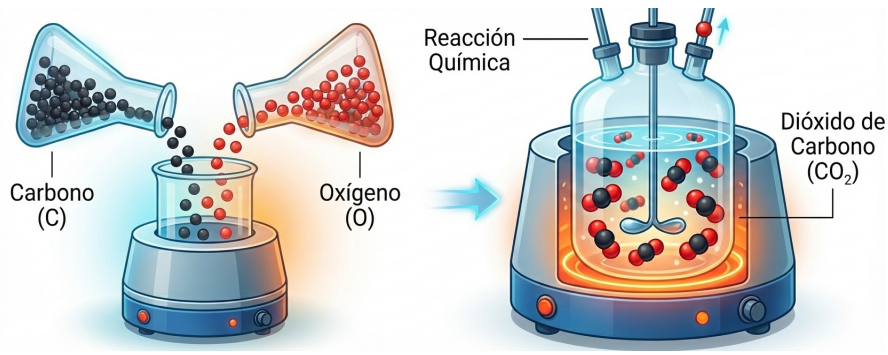


Figure (9.1): Chemical Reaction

9.1.2 How to read a chemical reaction

It is necessary to know how to write a chemical reaction given its description in words. To do this we follow the following rules. Let's imagine the following general reaction:

$$\mathrm{A\,(g) + B\,(s) \longrightarrow C\,(l) + D\,(aq)}$$

Here, the reactants are A and B, while the products are C and D.

1. If the reactants or compounds are written with their name, we must first obtain their formula.
2. If there is a $+$ symbol in the reactants, it is read as 'reacts with'. For example, in the equation above we can say that A reacts with B.
3. The symbol $\rightarrow$ is read as "to form" or "to give".
4. If the $+$ symbol is in the products, that $+$ symbol is read as "and". In the previous case we can say that C "and" D are obtained.
5. Next to A appears (g), next to B appears (s), next to C appears (l), while next to D appears (aq). These words indicate the state to matter to each compound and have the following meaning:
   - (s): solid.
   - (l): liquid.
   - (g): gas.
   - (aq): aqueous

9.1.3 How a chemical reaction occurs

For a chemical reaction to occur, the following steps must be followed:

1. The atoms in the reactant molecules must separate.
2. Once the atoms are separated, they collide with each other. According to **collision theory**, chemical reactions occur due to collisions between atoms. For a collision to be effective and produce a chemical reaction, the following must be met:
   - The reactants must have sufficient energy.
   - The reactants must collide with proper orientation.
3. Once an effective collision occurs, the corresponding products are obtained.

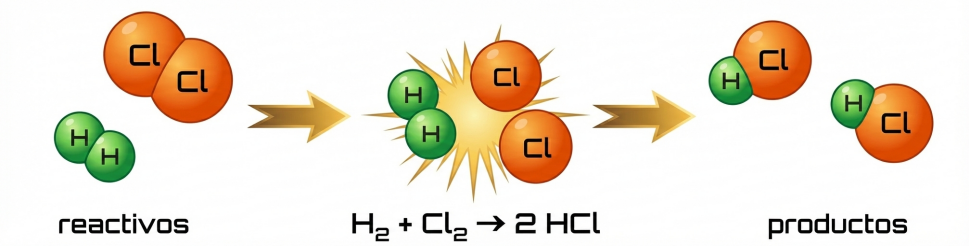


Figure (9.2)

Exercise 9.1 Cu represents the element copper, S the element sulphur and CuS the sulphide to copper(II). Use your own words to explain what this chemical equation represents ■

Exercise 9.2 Calcium carbonate acid (CaCO3) decomposes at high temperatures into calcium oxide (CaO) and carbon dioxide (CO2). This process is known as calcination. Write the chemical equation for this reaction. ■

Exercise 9.3 What's the name to the theory that explains how chemical reactions occur? ■

Exercise 9.4 Explain the following chemical equations in words:

$$CuCO_3\,(s) \longrightarrow CUO\,(s) + CO_2\,(g)$$

$$H_2\,(g) + O_2\,(g) \longrightarrow H_2O\,(l)$$

■

9.2 Lavoisier's Principle. Law to Conservation to Mass

In a chemical reaction (in a closed container) the sum to the masses to the reactants must equal the sum to the masses to the products, that is, **mass is conserved**.

Consider the following reaction:

$$AgNO_3 + HCl \longrightarrow AgCl + HNO_3$$

If we have 17 g to $AgNO_3$ and it reacts with 3,65 g to HCl, 14,35 g to AgCl and 6,3 g to HNO_3 will form.

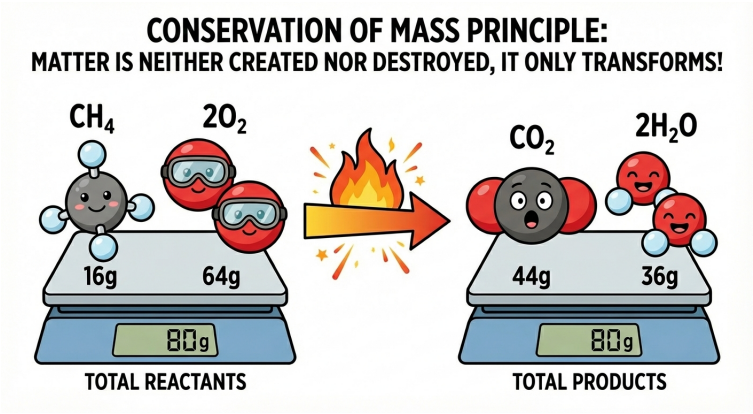


Figure (9.3): Law to conservation to mass

9.2.1 Chemical reactions and the law to conservation to mass

Every chemical reaction must follow Lavoisier's principle and, therefore:

We must have the same number to atoms to each element in the reactants (left side) and in the products (right side). When we have this situation we say that the reaction is **balanced**

If the chemical reaction is not balanced, we must balance it so that Lavoisier's principle is fulfilled.

The numbers placed in front to each chemical formula are known as **stoichiometric coefficients**.



$$2\,H_2 + O_2 \longrightarrow 2\,H_2O \tag{9.1}$$

These stoichiometric coefficients tell us the number to molecules or moles to each compound that react with each other to obtain the corresponding number to molecules or moles to products.

These stoichiometric coefficients tell us that:

- **IMPORTANT!!!!**: no number appears in front too chO2, when we don't have a number we must assume it is a 1. This means that 1 mole to O_2 will react.
- 2 molecules to hydrogen (H_2) react with 1 molecule toooxygen (O_2) to form 2 molecules to water (H_2O).
- 2 moles to hydrogen (H_2) react with 1 mole to oxygen (O_2) to form 2 moles to water (H_2O).
- These coefficients also tell us that 2 moles to H_2 are related to 1 mole to O_2 and in turn to 2 moles too chH2O.

9.2.2 Steps for balancing (Trial and Error Method)

1. **Count:** Count the atoms to each element on both sides to the arrow.
2. **Balance with coefficients:** Place large numbers in front to molecules to multiply the number to atoms.
3. **Recommended order:**
   - Metals and Non-metals.
   - Hydrogen (H).
   - Oxygen (O).

Step-by-Step Example: Combustion to Methane We want to balance the following reaction:

$$CH_4 + O_2 \longrightarrow CO_2 + H_2O$$

Step 1: Count initial atoms

- Reactants: $C = 1, H = 4, O = 2$
- Products: $C = 1, H = 2, O = 3$ (2 from CO_2 and 1 from H_2O)

Step 2: Balance Hydrogen

Since there are 4 H on the left and only 2 on the right, we put a **2** in front to water:

$$CH_4 + O_2 \longrightarrow CO_2 + 2\,H_2O$$

Now we have: $H = 2 \times 2 = 4$. Balanced!

Step 3: Balance Oxygen

Now on the right there are: 2 (from CO_2) + 2 (from $2\,H_2O$) = 4 atoms to O. On the left there are only 2. We put a **2** in front to O_2:

$$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$$

Final Result The balanced equation is:

$$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$$

Exercise 9.5 In your notebook, balance the chemical equations.

a)

$$Na\,(s) + O_2\,(g) \rightarrow Na_2O_2\,(s)$$

b)

$$H_2\,(g) + Cl_2\,(g) \rightarrow HCl\,(g)$$

c)

$$Ca\,(s) + O_2\,(g) \rightarrow CaO\,(s)$$

d)

$$Al\,(s) + Br_2\,(l) \rightarrow AlBr_3\,(s)$$

■

Exercise 9.6 Explain how to balance chemical equations ■

Exercise 9.7 Hydrogen reacts with oxygen to give water. What mass to liquid water is obtained if 2 g to hydrogen reacts exactly with 16 g to oxygen? ■

Exercise 9.8 We react iron with sulphur to obtain iron sulphide:

$$S(s) + Fe(s) \rightarrow FeS(s)$$

We know that 32 g to sulphur reacts exactly with 56 g to iron.

a) What mass to iron sulphide is obtained?

b) What's the percentage to sulphur and iron in iron sulphide?

■

Exercise 9.9 In an experiment, we verify that 3.2 g to sulphur (S) reacts fully with 20 g to mercury (Hg) to give mercury (II) sulphide (HgS). Balance the reaction and do the calculations for the questions below.

S(s) + Hg(l)→HgS(s)

a) the amount to mercury sulphide obtained.
b) the mass to sulphur required to fully react with 5 g to mercury.
c) the mass to mercury needed to fully react with 10 g to sulphur.
d) the percentage to sulphur and mercury in the mercury(II) sulphide.

■

Exercise 9.10 Write the chemical equation for the reaction to solid magnesium with hydrogen chloride (HCl) in an aqueous solution to give solid magnesium chloride (MgCl2) and hydrogen gas (H2). Balance the chemical equation. ■

Exercise 9.11 Balance these reactions.

a) Ca (s) + O2 (g) → CaO (s)
b) CuO (s) + C (s) → Cu (s) + CO2 (g)
c) Na (s) + Cl2 (g) → NaCl (s)
d) Al (s) + O2 (g) → Al2O3 (s)
e) Fe (s) + O2 (g) → FeO (s)

■

9.3 Calculations with chemical reactions

Calculations in chemical reactions will allow us to know how much quantity (whether in grams or moles) to products will be obtained or how much quantity to reactants we will need. As an example we will use the reaction: hydrogen reacts with oxygen to form water.

To perform the calculations correctly, the following steps must be followed:

1. Identify the reactants and products. In this case the reactants will be H_2 and O_2 while we will have a single product which is H_2O.
2. If in the reaction they say hydrogen, nitrogen, oxygen, fluorine, chlorine, bromine, or iodine, we will always write them as H_2, N_2, O_2, F_2, Cl_2, Br_2, or I_2. They may also be named as molecular nitrogen or dinitrogen. For the rest to the elements mentioned above it is the same.
3. Write the chemical reaction using their corresponding chemical formulas.

$$H_2 + O_2 \longrightarrow H_2O$$

4 Balance the chemical reaction if necessary.

$$2\,H_2 + O_2 \longrightarrow 2\,H_2O$$

5 If we are given quantities in grams, we must convert them all to moles (we will represent moles with the letter $\mathfrak{n}$). If we are given for example 10 g to H_2O, we would have to first calculate the molar mass to water which in this case is 18 $\frac{g}{mol}$

- Using the formula:

$$\mathfrak{n}_{H_2O} = \frac{\text{grams too ubstance}}{\text{molar mass to compound}} = \frac{10\,g}{18\,\frac{g}{mol}} = 0{,}56\,\text{mol to } H_2O$$

- We can also use conversion factors:

$$10\ \cancel{\text{g to } H_2O} \cdot \frac{1\,\text{mol to } H_2O}{18\ \cancel{\text{g to } H_2O}} = 0{,}56\,\text{mol to } H_2O$$

6 Now we calculate the moles needed for the rest to the compounds using the stoichiometric coefficients. Now we will calculate the moles to H_2 and O_2:
Calculation to moles to H_2

$$0.56\ \cancel{\text{mol to } H_2O} \cdot \frac{2\,\text{mol to } H_2}{1\ \cancel{\text{mol to } H_2O}} = 1{,}12\,\text{mol to } H_2$$

Calculation to moles to O_2

$$0.56\ \cancel{\text{mol to } H_2O} \cdot \frac{1\,\text{mol to } O_2}{2\ \cancel{\text{mol to } H_2O}} = 0{,}28\,\text{mol to } O_2$$

7 Now we calculate in grams the quantities to all those compounds requested
Let's imagine now that we calculate the amount in grams too chH2 and O_2. To do this we will use the moles calculated previously and the molar mass to each compound.
Calculation to molar masses:

- H_2: 2×1 amu $= 2\,u = 2\,\frac{g}{mol}$
- O_2: 2×16 amu $= 32\,u = 32\,\frac{g}{mol}$

Now we perform the calculations we mentioned earlier, we will do it in two different ways (only one way is necessary) using the mole formula and conversion factors.

- Calculation to grams to H_2 and O_2 using the formula:
  We know that $\mathfrak{n} = \frac{\text{grams too ompound}}{\text{molar mass to compound}}$
  if we solve for the grams to the compound we will have:
  grams to compound $= \mathfrak{n} \cdot$ molar mass too ompound
  grams to $H_2 = 1.12 \cdot 2 = 2{,}24\,g$ to H_2.
  grams to $O_2 = 0.28 \cdot 32 = 8{,}96\,g$ to O_2.
- Calculation to grams to H_2 and O_2 using conversion factors. For this we will start now

from the moles obtained to H_2 and O_2 calculated in the previous section and using the molar masses to each compound.

Amount to H_2:

$$1.12 \text{ mol to } H_2 \cdot \frac{2\,\text{g to } H_2}{1 \text{ mol to } H_2} = 2{,}24\,\text{g to } H_2$$

Amount to O_2:

$$0.28 \text{ mol to } O_2 \cdot \frac{32\,\text{g to } H_2}{1 \text{ mol to } O_2} = 0{,}96\,\text{g to HO}$$

■ **Example 9.1** Two elements such as sulfur and copper react to form copper(II) sulfide. What mass to copper(II) sulfide is obtained if we start from 64 g to sulfur? What amount in grams to copper do we need?

1. We identify the reactants and products. In this case, the reactants are S and Cu while the product is CuS.
2. We write the chemical reaction:

$$S + Cu \longrightarrow Cu_2S$$

3. We balance the reaction and it would be:

$$S + 2\,Cu \longrightarrow Cu_2S$$

4. Now we calculate the moles to all the quantities that the exercise asks for.

$$64 \text{ g to S} \cdot \frac{1\,\text{mol to S}}{32 \text{ g to S}} = 2\,\text{mol to S}$$

5. Since they are asking for the amount to CuS and Cu, we calculate the moles to each species with stoichiometric calculations.
   - For CuS:

$$2 \text{ mol to S} \cdot \frac{1\,\text{mol to CuS}}{1 \text{ mol to S}} = 2\,\text{mol to S}$$

   - For Cu:

$$2 \text{ mol to S} \cdot \frac{2\,\text{mol to Cu}}{1 \text{ mol to S}} = 4\,\text{mol to S}$$

6. Now we calculate the masses in grams to each species requested.
   - For CuS:

$$2 \text{ mol to CuS} \cdot \frac{95{,}5\,\text{g to CuS}}{1 \text{ mol to CuS}} = 191\,\text{g to CuS}$$

- For Cu:

$$4 \cancel{\text{ mol to Cu}} \cdot \frac{63{,}5\,\text{g to Cu}}{1 \cancel{\text{ mol to Cu}}} = 254\,\text{g to Cu}$$

These same calculations can be done using the formula

$$n = \frac{\text{grams}}{\text{molar mass}} \longrightarrow \text{grams} = n \cdot \text{molar mass}$$

- For CuS:

$$\text{grams to CuS} = 2 \cdot 95.5 = 191\,\text{g to CuS}$$

- For CuS:

$$\text{grams to Cu} = 4 \cdot 63.5 = 254\,\text{g to Cu}$$

■

Exercise 9.12 What mass to hydrogen is formed when 27 g to aluminium react with excess hydrochloric acid (HCl) according to the equation?

$$2\,\text{Al} + 6\,\text{HCl} \rightarrow 2\,\text{AlCl}_3 + 3\,\text{H}_2$$

■

Exercise 9.13 Calcium carbonate (CaCO3), or limestone, decomposes at high temperatures into calcium oxide (CaO), or lime, and carbon dioxide, according to this equation:

$$\text{CaCO}_3(\text{s}) \rightarrow \text{CaO(s)} + \text{CO}_2(\text{g})$$

If we heat 160 kg to calcium carbonate, we obtain 89.6 kg to calcium oxide. What mass to calcium oxide would we obtain from 80 kg to carbonate? ■

Exercise 9.14 Heating 0.5 g to pulverised iron under a stream to gaseous chlorine produces iron dichloride according to this reaction:

$$\text{Fe(s)} + \text{Cl}_2(\text{g}) \rightarrow \text{FeCl}_2(\text{s})$$

a) Calculate the mass to chlorine consumed.

b) Calculate the mass to iron dichloride produced.

■

Exercise 9.15 One gram to zinc burns in the air according to the equation:

$$Zn(s) + O_2(g) \rightarrow ZnO(s)$$

a) Balance the chemical equation.
b) How much oxygen does it consume?
c) How much zinc oxide does it produce?
d) What percentage to zinc oxide is zinc?
e) Is the law to conservation to mass fulfilled?

■

Exercise 9.16 Mercury and sulphur react to make mercury sulphide according to this formula:

$$S(s) + Hg(s) \rightarrow HgS(s)$$

If 20 g to Hg reacts:
a) how much sulphur does it react with?
b) what mass to mercury sulphide does it produce?

■

Exercise 9.17 Magnesium (Mg) reacts with oxygen (O2) to give magnesium oxide (MgO).
a) Write and balance this chemical equation.
b) If 48.6 g to magnesium reacts with the required amount to oxygen, how much magnesium oxide is obtained?

■

Exercise 9.18 Aluminium reacts with oxygen producing aluminium oxide. Calculate the mass to aluminium oxide produced when 15 g to aluminium reacts with excess oxygen. ■

Exercise 9.19 Aluminium reacts with oxygen producing aluminium oxide. Calculate the mass to aluminium oxide produced when 15 g too luminium reacts with excess oxygen. ■

Exercise 9.20 Ammonia (NH3) is formed by the reaction to nitrogen with hydrogen. If 420 g to nitrogen are available, how many grams to ammonia are formed? What volume to hydrogen under standard conditions is consumed in the reaction? ■

Exercise 9.21 Zinc sulfide reacts with oxygen producing zinc oxide and sulfur dioxide. With 168 L to oxygen under standard conditions:
a) what mass to zinc sulfide will react?
b) how many moles to zinc oxide will be produced?

c) what volume to sulfur dioxide under standard conditions will be obtained?

■

Exercise 9.22 Methane (CH_4) reacts with oxygen producing carbon dioxide and water. With 20 mol to methane:

a) what volume to oxygen, under standard conditions, reacts?

b) what mass to carbon dioxide is formed? how many moles to water are produced?

■

Exercise 9.23 By the effect to heat, 245 g to potassium chlorate ($KClO_3$) decompose. Calculate the mass to KCl and the moles to oxygen that are formed. ■

Exercise 9.24 Calculate the volume to carbon dioxide (under standard conditions) and the mass to water produced in the combustion too100 g to butane (C4H10). How many moles to oxygen are consumed? ■

Exercise 9.25 Ammonia (NH3) reacts with oxygen producing nitrogen monoxide and water. If 500 cm^3 to nitrogen monoxide have been obtained, measured under standard conditions, how many grams to ammonia and how many moles to oxygen will have been consumed? ■

Exercise 9.26 Calcium carbonate (CaCO3) reacts with hydrochloric acid producing calcium chloride, carbon dioxide, and water.

a) What mass to calcium carbonate will be necessary for 100 moles to hydrochloric acid to react?

b) What volume to carbon dioxide will be produced measured at STP?

■

9.4 Reaction rate. Factors affecting the rate

Definition 9.2 — Reaction rate. Is the rate at which **reactants disappear** or the rate at which **products appear**. The unit to reaction rate will be amount to substance per unit to time.

The factors that will make a reaction faster or slower are the following:

1. **Concentration to compounds**, that is, the amount we have to each substance. By increasing the concentration, the reaction rate will also increase.
2. **Contact surface area**. Since reactions occur by collisions, increasing the surface area to that collision will also increase the reaction rate.
3. **Temperature**. By increasing the temperature, the reaction rate will also increase.
4. **Presence to catalysts**. A catalyst is a substance that is added to increase the reaction rate.

9.5 Vocabulary

■ **Vocabulary 9.1 — Chemical Reaction.** Process in which initial substances or compounds are transformed into other final substances with a different chemical composition.
Translation to Spanish: Reacción química

■ **Vocabulary 9.2 — Reactants.** Initial substances written to the left to the arrow in a chemical reaction indicating what undergoes a change.
Translation to Spanish: Reactivos

■ **Vocabulary 9.3 — Products.** Final substances written to the right to the arrow representing the result to a chemical reaction process.
Translation to Spanish: Productos

■ **Vocabulary 9.4 — Chemical formula.** Notation using chemical symbols and subscripts to represent the types and number to atoms in a substance.
Translation to Spanish: Fórmula química

■ **Vocabulary 9.5 — Collision theory.** Scientific explanation stating that chemical reactions occur due to collisions between atoms with enough energy and correct orientation.
Translation to Spanish: Teoría de colisiones

■ **Vocabulary 9.6 — Effective collision.** A specific collision between reactants that has sufficient energy and proper orientation to produce a chemical reaction.
Translation to Spanish: Colisión eficaz

■ **Vocabulary 9.7 — Calcination.** Chemical process where calcium carbonate decomposes at high temperatures into calcium oxide and carbon dioxide gas.
Translation to Spanish: Calcinación

■ **Vocabulary 9.8 — Law to Conservation to Mass.** Principle stating that in a chemical reaction, the sum too asses to reactants must equal the sum to products.
Translation to Spanish: Ley de conservación de la masa

■ **Vocabulary 9.9 — Balanced reaction.** State to a chemical equation when there is the same number too toms to each element on both sides.

Translation to Spanish: Reacción ajustada

■ **Vocabulary 9.10 — Stoichiometric coefficients.** Numbers placed in front to chemical formulas to indicate the number to molecules or moles that react.
Translation to Spanish: Coeficientes estequiométricos

■ **Vocabulary 9.11 — Trial and Error Method.** Procedure for balancing equations by counting atoms and placing coefficients until both sides to the reaction are equal.
Translation to Spanish: Método de tanteo

■ **Vocabulary 9.12 — Molar mass.** Value calculated as the grams to a substance divided by the number to moles in a specific compound.
Translation to Spanish: Masa molar

■ **Vocabulary 9.13 — Mole.** Standard unit to measurement for amount to substance, often represented by the letter n in chemical calculations.
Translation to Spanish: Mol

■ **Vocabulary 9.14 — Solid.** State to matter where particles are closely packed, represented by the symbol (s) in chemical equations.
Translation to Spanish: Sólido

■ **Vocabulary 9.15 — Liquid.** State to matter where particles flow freely, represented by the symbol (l) in a chemical reaction description.
Translation to Spanish: Líquido

■ **Vocabulary 9.16 — Gas.** State to matter where particles expand to fill space, represented by the symbol (g) in chemical equations.
Translation to Spanish: Gas

■ **Vocabulary 9.17 — Aqueous.** Term describing a substance dissolved in water, represented by the symbol (aq) in chemical reaction formulas.
Translation to Spanish: Acuoso

■ **Vocabulary 9.18 — Conversion factors.** Mathematical tools used in chemistry to convert between different units, such as grams to moles or vice versa.

Translation to Spanish: Factores de conversión

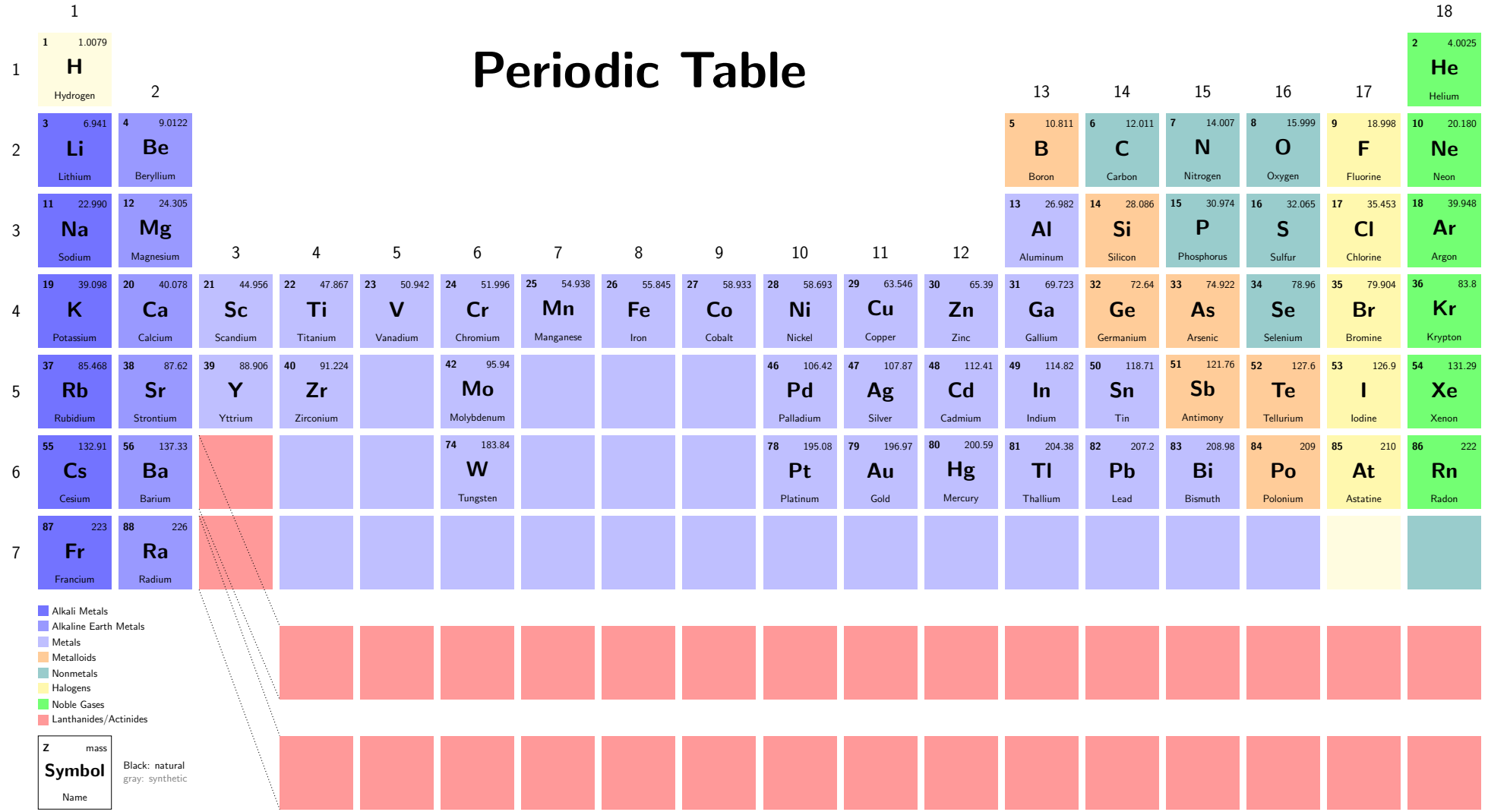


Figure (9.4): Periodic Table of Elements

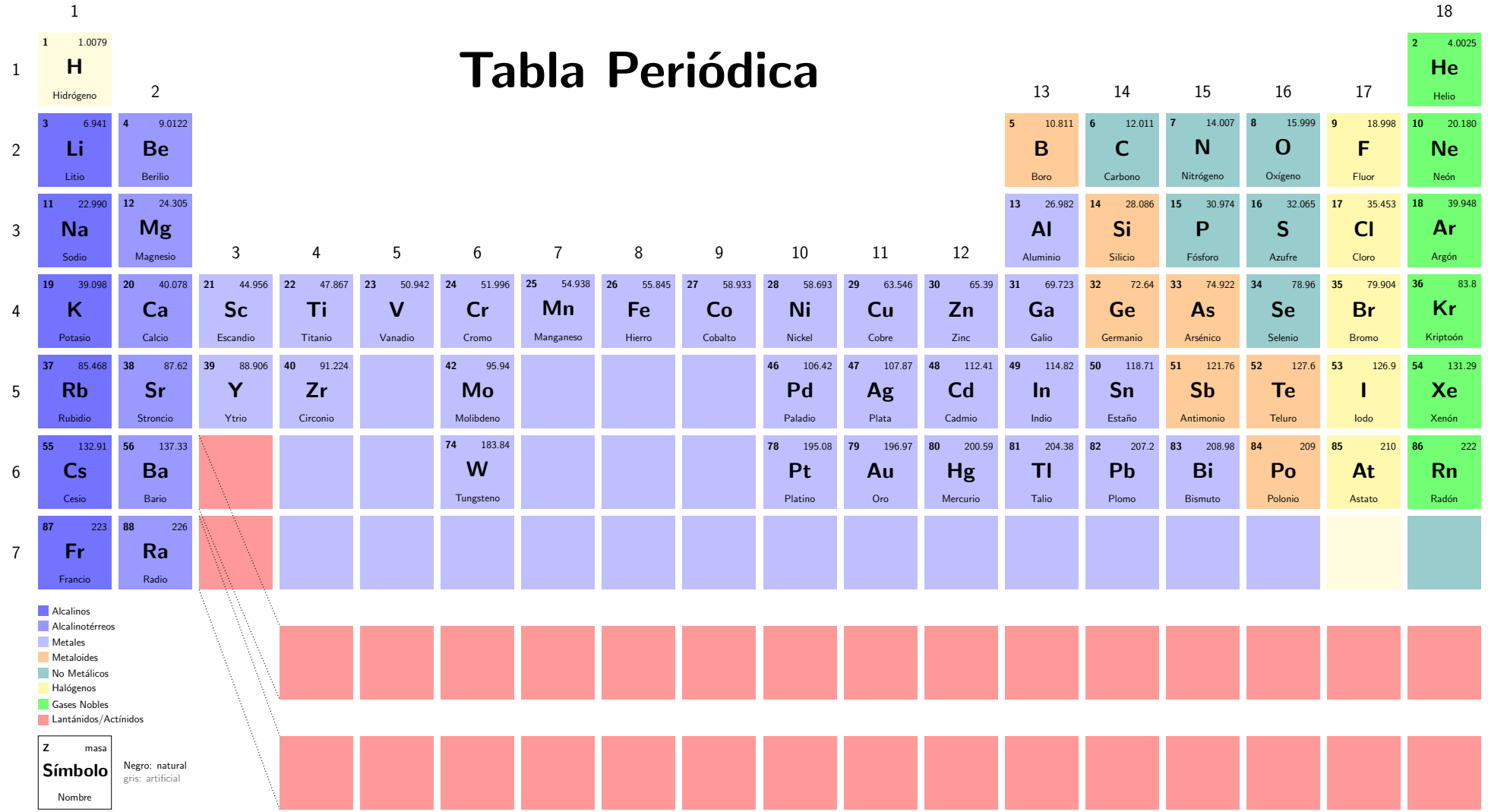


Figure (9.5): Tabla Periódica de los elementos

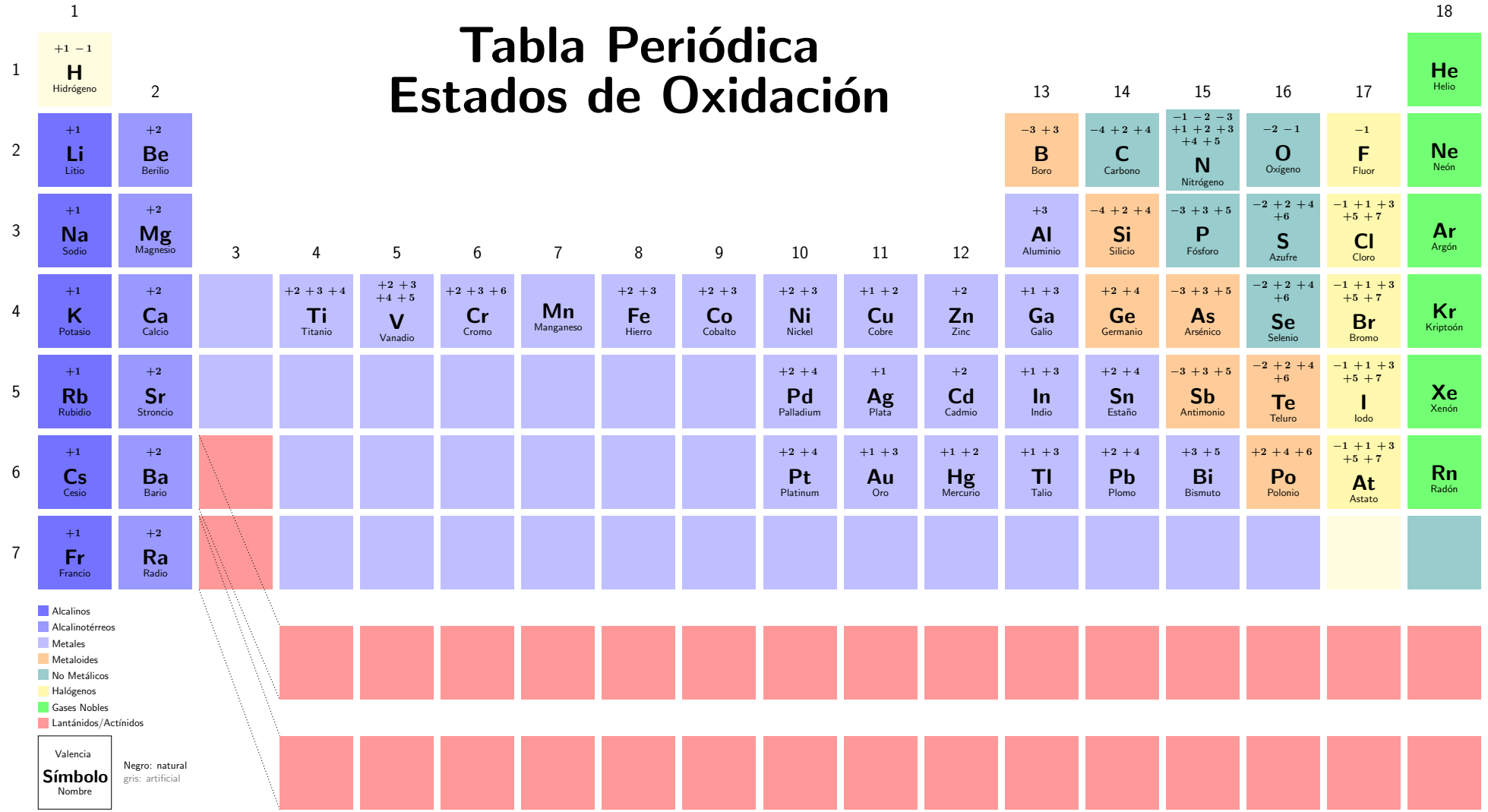

Tabla Periódica
Estados de Oxidación
1
2
3
4
5
6
7
18
13
14
15
16
17
+1 −1 H Hidrógeno
He Helio
+1 Li Litio
+2 Be Berilio
−3 +3 B Boro
−4 +2 +4 C Carbono
−1 −2 −3 +1 +2 +3 +4 +5 N Nitrógeno
−2 −1 O Oxígeno
−1 F Fluor
Ne Neón
+1 Na Sodio
+2 Mg Magnesio
3
4
5
6
7
8
9
10
11
12
+3 Al Aluminio
−4 +2 +4 Si Silicio
−3 +3 +5 P Fósforo
−2 +2 +4 +6 S Azufre
−1 +1 +3 +5 +7 Cl Cloro
Ar Argón
+1 K Potasio
+2 Ca Calcio
+2 +3 +4 Ti Titanio
+2 +3 +4 +5 V Vanadio
+2 +3 +6 Cr Cromo
Mn Manganeso
+2 +3 Fe Hierro
+2 +3 Co Cobalto
+2 +3 Ni Nickel
+1 +2 Cu Cobre
+2 Zn Zinc
+1 +3 Ga Galio
+2 +4 Ge Germanio
−3 +3 +5 As Arsénico
−2 +2 +4 +6 Se Selenio
−1 +1 +3 +5 +7 Br Bromo
Kr Kriptoón
+1 Rb Rubidio
+2 Sr Stroncio
+2 +4 Pd Palladium
+1 Ag Plata
+2 Cd Cadmio
+1 +3 In Indio
+2 +4 Sn Estaño
−3 +3 +5 Sb Antimonio
−2 +2 +4 +6 Te Teluro
−1 +1 +3 +5 +7 I Iodo
Xe Xenón
+1 Cs Cesio
+2 Ba Bario
+2 +4 Pt Platinum
+1 +3 Au Oro
+1 +2 Hg Mercurio
+1 +3 Tl Talio
+2 +4 Pb Plomo
+3 +5 Bi Bismuto
+2 +4 +6 Po Polonio
−1 +1 +3 +5 +7 At Astato
Rn Radón
+1 Fr Francio
+2 Ra Radio
Alcalinos
Alcalinotérreos
Metales
Metaloides
No Metálicos
Halógenos
Gases Nobles
Lantánidos/Actínidos
Valencia
Símbolo
Nombre
Negro: natural
gris: artificial

Figure (9.6)